

Introduction to Bioimage Analysis with Fiji/ImageJ

Pasteur NEUBIAS Course on Bioimage Analysis

Early Career Investigators Track

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Who has done image analysis
before?

Who has used Fiji/ImageJ before?

What are your backgrounds?

Plan for today

I/ Fiji

II/ Fundamentals

III/ Quantification

Hands-on 1: Count nuclei and determine which ones are green

Hands-on 2: Quantify the number of red spots per nucleus

IV/ Intro to automation

Plan

I/ Fiji

Image analysis software: the usual suspects

Flavors of ImageJ

Installing and updating

Plugins

Opening images

I/ Fiji: installing and updating, installing plugins

BIA software: the usual suspects

Open source



Fiji/ImageJ

This workshop



Icy



Ilastik



CellProfiler
cell image analysis software

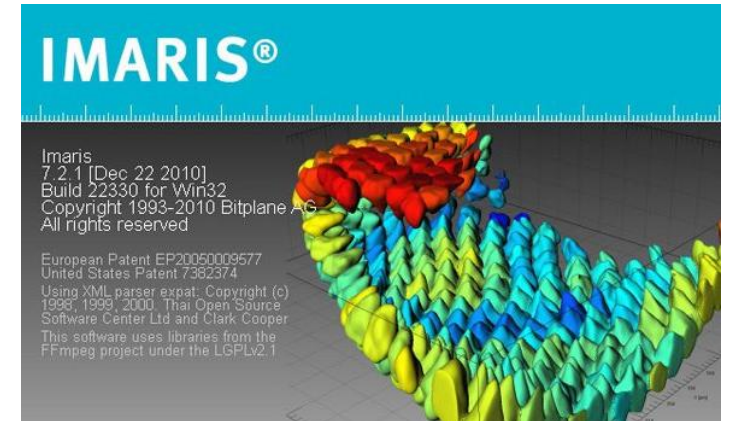


Napari



QuPath

Commercial



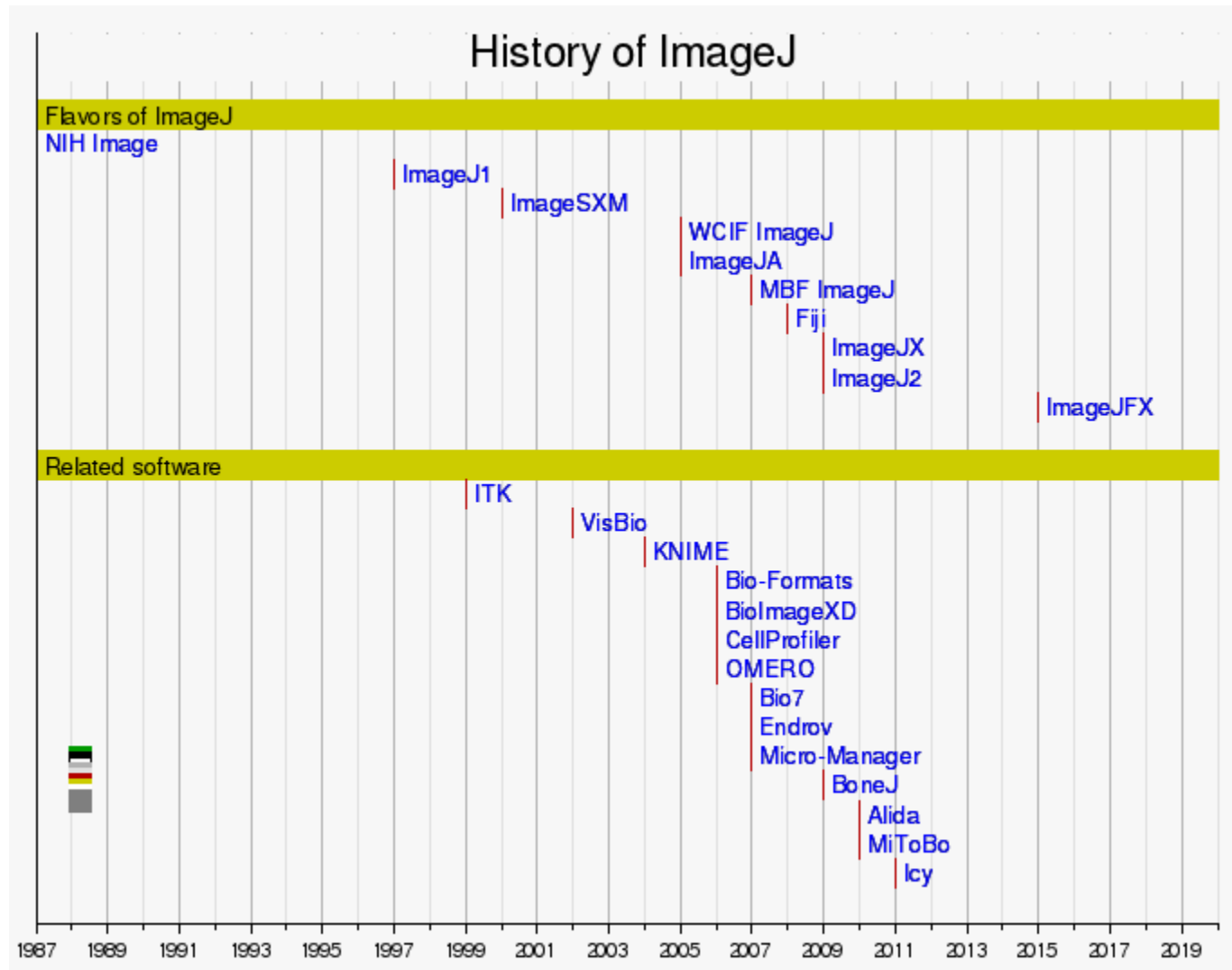
Haase, Robert, Elnaz Fazeli, David Legland, Michael Doube, Siân Culley, Ilya Belevich, Eija Jokitalo, Martin Schorb, Anna Klemm, et Christian Tischer. « A Hitchhiker's Guide through the Bio-Image Analysis Software Universe ». *FEBS Letters* 596, n° 19 (2022): 2472-85. <https://doi.org/10.1002/1873-3468.14451>.

I/ Fiji: Flavors of ImageJ

There are a many different derivatives of ImageJ with very similar names, and some confusion is inevitable. Below is a table which should help to clarify the purpose of each. For the historical context of these projects, see [History](#) below.

	Name	Author/Maintainer(s)	Description	Initiated	Status
	ImageJ	Wayne Rasband	The original version of ImageJ which has been in development since 1997. It has a strong, established user base, with thousands of plugins and macros for performing a wide variety of tasks. Sometimes referred to as the original ImageJ or ImageJ 1.x for technical clarity, or informally with the shorthands ImageJ1 or IJ1 .	1997	Active
	ImageJ2	ImageJ2 developers	A new version of ImageJ targeting scientific multidimensional image data. It is a complete rewrite of the original ImageJ , but includes the original ImageJ with a compatibility layer, so that old-style plugins and macros can run the same as always. ImageJ2 provides several significant new features, such as an automatic updater , and improved scripting capabilities.	Dec. 2009	Active
	Fiji	Fiji contributors	Fiji is Just ImageJ , with extras. It is a distribution of ImageJ and ImageJ2 with many plugins useful for scientific image analysis in fields such as life sciences. It is actively maintained, with updates released often.	Dec. 2007	Active
	ImageJ.JS	Wei Ouyang	ImageJ.JS is a web version of ImageJ that runs in the browser without installation, compiled from Java to JavaScript using the Cheerpj compiler and integrated with the ImJoy plugin system. It's accessible from https://ij.imjoy.io and also supports mobile devices and tablets.	2020	Active

I/ Fiji: Flavors of ImageJ



I/ Fiji: Installation

<https://fiji.sc/>



Fiji

Fiji is an image processing package — a "batteries-included" distribution of [ImageJ](#), bundling many plugins which facilitate scientific image analysis.

 **Download**

latest | Windows | x86-64 | FR

Distribution: Latest Stable

Mirror: US UK FR

[More Downloads](#) [Cite](#) [Contribute](#)

Why Fiji?



Easy to Use

Fiji is easy to use and install - in one-click, Fiji installs all of its plugins, features an automatic updater, and offers comprehensive documentation.



Powerful

Fiji bundles together many popular and useful ImageJ plugins for image analysis into one installation, and automatically manages their dependencies and updating.



Free & Open Source

Like ImageJ itself, Fiji is an [open source](#) project hosted on [GitHub](#), developed and written by the community.

<https://imagej.net/contribute/citing>

<https://imagej.net/software/fiji/downloads>



Fiji Downloads

Fiji is a distribution of ImageJ which includes many useful plugins [contributed by the community](#).

Operating system		Download links
	Linux (64-bit)	imagej.net (USA) micron.ox.ac.uk (Europe)
	macOS (Apple silicon)	imagej.net (USA)
	macOS (Intel)	imagej.net (USA) micron.ox.ac.uk (Europe)
	Windows (64-bit)	imagej.net (USA) micron.ox.ac.uk (Europe)
	Windows (32-bit)	imagej.net (USA) micron.ox.ac.uk (Europe)
	No JRE	imagej.net (USA) micron.ox.ac.uk (Europe)

- Download for your OS
- Unzip (!\ Somewhere you have full permissions!)

I/ Fiji: Installation

Explore your Fiji directory

To launch Fiji,
double click on the .exe
You can also create a
shortcut on your desktop.

Nom	Type	Taille compressée
config	Dossier de fichiers	
Contents	Dossier de fichiers	
Fiji.app	Dossier de fichiers	
images	Dossier de fichiers	
jars	Dossier de fichiers	
java	Dossier de fichiers	
lib	Dossier de fichiers	
licenses	Dossier de fichiers	
luts	Dossier de fichiers	
macros	Dossier de fichiers	
plugins	Dossier de fichiers	
retro	Dossier de fichiers	
scripts	Dossier de fichiers	
db.xml.gz	7-zip.gz	
fiji-windows-x64.exe	Application	
ImageJ-win64.exe	Application	
README.md	Fichier MD	
WELCOME.md	Fichier MD	

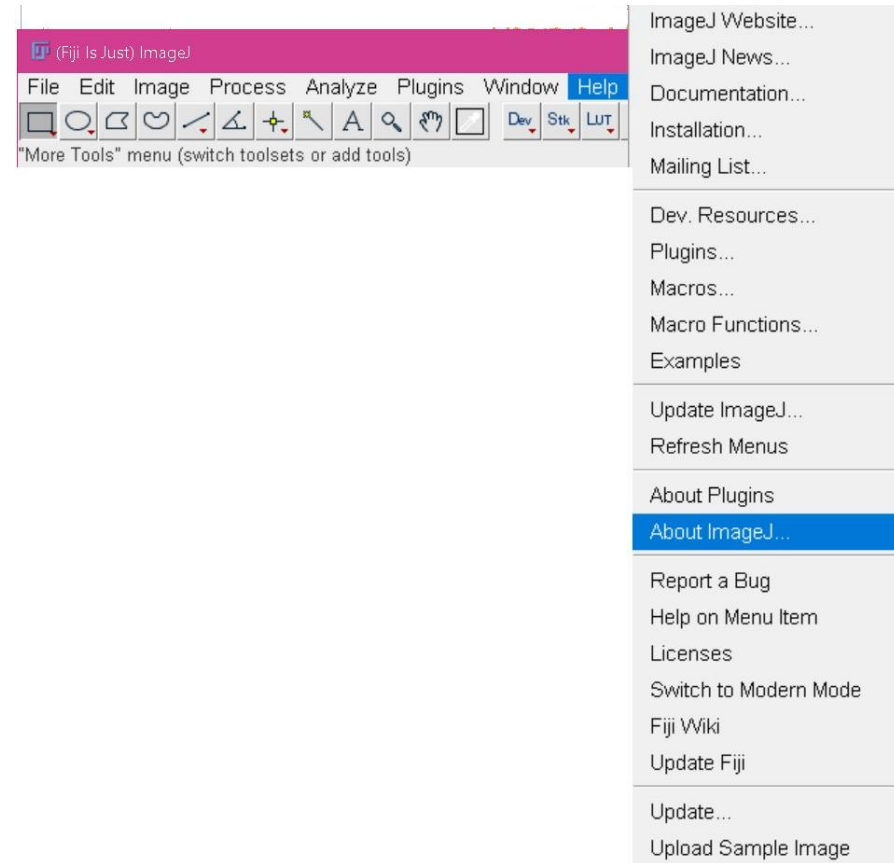
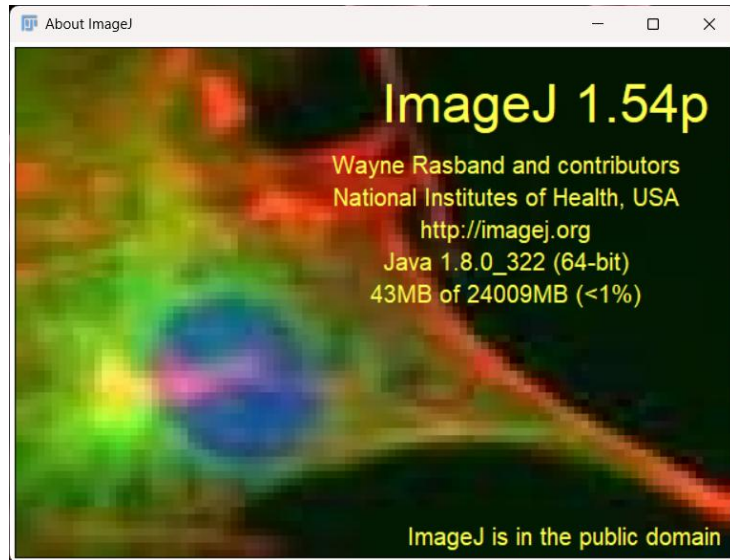


Fiji.app (18 éléments)

Sélectionnez un seul fichier pour obtenir plus d'informations et partager votre contenu cloud.

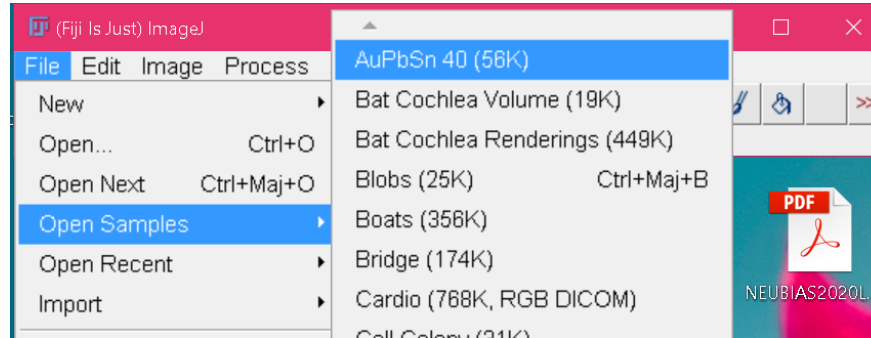
I/ Fiji: Installation

Check the version: *Help > About ImageJ*

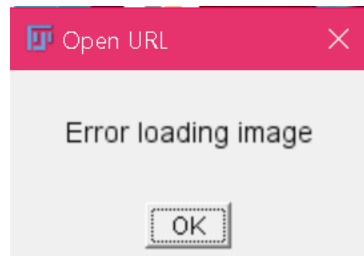


I/ Fiji: Installation

Try to do *File > Open Samples > AuPnSn 40 (56k)*



Do you get



?

I/ Fiji: Installation: Plugins

Extending Fiji: plugins: <https://imagej.net/list-of-extensions>,
<https://imagej.net/list-of-update-sites/>

▼ Categories filter

☐ All ☒ Any

☒ 3D	☒ Deep Learning	☒ ImgLib	☒ Noise	☒ Stacks
☒ Analysis	☒ Denoising	☒ Import-Export	☒ OME	☒ Stitching
☒ Annotation	☒ Development	☒ Integral Image	☒ Object Detection	☒ Super-resolution
☒ Automation	☒ Digital Volume Flattening	☒ Interactive	☒ Ops	☒ Time Signal Analysis
☒ Benchmark	☒ Digital Volume Unrolling	☒ Kymograph	☒ Optic Flow	☒ Tissue
☒ Binary	☒ Entropy	☒ Lifetime	☒ Particle Analysis	☒ Track analysis
☒ CellProfiler	☒ Example Data	☒ MATLAB	☒ Pattern Recognition	☒ Tracking
☒ Channel alignment	☒ Fast Fourier Transformation	☒ MRI	☒ Perfusion	☒ TrakEM2
☒ Cilia	☒ Feature Extraction	☒ Machine Learning	☒ Photogrammetry	☒ Transform
☒ Classification	☒ Filtering	☒ Macro	☒ Photography	☒ Tutorials
☒ Colocalization	☒ Fractal Dimension	☒ MacroExtensions	☒ Plotting	☒ Uncategorized
☒ Color Processing	☒ Frequency Analysis	☒ Mathematical Morphology	☒ Projection	☒ Unmaintained
☒ Complexity	☒ Gut	☒ Microscopy	☒ Registration	☒ User Interface
☒ Cookbook	☒ Help	☒ Microtubules	☒ SciJava	☒ Utilities
☒ Curvature	☒ Image Annotation	☒ Montage	☒ Scripting	☒ Visualization
☒ Curve Tracing	☒ ImageJ2	☒ Neuroanatomy	☒ Segmentation	
☒ Deconvolution	☒ ImageScience	☒ Neuron	☒ Skeleton	

 Save as animated gif Import-Export	 Descriptor based registration Uncategorized	 Make Screencast Scripting	 Sigma Filter Filtering
 Bio-Formats Import-Export, SciJava, OME	 Detrendr Uncategorized	 Malaria Sporozoites (9.2MB) Uncategorized	 SIMcheck Uncategorized
 DICOM Import-Export	 DHM Utilities Uncategorized	 Getting started with MaMuT Tracking, Tutorials	 Simulate Color Blindness Color Processing

I/ Fiji: Installation: Plugins: *Hands-on!*

Example of plugin installation: Stardist



Step 1

Open Fiji Updater on Fiji

Help > Update...

Step 2

Click on Manage update sites and check the following sites

Update sites : CSBDeep, StarDist and TensorFlow

Tip: use the Search bar!

Step 3

Close Manage update sites and Click on Apply changes

Step 4

Restart Fiji/ImageJ

Sometimes a simple Help> Refresh can do the job

Step 5

The plugin is ready to use

It should appear under Plugins>StarDist

<https://imagej.net/plugins/stardist>

I/ Fiji: Installation: Plugins: *Hands-on!*

1. **Update...**

2. Please wait

3. **OK**

4. **Manage update sites**

5. **Bio-Formats**

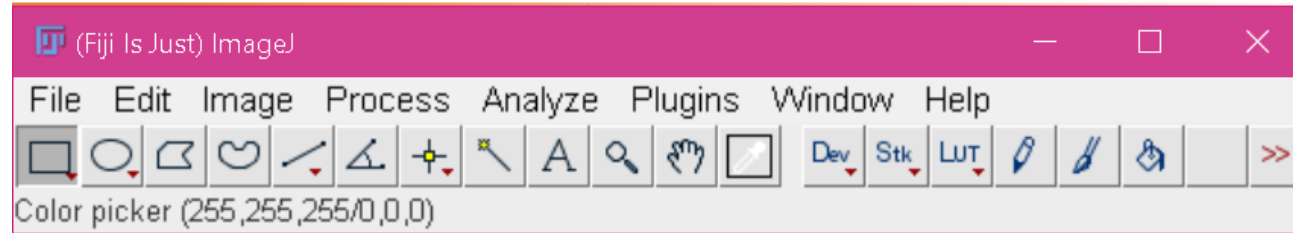
6. **Close**

7. **Apply changes**

Name	Status/Action	Update Site
plugins/Algorithm_Launcher.jar	Uninstall it	Fiji
plugins/BeanShell_Interpreter.jar	Uninstall it	Fiji
plugins/CLI.jar	Uninstall it	Fiji
plugins/Clojure_Interpreter.jar	Uninstall it	Fiji
plugins/Fiji_Updater.jar	Uninstall it	Fiji
plugins/JRuby_Interpreter.jar	Uninstall it	Fiji
plugins/JavaScript.jar	Uninstall it	Fiji
plugins/Jython_Interpreter.jar	Uninstall it	Fiji
plugins/Refresh_Javas.jar	Uninstall it	Fiji
plugins/IJ-ImageIO.jar	Uninstall it	Fiji
jars/ant-junit.jar	Uninstall it	Fiji
jars/ant-launcher.jar	Uninstall it	Fiji
jars/ant-nopds.jar	Uninstall it	Fiji
jars/ant.jar	Uninstall it	Fiji
jars/batik.jar	Uninstall it	Fiji
jars/bcmail-jdk14-138.jar	Uninstall it	ImageJ
jars/bcprov-jdk14-138.jar	Uninstall it	ImageJ
jars/bio-formats/formats-common.jar	Uninstall it	Bio-Formats
jars/bio-formats/mbdtools-java.jar	Uninstall it	Bio-Formats
jars/fake.jar	Uninstall it	Fiji
jars/fiji-scripting.jar	Uninstall it	Fiji
jars/itext.jar	Uninstall it	ImageJ
jars/jack.jar	Uninstall it	Fiji
jars/java-cup.jar	Uninstall it	Fiji
jars/is.jar	Uninstall it	Fiji

I/ Fiji: Exploring the menu

Going through the menu



I/ Fiji: Opening an image: *Hands-on!*

Two methods:

- File > open
- Drag and drop onto the Fiji bar

Step 1

Open boats.tif

Step 2

Move your cursor around on the image

Question

What do you notice?

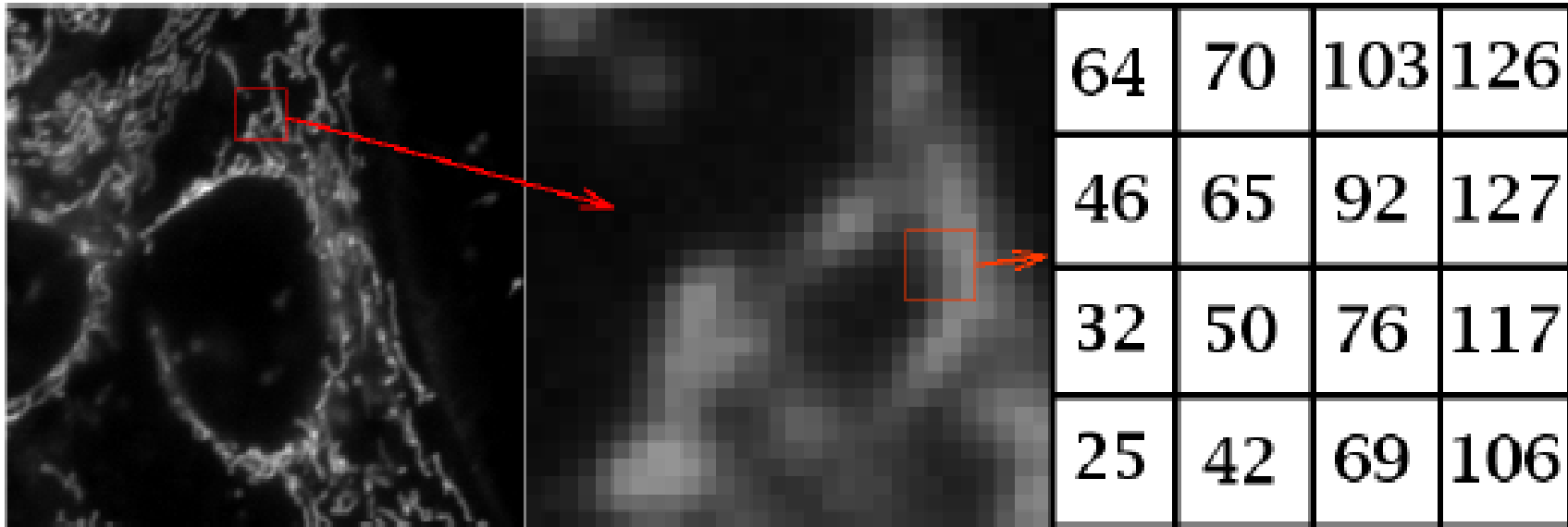
Rule number 1 in Fiji:

Anything you will do on an image will apply on the **active image**, i.e. the last image...

- which you have opened,
- or clicked on,
- or which was created by a process!

II/ Fundamentals

II/ Fundamentals: What is a digital image?

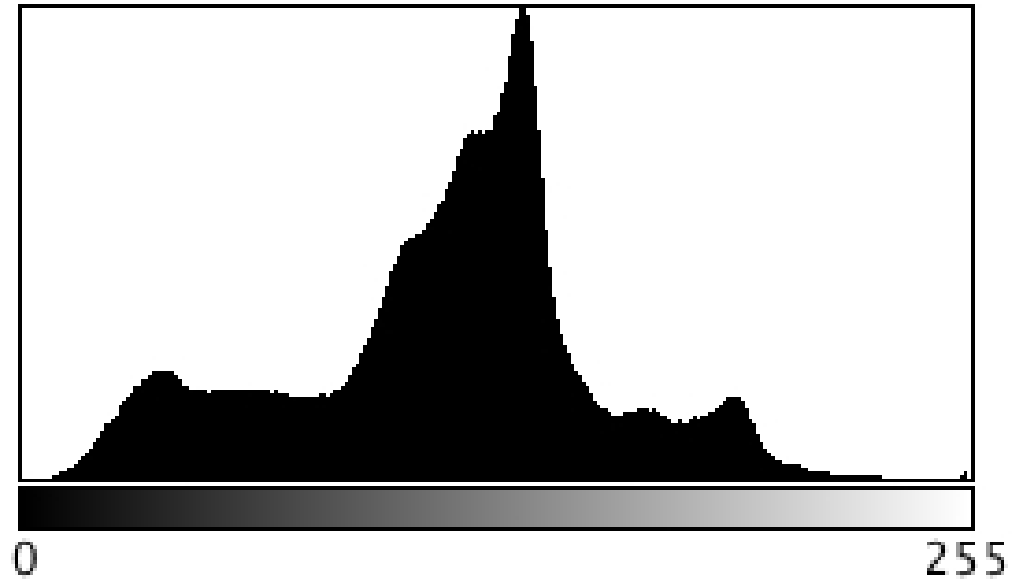
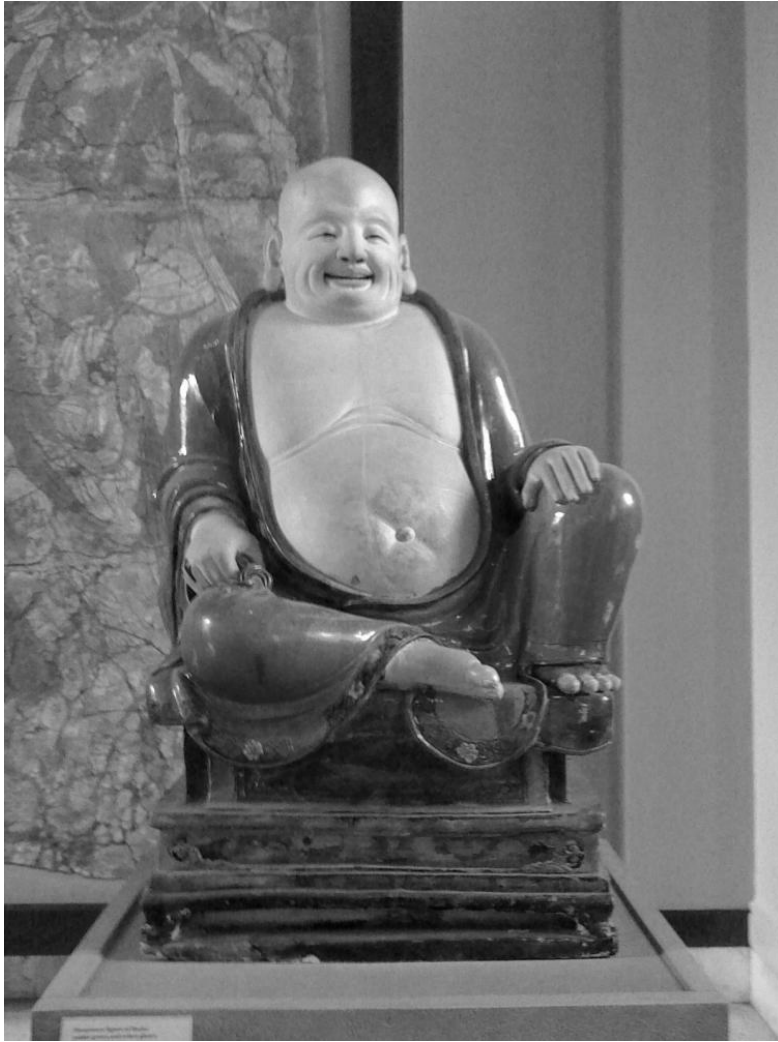


It is a matrix! There are only numbers, that we're going to name **intensity**

The cells are pixels (PICTure ELelement)

In 3D = voxels (VOlumoc Element)

II/ Fundamentals: Histogram

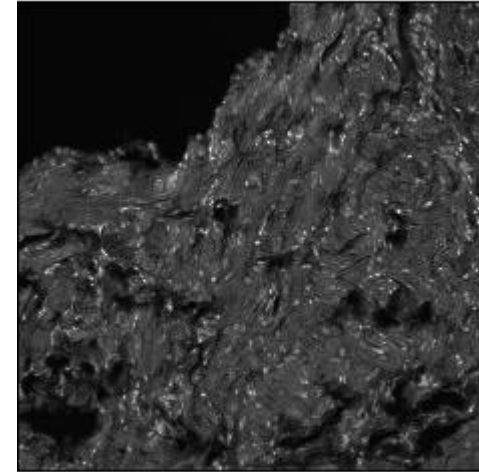
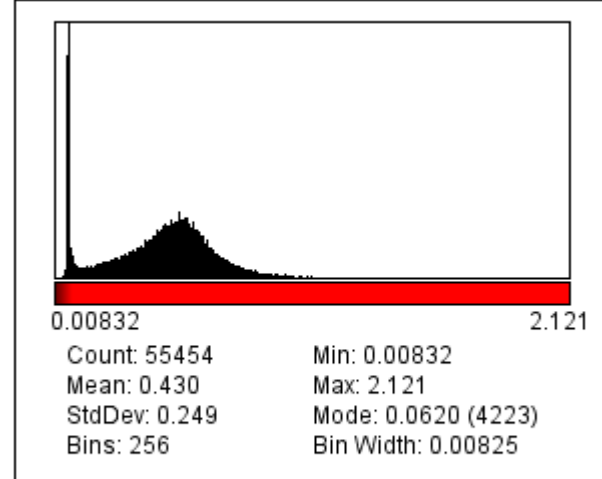


Count: 1920000
Mean: 113.658
StdDev: 42.343
Bins: 256

Min: 0
Max: 255
Mode: 135 (33747)
Bin Width: 0.996

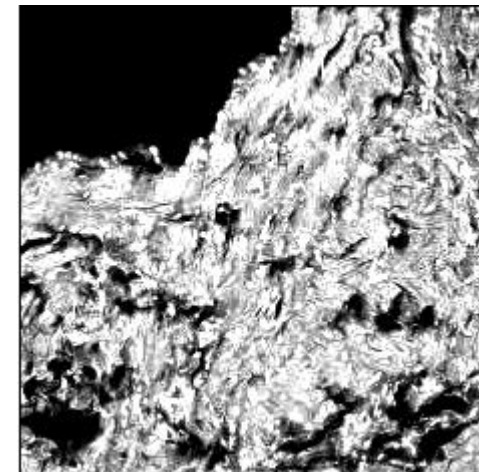
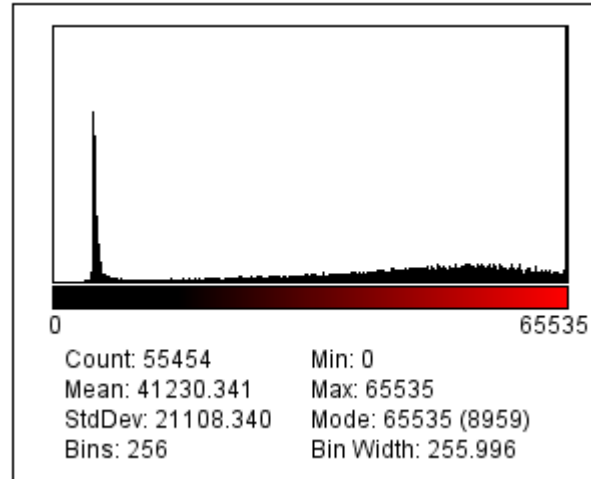
II/ Fundamentals: Saturation

300x240 pixels; RGB; 281K



value=0.668
count=410

300x240 pixels; RGB; 281K



II/ Fundamentals: Display: Brightness Contrast

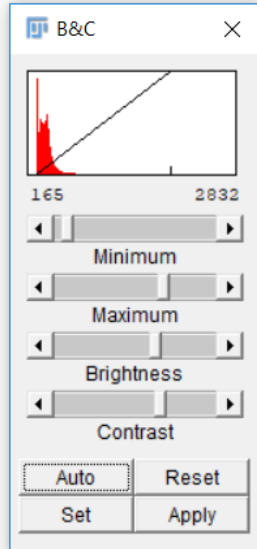


Image > Adjust > Brightness/Contrast...
Shift + C

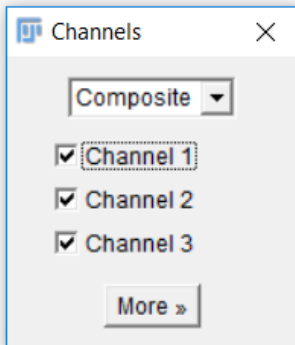
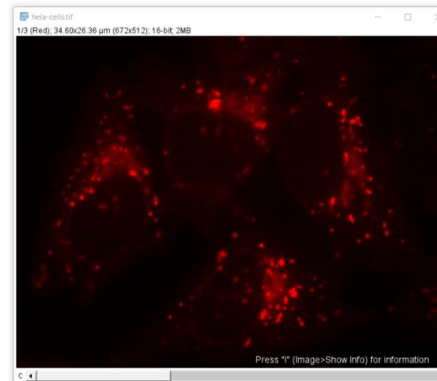
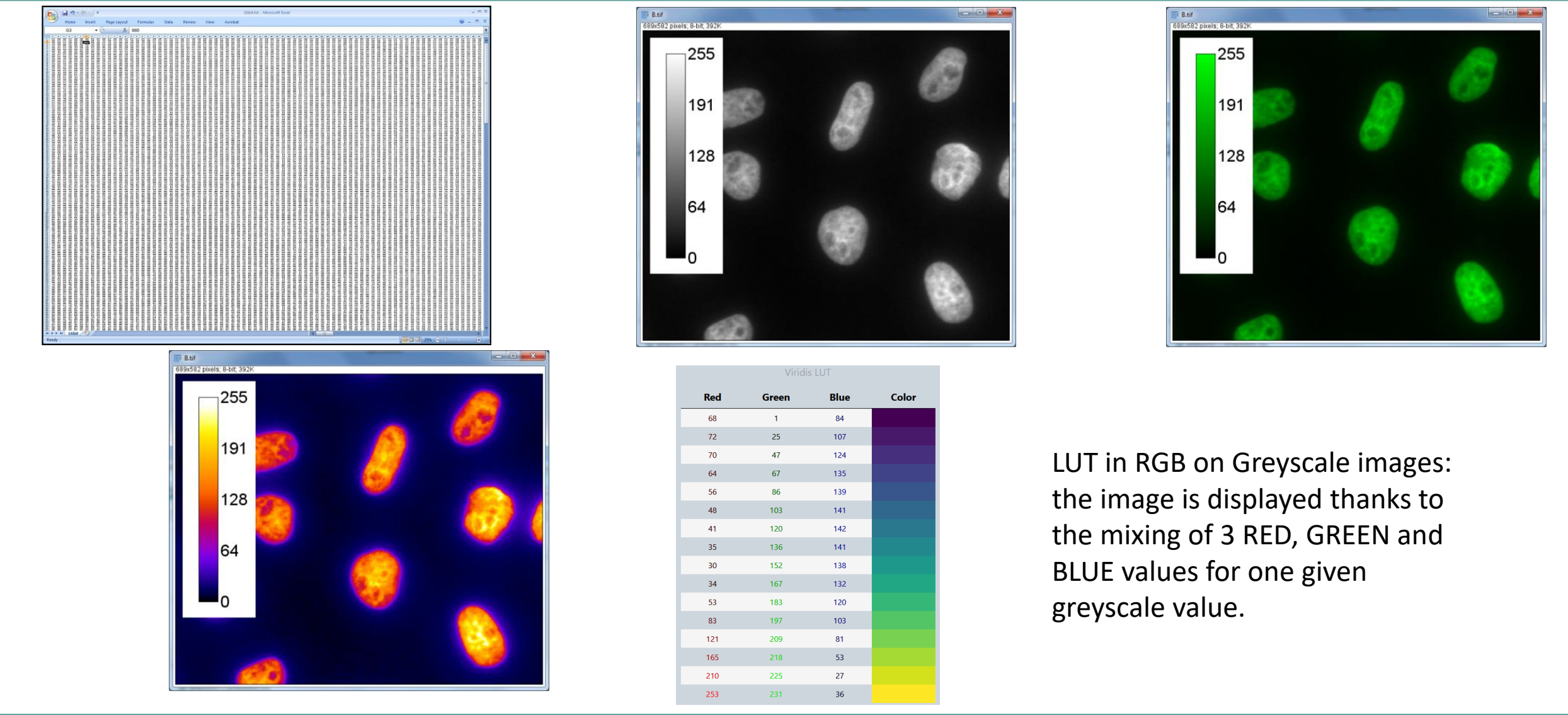


Image > Color > Channels Tool...
Shift + Z

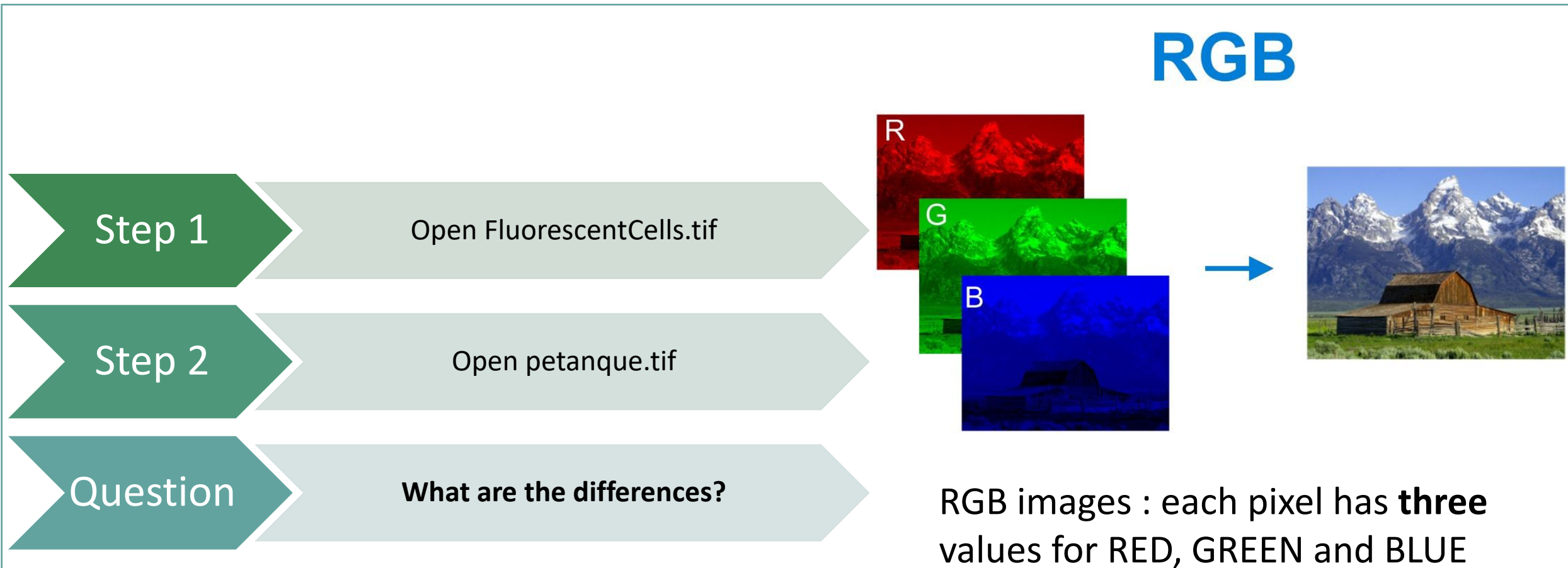


II/ Fundamentals: Display: Lookup Table (LUT) or colormap



LUT in RGB on Greyscale images:
the image is displayed thanks to
the mixing of 3 RED, GREEN and
BLUE values for one given
greyscale value.

II/ Fundamentals: RGB images



II/ Fundamentals: *Read the information about an image*

 **File>Open Samples> Mitosis**

Aim: Get the dimensions, the sampling or pixel size (voxel in 3D), the bit depth and the dynamic range of mitosis

Step 1

Get the dimensions (in X? Y? Z? C? T?)

*Observe the Information above the image.
Alternatively Image> Show Info*

Step 2

Get the bit depth

*Observe the Information above the image.
Alternatively Image> Show Info*

Step 3

Get the pixel size/ voxel size

Image>Properties

Step 4

Get the dynamic range

*Process>Histogram (min, max), Alternatively
Image>Show Info*

II/ Fundamentals: *Read the information about an image*



File>Open Samples> Mitosis

Aim: Get the dimensions, the sampling or pixel size (voxel in 3D), the bit depth and the dynamic range of mitosis

Step 1

Get the dimensions (in X? Y? Z? C? T?)

X=171, Y=196, Z=5, C=2, T=51, 0,14286 seconds between frames

Step 2

Get the bit depth

Bit-depth = 16 bits

Step 3

Get the pixel size/ voxel size

Pixel(Voxel in 3D) Size XYZ= 0.0885x0.0885x1 μm^3

Step 4

Get the dynamic range

c1 1582-6440; c2 1614-15787

Bonus: C1 and C2 content? Drosophila S2 cell expressing GFP-Aurora B and mCherry-tubulin

II/ Fundamentals: Show Info

Drosophila S2 cell expressing GFP-Aurora B and mCherry-tubulin fusion protein undergoing mitosis. Courtesy of Eric Griffiths.

(Fiji Is Just) ImageJ 2.0.0-rc-69/1.52i; Java 1.8.0_172 [64-bit]; Windows 10 10.0; 3145MB of 12141MB (25%)

Title: mitosis.tif

Width: 15.1335 μm (171)

Height: 17.3460 μm (196)

Depth: 5 μm (5)

Size: 33MB

Resolution: 11.2994 pixels per μm

Voxel size: 0.0885x0.0885x1 μm^3

ID: -26

Bits per pixel: 16 (unsigned)

Display ranges

1: 1582-6440

2: 1614-15787

Frame: 7/510

Frame interval: 0.14286 sec

Channel: 1/2

Slice: 4/5

Frame: 1/51

Composite mode: "composite"

No threshold

ScaleToFit: false

Uncalibrated

URL: <http://imagej.nih.gov/ij/images/Spindly-GFP.zip>

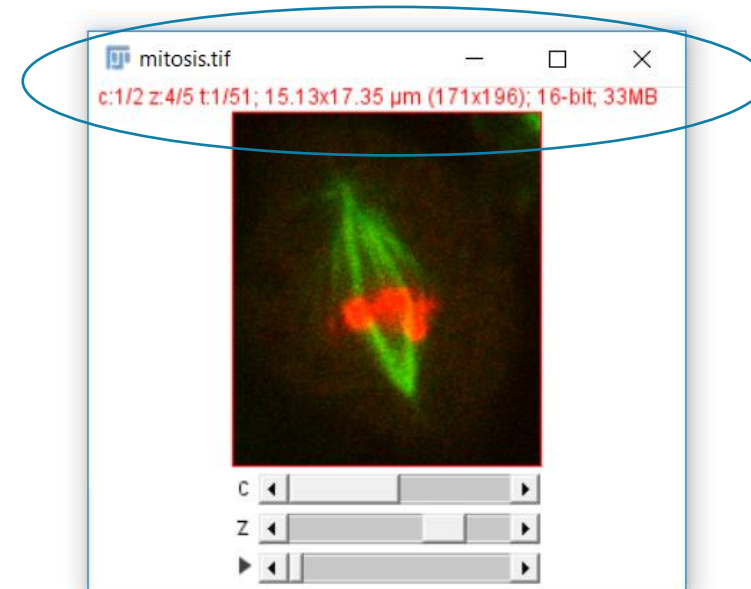
Screen location: 1171,328 (2048x1152)

Coordinate origin: 0,0,0

No overlay

No selection

Image > Show Info (i)



II/ Fundamentals: Hyperstacks

Image 2D

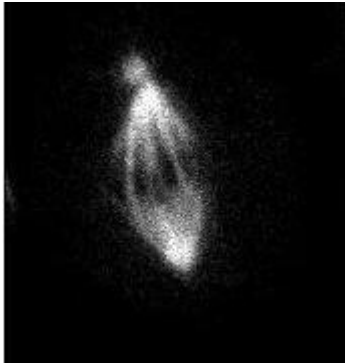


Image 3D:

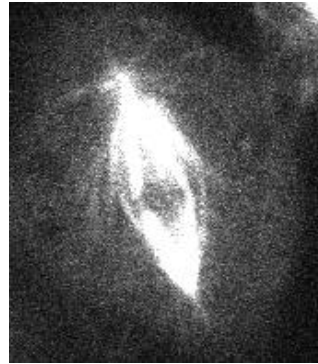
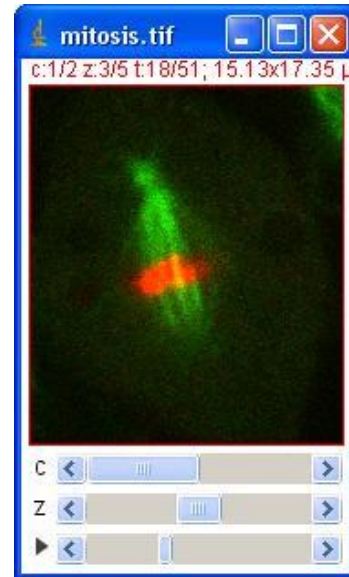
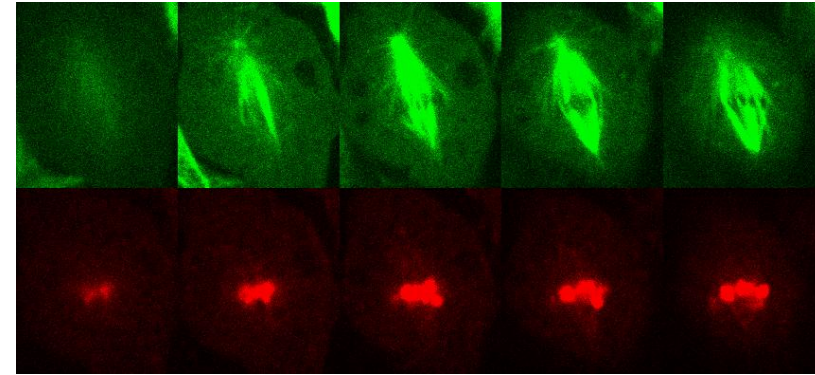


Image 4D+: Hyperstack

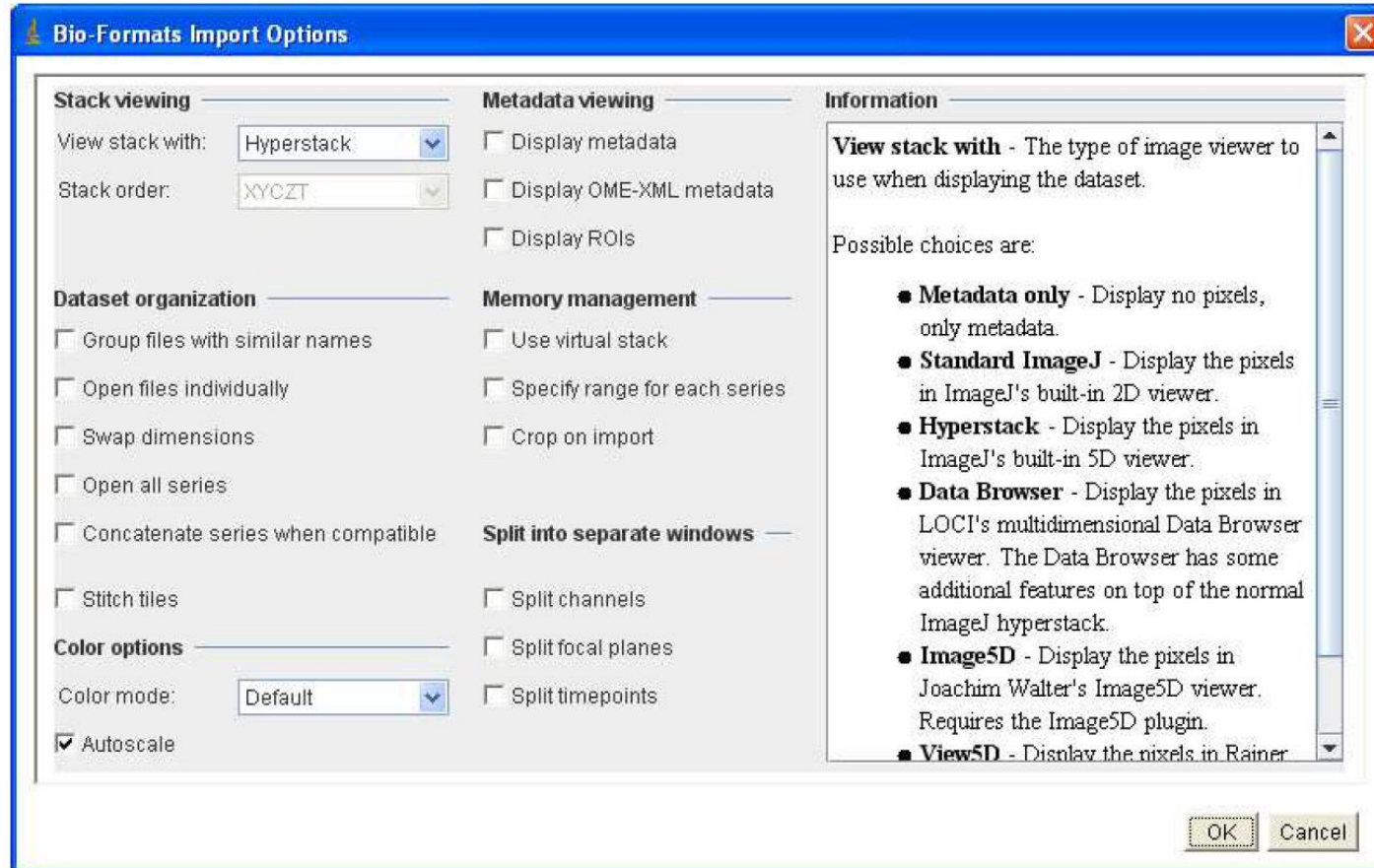


Hyperstack = 5 dimensions

- x
- y
- z (slice)
- channel
- time (Frame)

II/ Fundamentals: Hyperstacks

Can be opened with Bio-Formats



II/ Fundamentals: Calibration: *Hands-on!*

Step 1

Open petanque.tif

Can you measure the distance between the ball and the jack? In cm?

Step 2

Draw a line on something you know the size of

Hint: he's holding a 20 cm ruler ;)

Step 3

Calibrate the image!

Analyze > Set scale

Step 4

Alternative method

If you know the pixel size, input it directly in Image>Properties

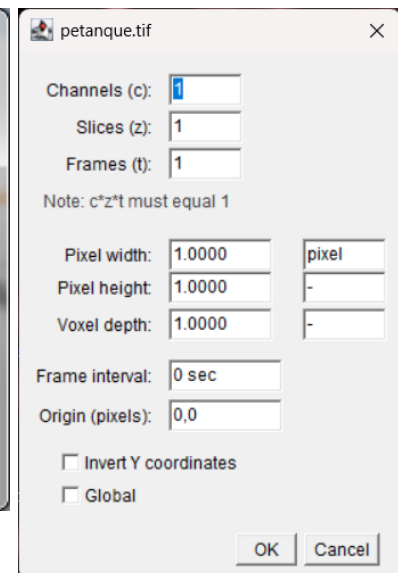
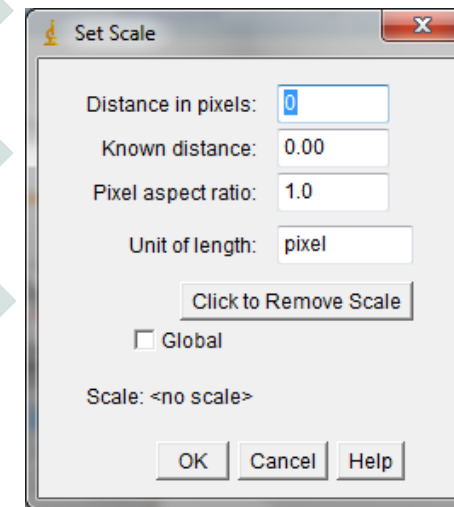
Step 5

Measure the distance in cm between the ball and the jack!



Analyze>Set scale

Image>Properties...



II/ Fundamentals: Recap: commands

Image > Adjust > Brightness/Contrast (**Shift + c**)

Image > Show info (**i**)

Image > Properties (**Shift + p**)

Image > Color > Channels Tool (**Shift + z**)

Analyze > Histogram (**h**)

Image > Lookup Tables

List of shortcuts: <https://imagej.net/ij/docs/shortcuts.html>

II/ Fundamentals: Other basic commands

Edit > Invert (**Shift + i**)

Image > Duplicate (**Shift + d**)

Image > Crop (**Shift + x**)

Image > Rename

Image > Transform > Rotate/Flip ...

Image > Adjust > Size

Image > Color > Split Channels/Merge Channels

Selection > Select All (**a**)

Selection > Select None (**Shift + a**)

Selection > Restore (**Shift + e**)

Selection > Add to Manager (**t**)

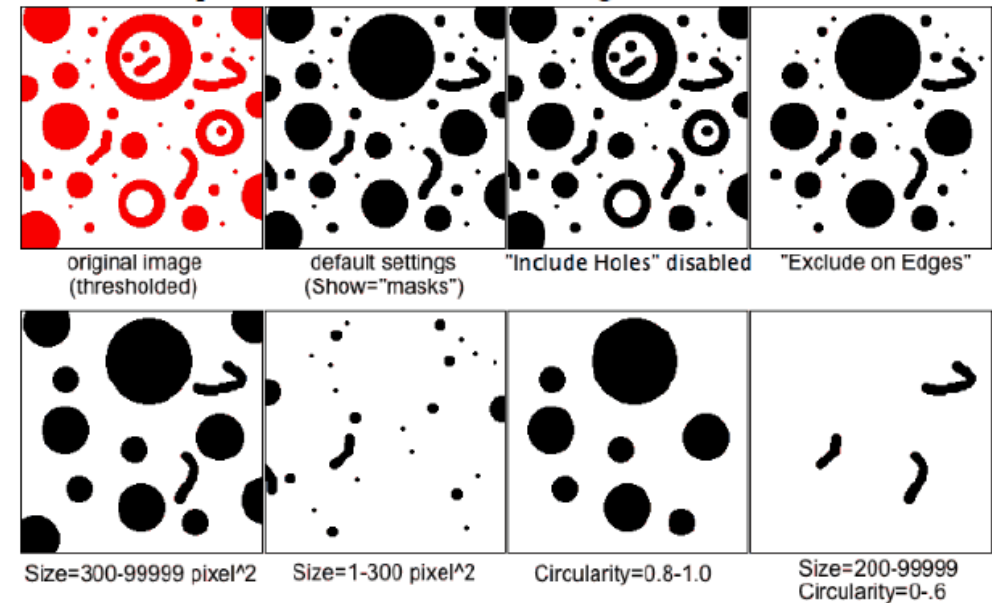
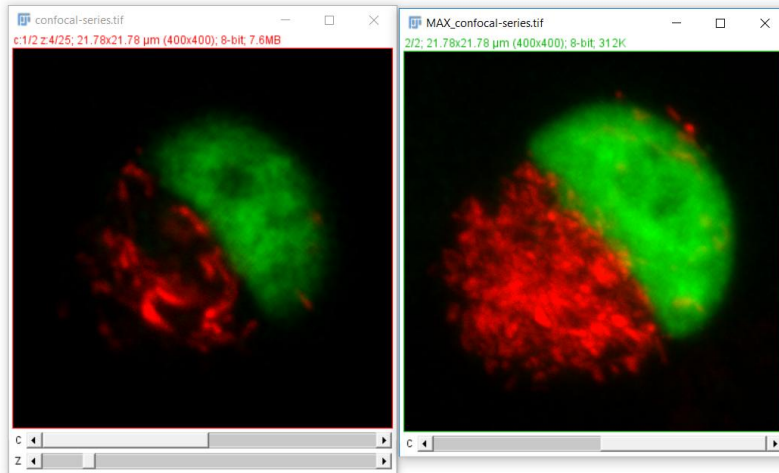
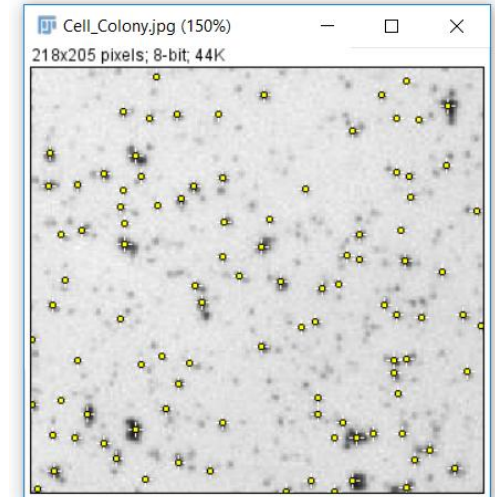
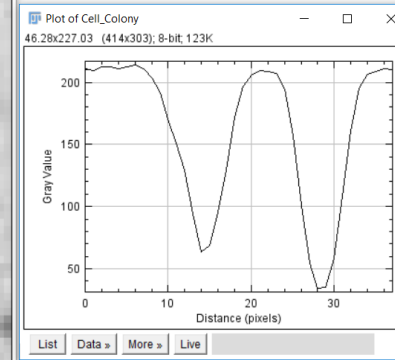
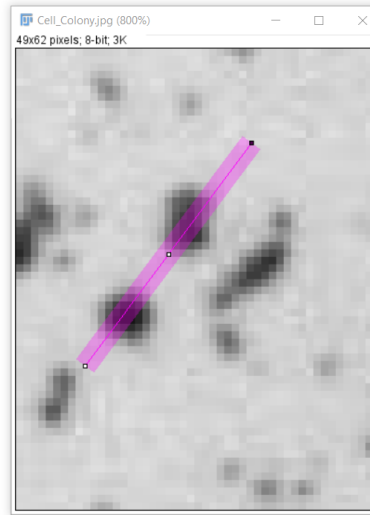
II/ Fundamentals: still a lot left to learn!

Image > stacks > Z project

Process > Find maxima

Analyse > Plot profile

Analyse > Analyse particles



Plan

I/ Fiji

II/ Fundamentals

III/ Quantification

Hands-on 1: Count nuclei and determine which ones are green

Hands-on 2: Quantify the number of red spots per nucleus

IV/ Intro to automation

III/ Quantification: Overview

Concepts we are going to go over:

- Segmentation
- Thresholding
- Mask/ binary image
- ROI: Region of Interest
- Binary operations on masks
- Filters
- Maths on images

Tools we are going to use:

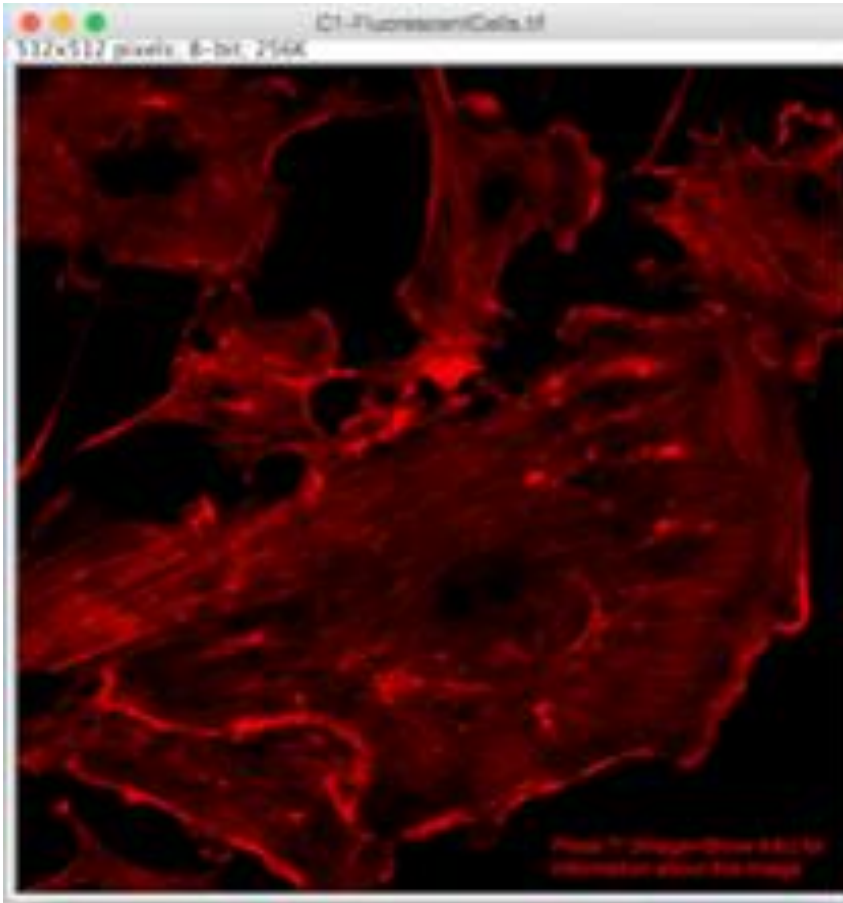
- Threshold
- Analyze Particles
- Watershed
- Z Projection
- Find Maxima



III/ Quantification: Segmentation

Segmentation = division of an image into several parts, generally what we are interested in and what we are not

Binary image/Segmentation mask



III/ Quantification: Segmentation: Thresholding

Segmentation = division of an image into several parts, generally what we are interested in and what we are not

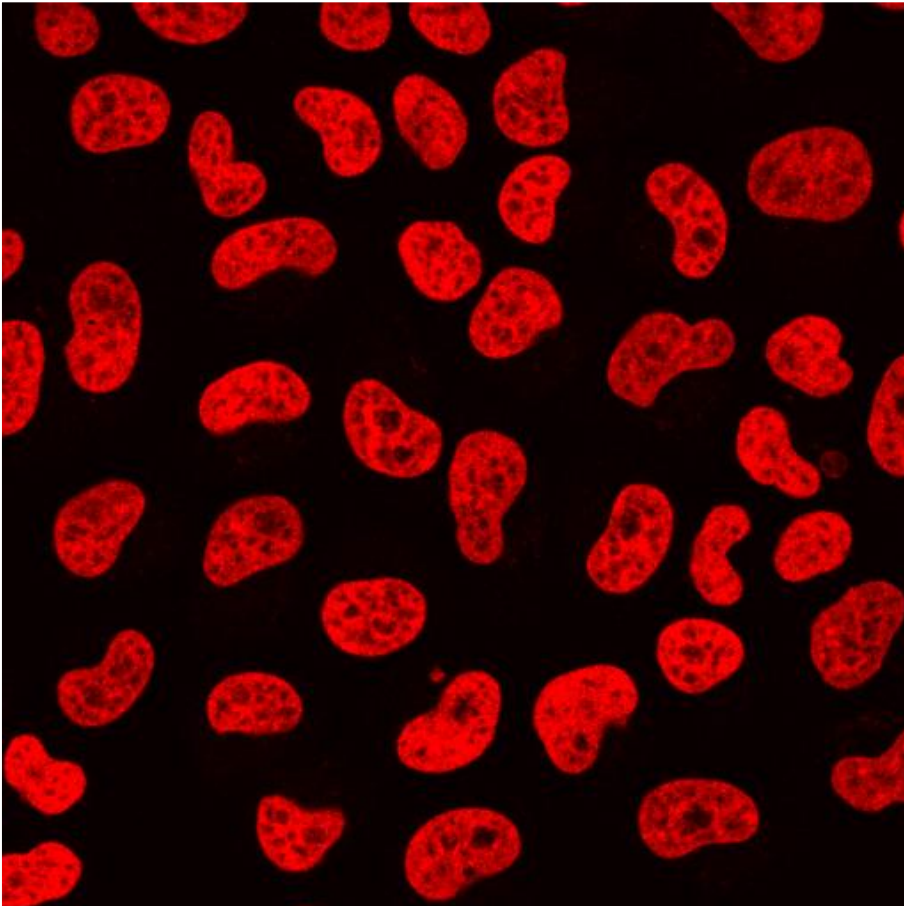


Image > Adjust > Threshold... (Ctrl + t)

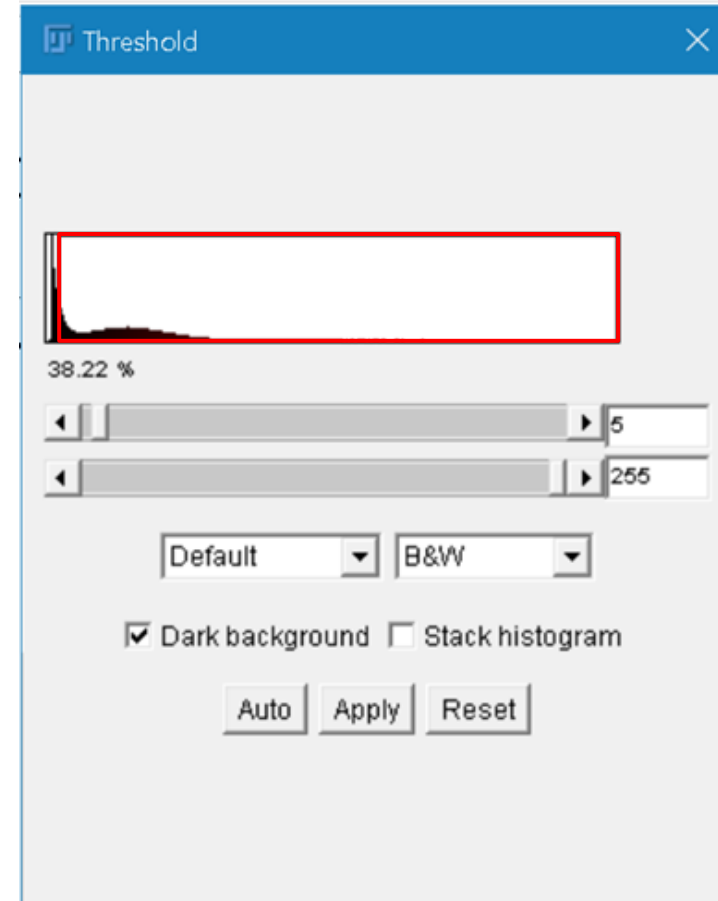
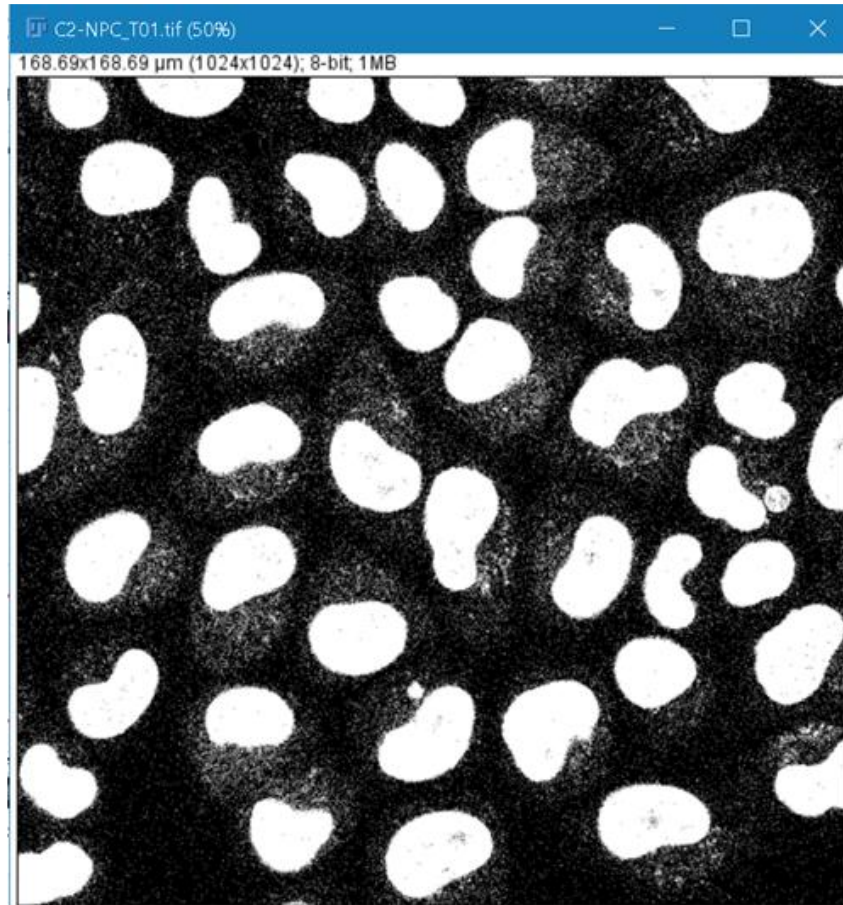


Count: 1048576
Mean: 13.777
StdDev: 19.322

Min: 1
Max: 168
Mode: 2 (578261)

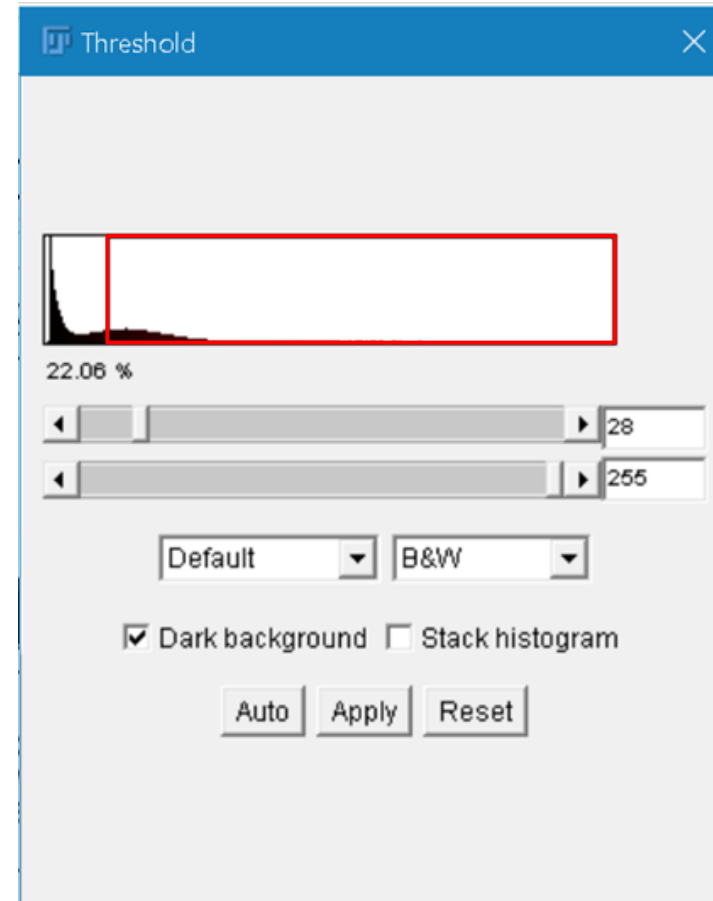
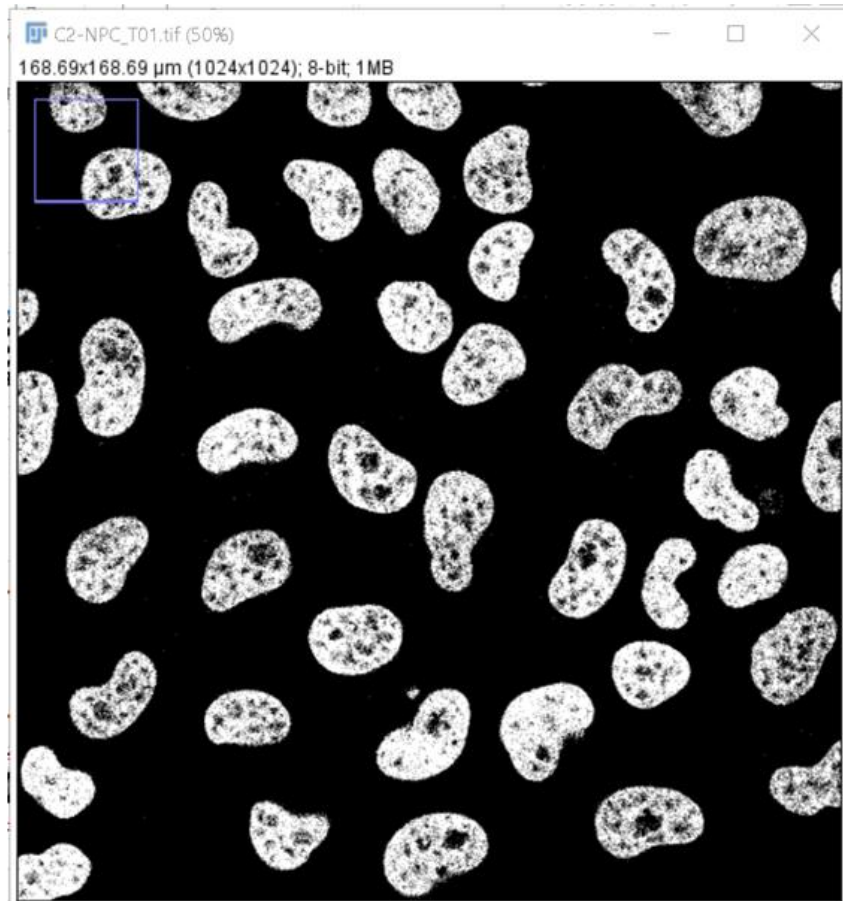
III/ Quantification: Segmentation: Thresholding

Segmentation = division of an image into several parts, generally what we are interested in and what we are not



III/ Quantification: Segmentation: Thresholding

Segmentation = division of an image into several parts, generally what we are interested in and what we are not



III/ Quantification: Segmentation: Thresholding



Exercise 1a

Step 1

File>Open Samples > Blobs

Step 2

Images> Look Up Tables > Invert LUT

Step 3

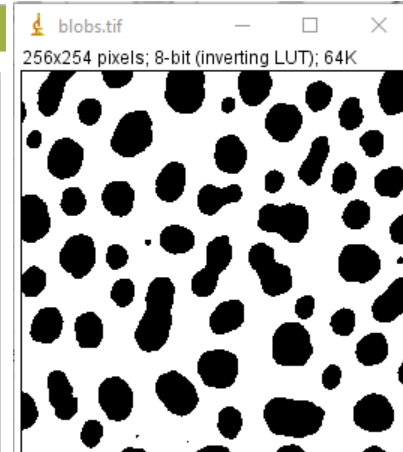
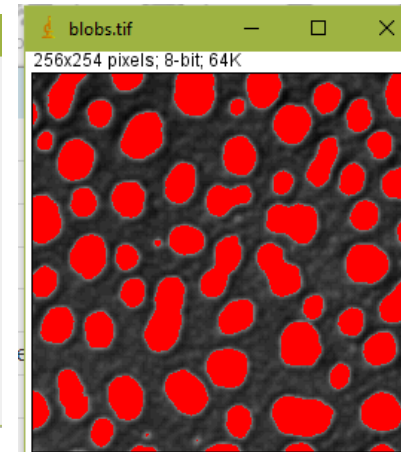
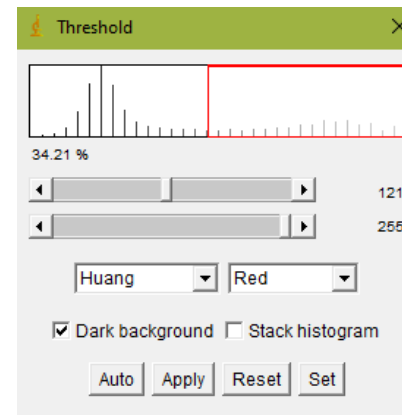
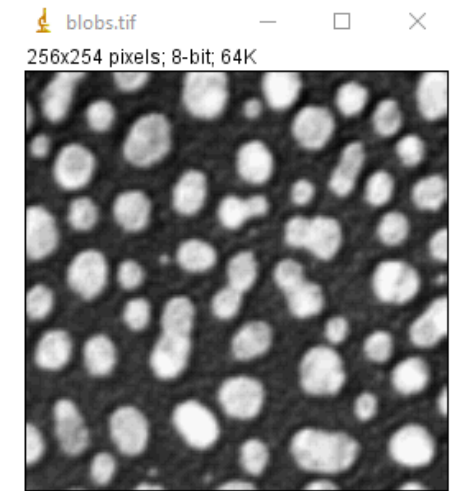
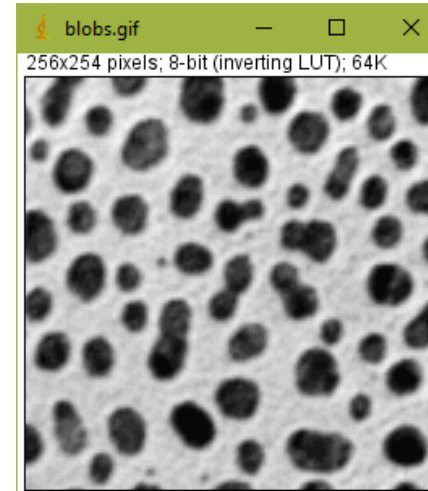
Image> Duplicate

Step 4

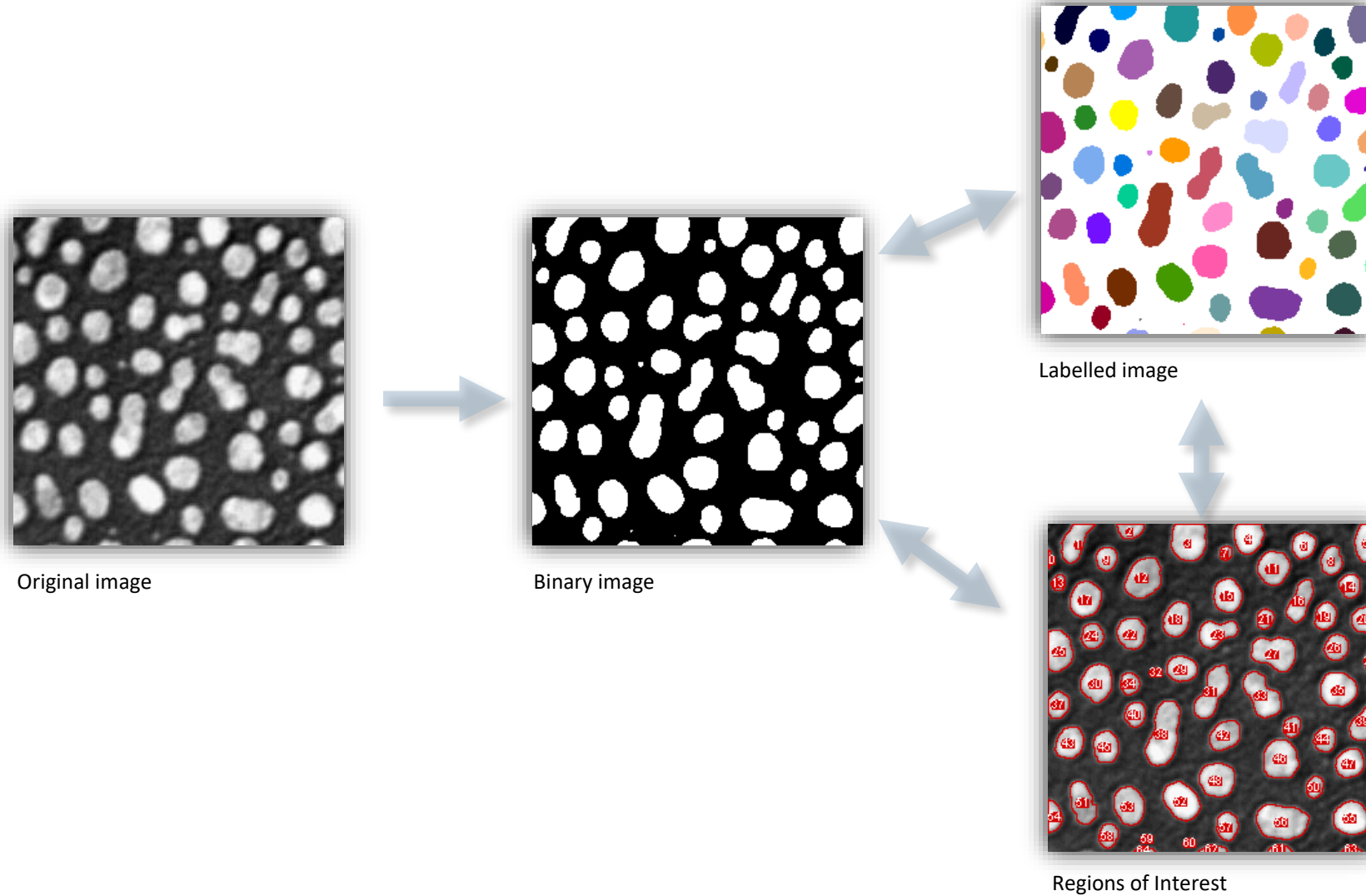
Image> Adjust> Threshold...

Step 5

Choose Threshold method and click on Apply



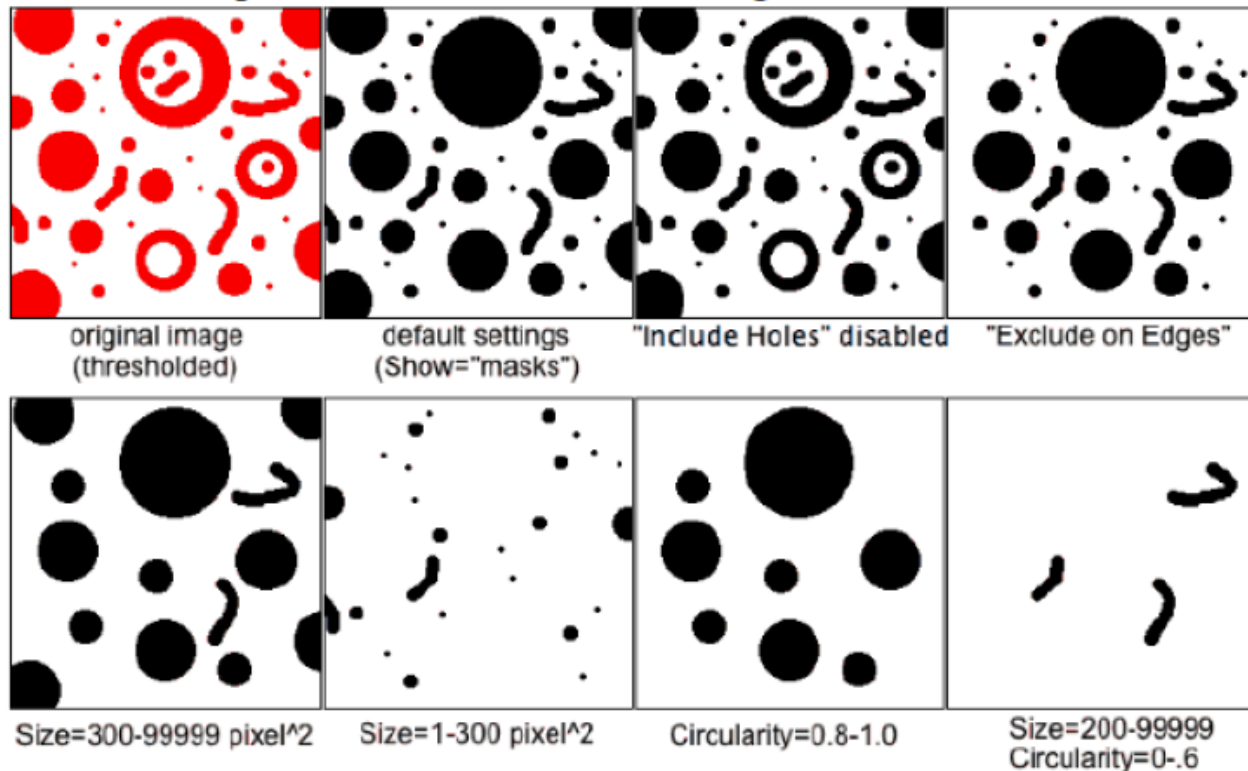
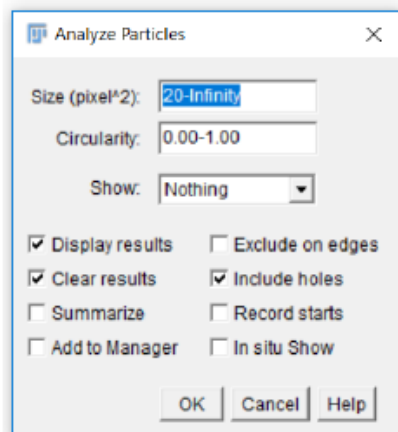
III/ Quantification: Segmentation: Outputs



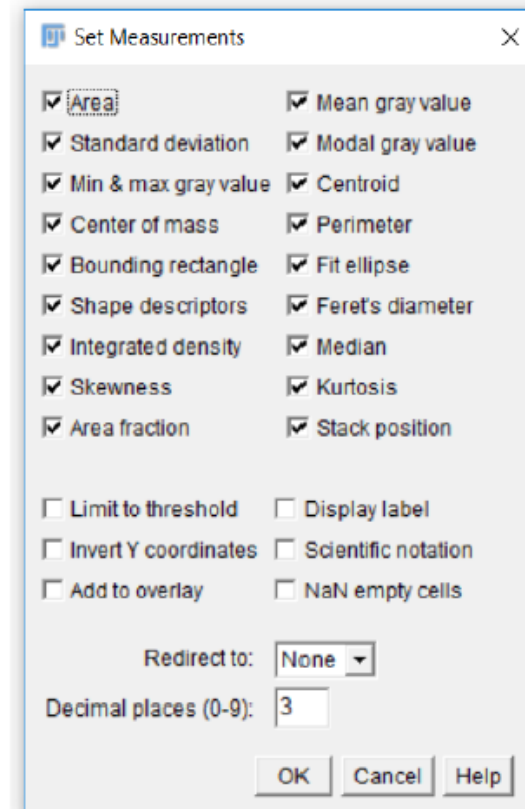
III/ Quantification: Analyze Particles



Analyze > Analyze Particles...

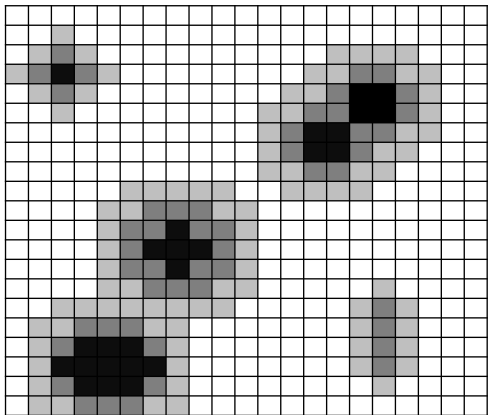


Analyze > Set measurements

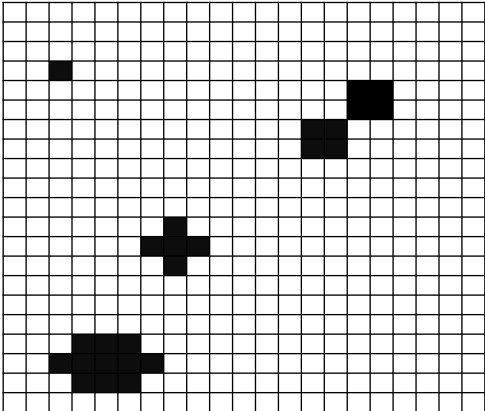


III/ Quantification: Analyze Particles: Connected components analysis

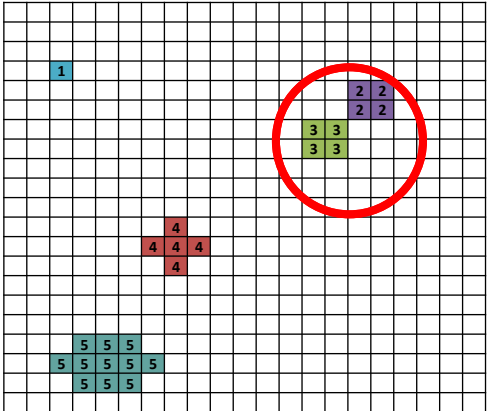
Original



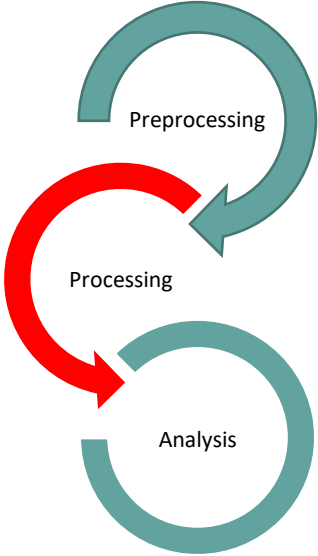
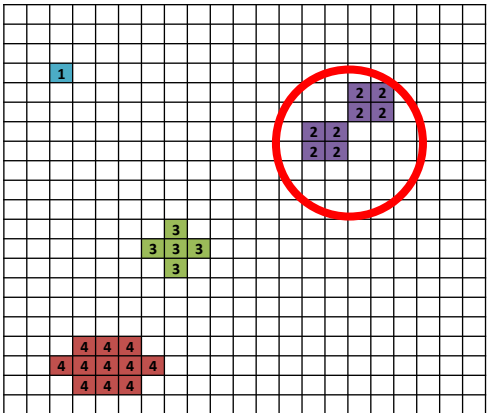
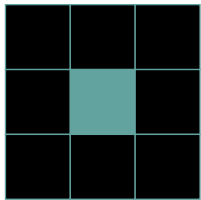
High threshold



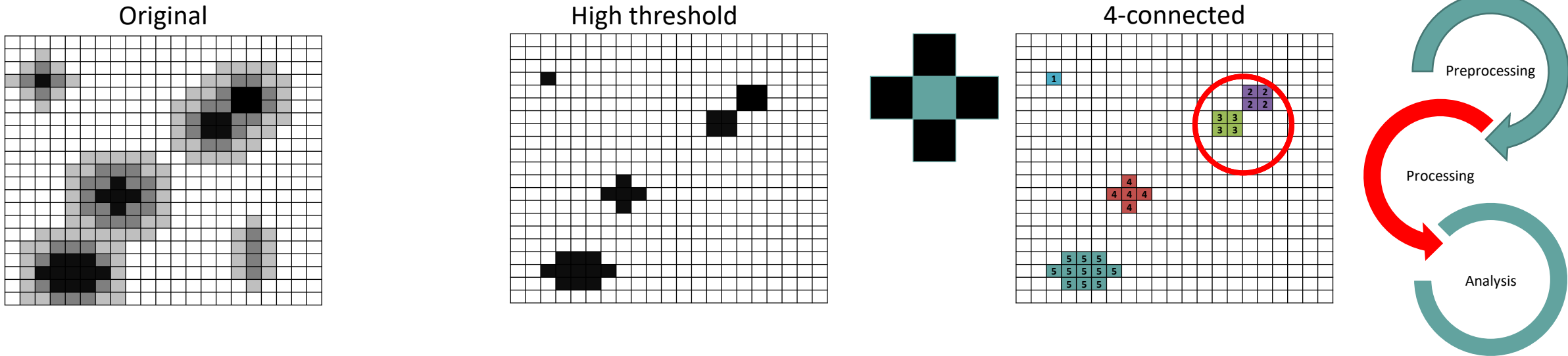
4-connected



8-connected



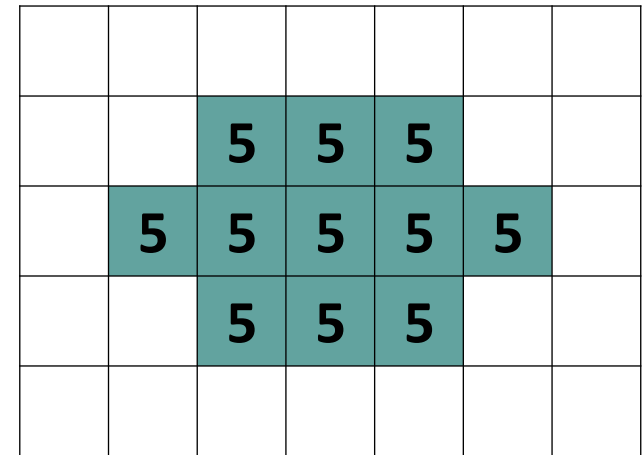
III/ Quantification: Analyze Particles: Connected components analysis



Example of parameters derived from label images:

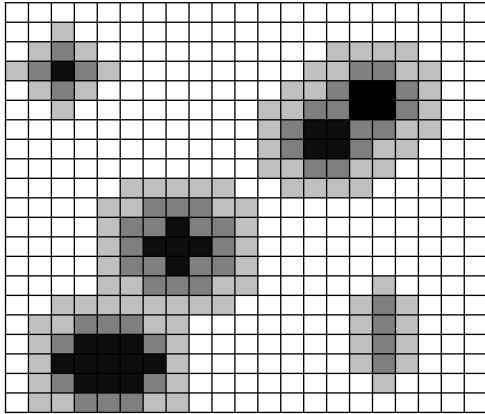
Area: number of pixels carrying the same label (e.g.: **11 pixels "5"**)

Perimeter: number of pixels carrying the same label, and missing at least one neighbour (e.g.: **8 pixels with 4-connected**)

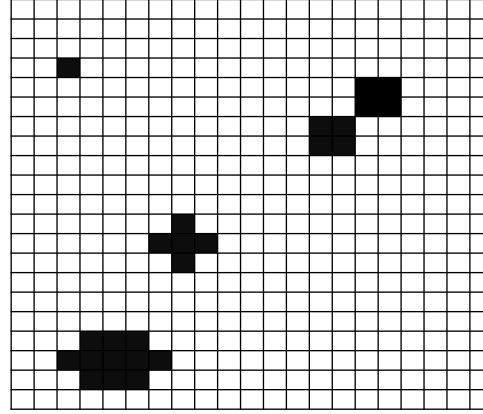


III/ Quantification: Analyze Particles: Connected components analysis

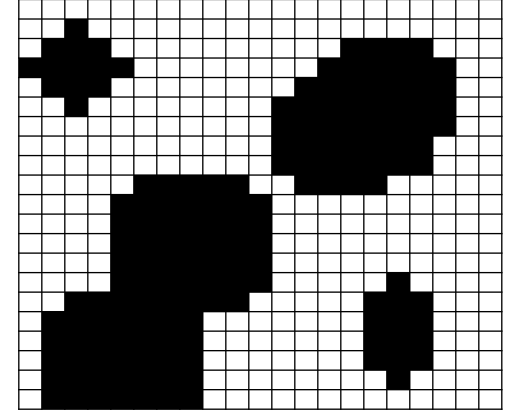
Original



High threshold



Low threshold



Always a bias!
But we should try to always have the
same



III/ Quantification: Analyze Particles: Measurements

Code: *Red = Geometry based*, *Blue = Intensity based*, *Orange = Mixed*

Area Area of selection in square pixels or in calibrated square units (e.g., mm², μm², etc.) if Analyze▷Set Scale...↓ was used to spatially calibrate the image.

Mean gray value Average gray value within the selection. This is the sum of the gray values of all the pixels in the selection divided by the number of pixels. Reported in calibrated units (e.g., optical density units, photon number, electron number...) if Analyze▷Calibrate...↓ was used to calibrate the image. For RGB images, the mean is calculated by converting each pixel to grayscale using the formula $\text{gray} = (\text{red} + \text{green} + \text{blue}) / 3$ or $\text{gray} = 0.299 \times \text{red} + 0.587 \times \text{green} + 0.114 \times \text{blue}$ if *Weighted RGB Conversions* is checked in Edit▷Options▷Conversions...↑

Standard deviation Standard deviation of the gray values used to generate the mean gray value. Uses the Results Table↑ heading **StdDev**.

Modal gray value Most frequently occurring gray value within the selection. Corresponds to the highest peak in the histogram. Uses the heading **Mode**.

Min & max gray level Minimum and maximum gray values within the selection.

Centroid The center point of the selection. This is the average of the x and y coordinates of all of the pixels in the image or selection. Uses the **X** and **Y** headings.



III/ Quantification: Analyze Particles: Measurements

Center of mass This is the brightness-weighted average of the x and y coordinates all pixels in the image or selection. Uses the **XM** and **YM** headings. These coordinates are the first order spatial moments.

Perimeter The length of the outside boundary of the selection. Uses the heading **Perim..** With IJ 1.44f and later, the perimeter of a composite selection is calculated by decomposing it into individual selections. Note that the composite perimeter and the sum of the individual perimeters may be different due to use of different calculation methods.

Bounding rectangle The smallest rectangle enclosing the selection. Uses the headings **BX**, **BY**, **Width** and **Height**, where **BX** and **BY** are the coordinates of the upper left corner of the rectangle.

Fit ellipse Fits an ellipse to the selection. Uses the headings **Major**, **Minor** and **Angle**. **Major** and **Minor** are the primary and secondary axis of the best fitting ellipse. **Angle** is the angle between the primary axis and a line parallel to the X-axis of the image. The coordinates of the center of the ellipse are displayed as **X** and **Y** if Centroid is checked. Note that ImageJ cannot calculate the major and minor axis lengths if *Pixel Aspect Ratio* in the Analyze▷Set Scale...↓ dialog is not 1.0. There are several ways to view the fitted ellipse:

- The Edit▷Selection▷Fit Ellipse↑ command replaces an area selection with the best fit ellipse.
- The DrawEllipse macro draws (destructively) the best fit ellipse and the major and minor axis.
- Select *Ellipses* from the *Show*: drop-down menu in the particle analyzer (Analyze▷Analyze Particles...↑) and it will draw the ellipse for each particle in a separate window.



III/ Quantification: Analyze Particles: Measurements

Shape descriptors Calculates and displays the following shape descriptors:

Circularity $4\pi \times ([Area])/([Perimeter]^2)$ with a value of 1.0 indicating a perfect circle. As the value approaches 0.0, it indicates an increasingly elongated shape. Values may not be valid for very small particles. Uses the heading **Circ**.

Aspect ratio The aspect ratio of the particle's fitted ellipse, i.e., $([Major Axis])/([Minor Axis])$. If **Fit Ellipse** is selected the **Major** and **Minor** axis are displayed. Uses the heading **AR**.

Roundness $4 \times ([Area])/(\pi \times [Major axis]^2)$ or the inverse of *Aspect Ratio*. Uses the heading **Round**.

Solidity $([Area])/([Convex area])$; Note that the Edit▷Selection▷Convex Hull↑ command makes an area selection convex.

Feret's diameter The longest distance between any two points along the selection boundary, also known as maximum caliper. Uses the heading **Feret**. The angle (0--180 degrees) of the Feret's diameter is displayed as **FeretAngle**, as well as the minimum caliper diameter (**MinFeret**). The starting coordinates of the Feret diameter (**FeretX** and **FeretY**) are also displayed (see also Feret's Diameter macro and Chamfer distances and Geodesic diameters plugin).



III/ Quantification: Analyze Particles: Measurements

Integrated density The sum of the values of the pixels in the image or selection. This is equivalent to the product of **Area** and **Mean Gray Value**. With IJ 1.44c and later, **Raw integrated density** (sum of pixel values) is displayed under the heading **RawIntDen** when **Integrated density** is enabled.

Median The median value of the pixels in the image or selection.

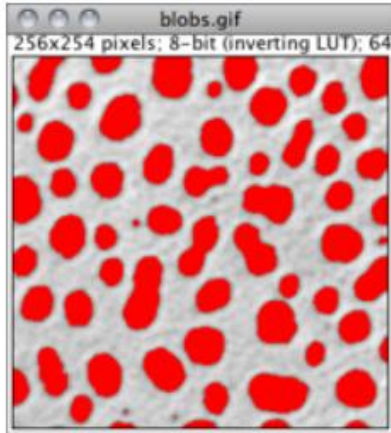
Skewness The third order moment about the mean. The documentation for the Moment Calculator plugin explains how to interpret spatial moments. Uses the heading **Skew**.

Kurtosis The fourth order moment about the mean. Uses the heading **Kurt**.

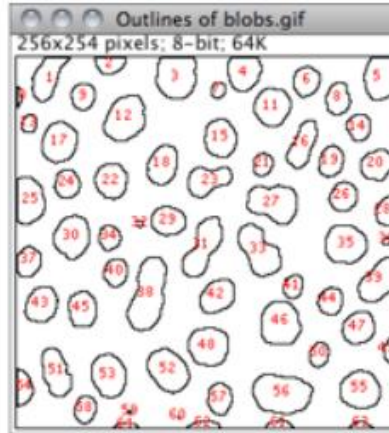
Area fraction For thresholded images is the percentage of pixels in the image or selection that have been highlighted in red using Image▷Adjust▷Threshold... [T]↑. For non-thresholded images is the percentage of non-zero pixels. Uses the heading **%Area**.

III/ Quantification: Analyze Particles: Outputs

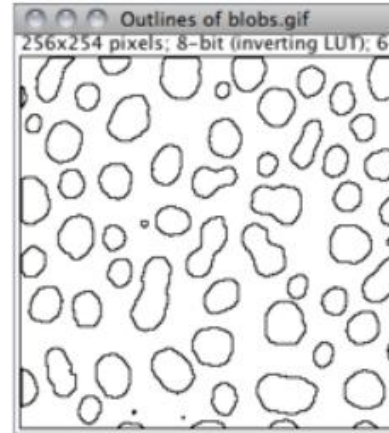
Original (thresholded)



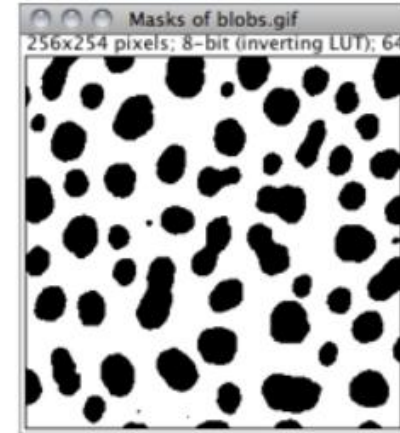
Outlines



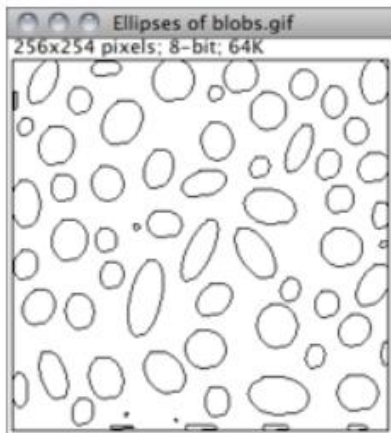
Bare Outlines



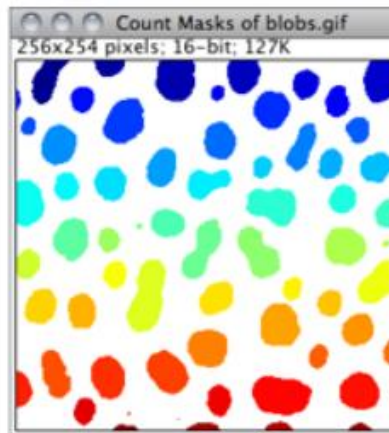
Masks



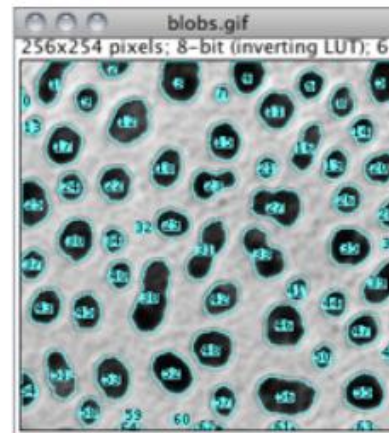
Ellipses



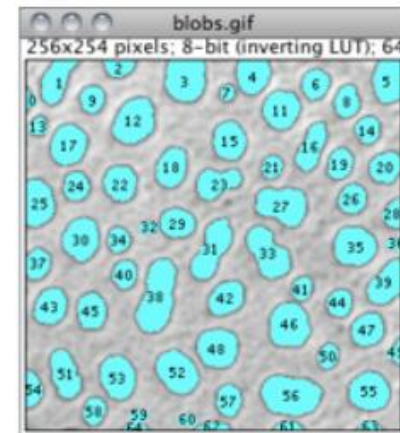
Count Masks



Overlay Outlines



Overlay Masks



III/ Quantification: Analyze Particles: *Hands-on!*



Exercise 1b

Step 6

Analyze > Analyze Particles

Step 7

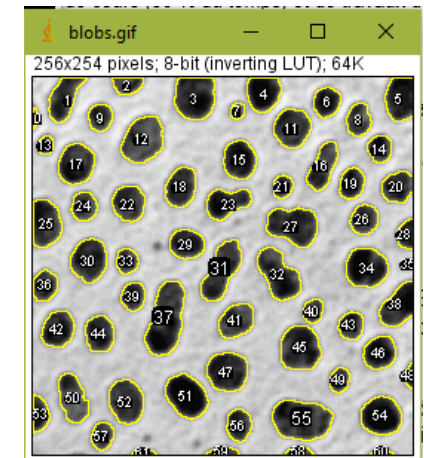
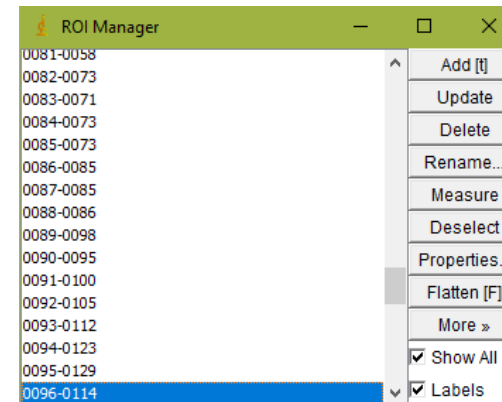
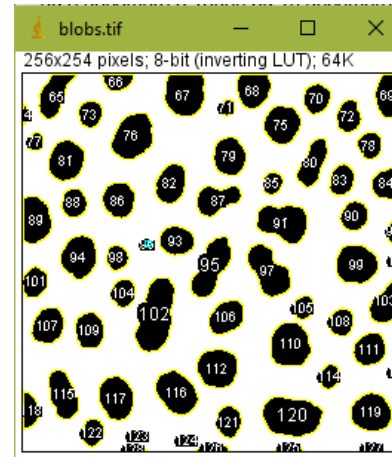
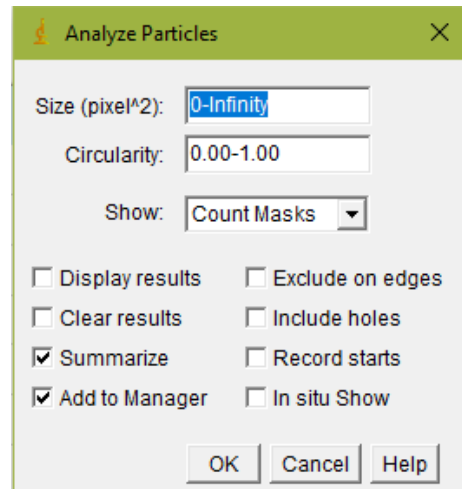
Do it again but exclude objects that are too small

Click on a too small ROI and click Measure in the ROI Manager

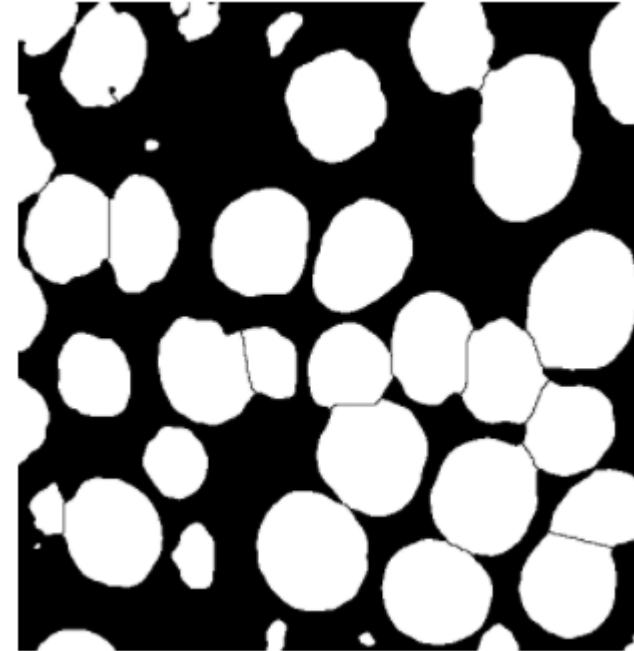
Step 8

Move the ROIs on another image

Open Blobs, then click on Show all in the ROI Manager



III/ Quantification: Watershed



Process > Binary > Watershed

III/ Quantification: Watershed

Exercise 1c

Step 9

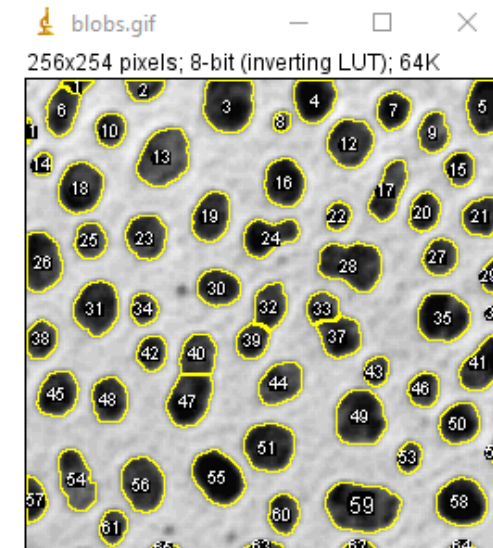
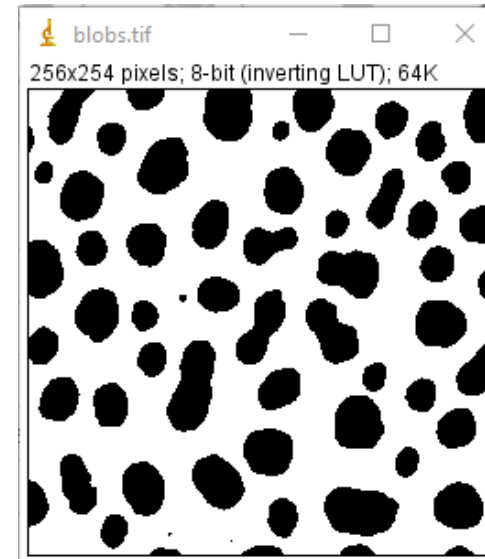
Process > Binary > Watershed

Step 10

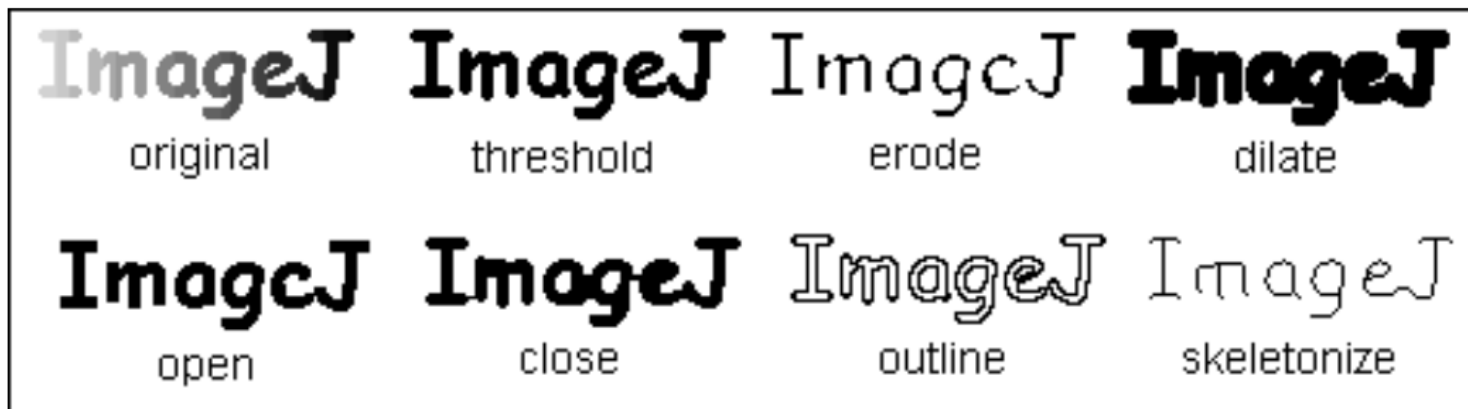
Redo Analyze > Particles and visualize the result on the original image

Step 11

Now try to exclude objects touching the borders



III/ Quantification: Binary operations



Original

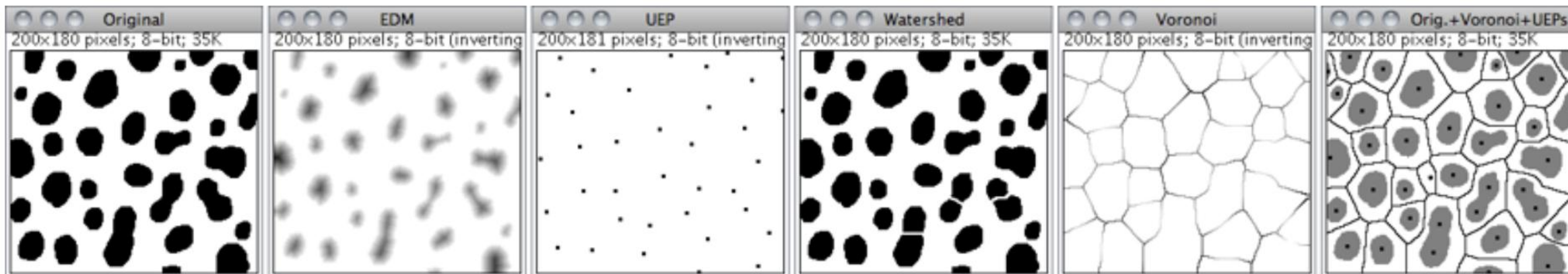
Distance Map↓

ULTIMATE POINTS↓

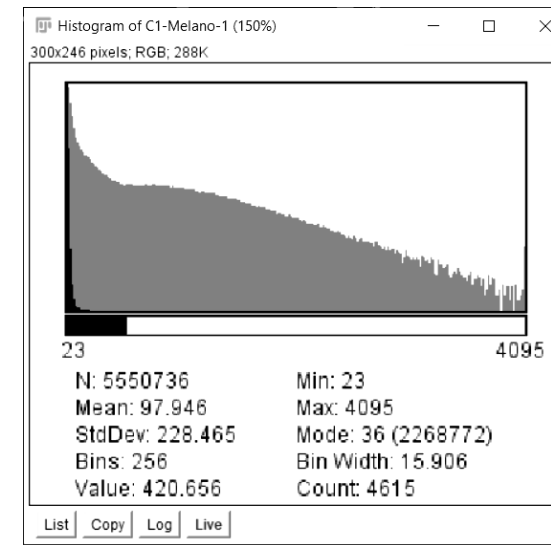
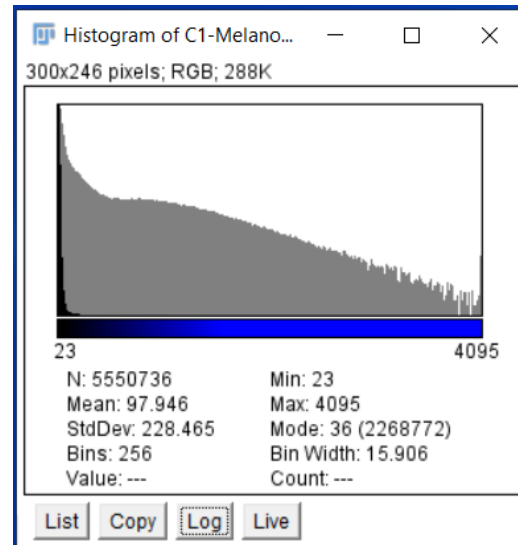
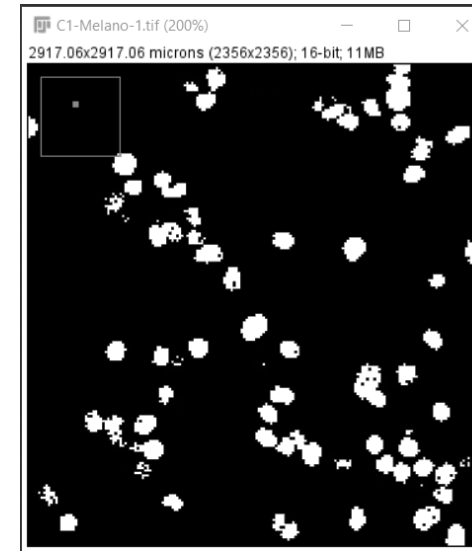
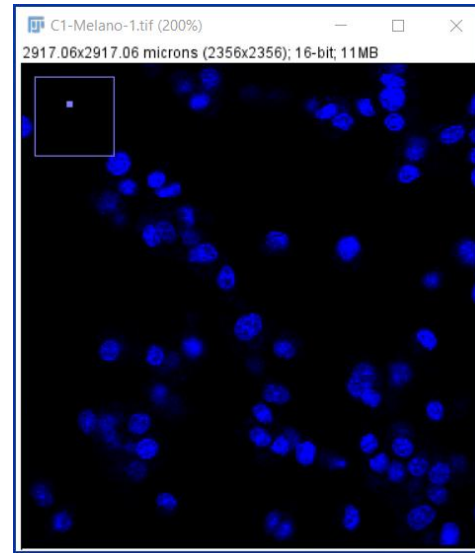
Watershed↓

Voronoi↓

Original + Voronoi↓ +
ULTIMATE POINTS↓



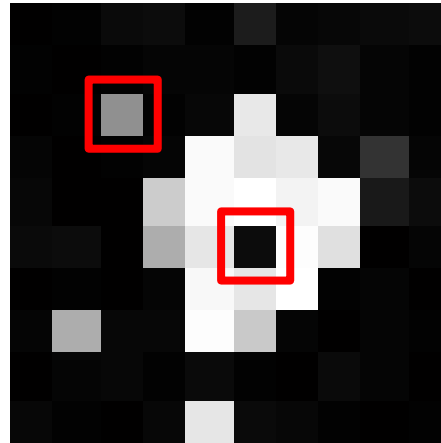
III/ Quantification: Filters



III/ Quantification: Filters

Application of a filter to smooth/denoise the image

0	1	4	5	1	11	2	3	4	5
1	0	1	2	2	1	4	6	2	1
0	1	56	1	3	90	2	5	2	1
2	0	1	2	97	88	90	3	20	2
3	0	0	79	97	99	94	97	10	5
4	5	0	67	89	5	98	87	0	2
0	1	0	2	96	89	99	1	2	0
2	67	3	3	98	78	2	0	2	1
0	2	3	1	4	1	0	4	2	0
3	1	0	3	89	4	3	1	0	1



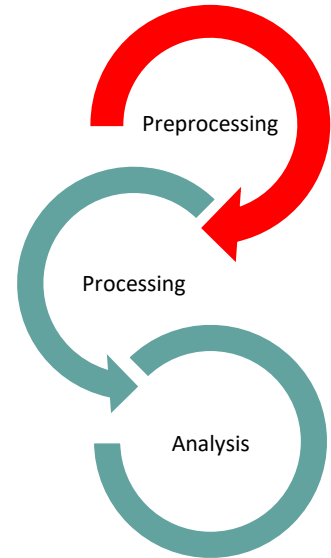
*

Filter Kernel

0.11111	0.11111	0.11111
0.11111	0.11111	0.11111
0.11111	0.11111	0.11111

$$\begin{aligned} &0 * 0.11111 + 1 * 0.11111 + 2 * 0.11111 + 1 * \\ &0.11111 + 56 * 0.11111 + 1 * 0.11111 + \\ &0 * 0.11111 + 1 * 0.11111 + 2 * 0.11111 \\ &= 7.11 \end{aligned}$$

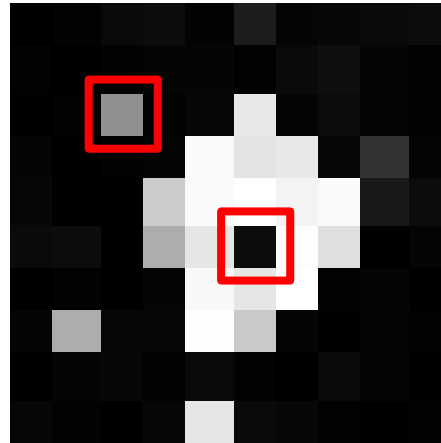
CONVOLUTION



IV/ Quantification: Filters

Application of a filter to smooth/denoise the image

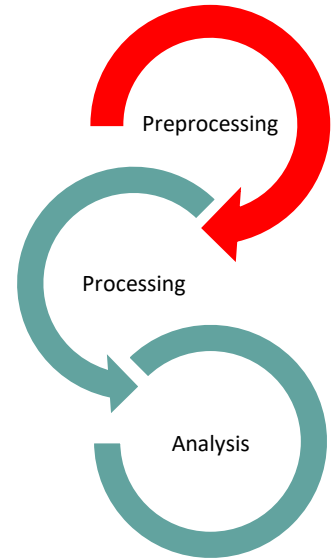
0	1	4	5	1	11	2	3	4	5
1	0	1	2	2	1	4	6	2	1
0	1	56	1	3	90	2	5	2	1
2	0	1	2	97	88	90	3	20	2
3	0	0	79	97	99	94	97	10	5
4	5	0	67	89	5	98	87	0	2
0	1	0	2	96	89	99	1	2	0
2	67	3	3	98	78	2	0	2	1
0	2	3	1	4	1	0	4	2	0
3	1	0	3	89	4	3	1	0	1



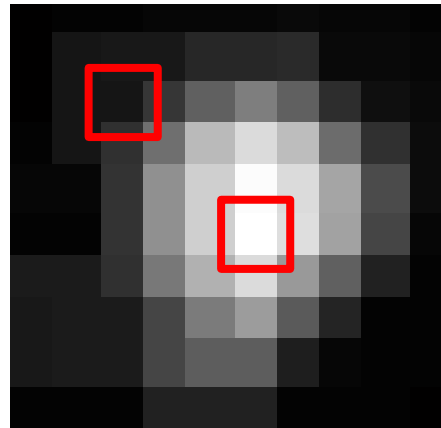
*

Filter Kernel

0.11111	0.11111	0.11111
0.11111	0.11111	0.11111
0.11111	0.11111	0.11111



0	1	1	2	2	2	3	2	2	1
0	7	8	8	13	13	14	3	3	2
0	7	7	18	32	42	32	15	5	3
1	7	18	37	62	73	63	36	16	4
2	2	17	48	69	84	73	55	25	4
1	1	17	48	69	85	74	54	23	2
9	9	16	40	59	73	51	32	11	1
8	9	9	23	41	52	30	12	1	1
8	9	9	23	31	31	10	2	1	1
1	1	1	11	11	11	1	1	1	0



Median Filter

15	18	14
29	27	13
12	19	21

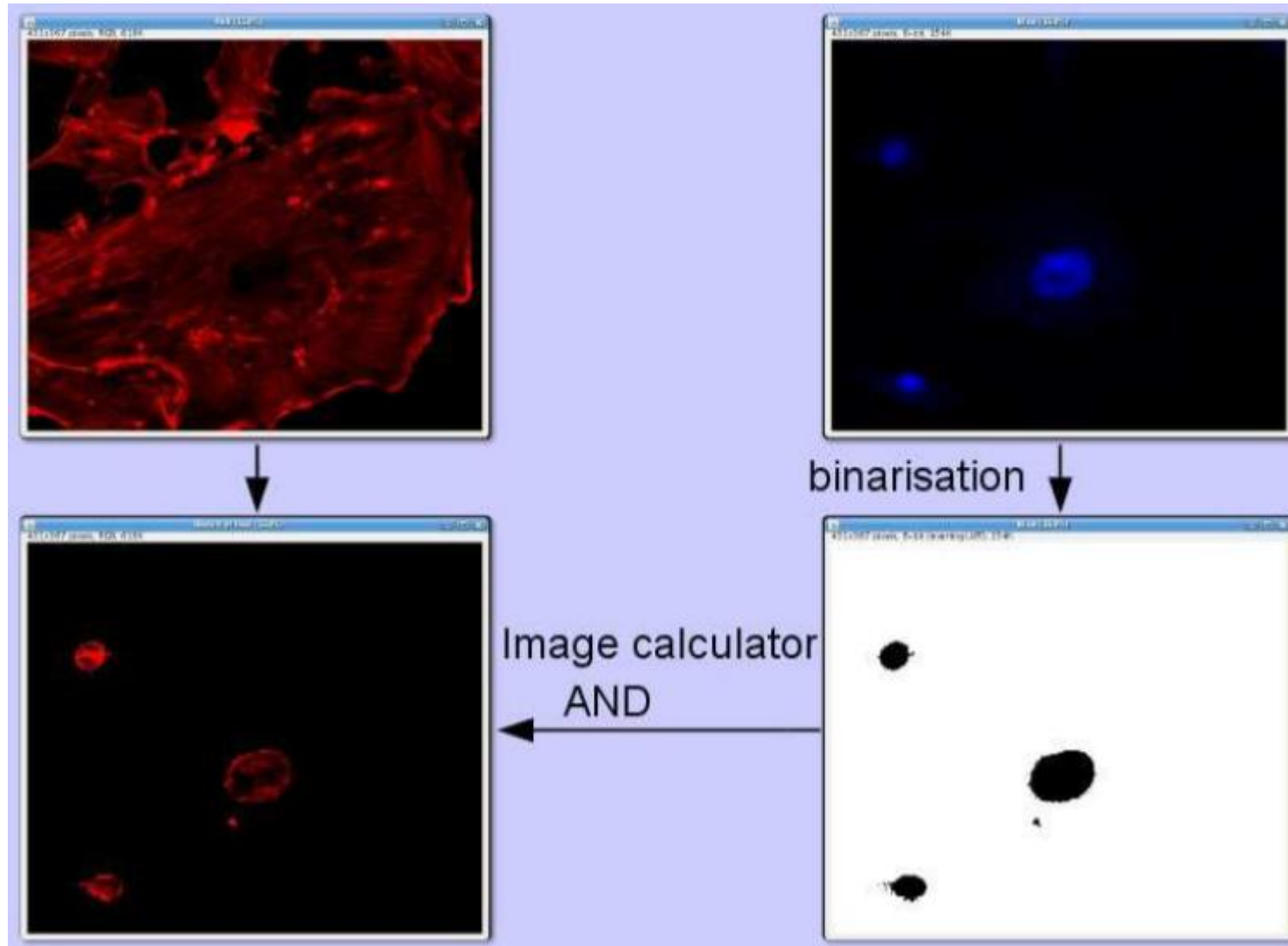


15	18	14
29	18	13
12	19	21

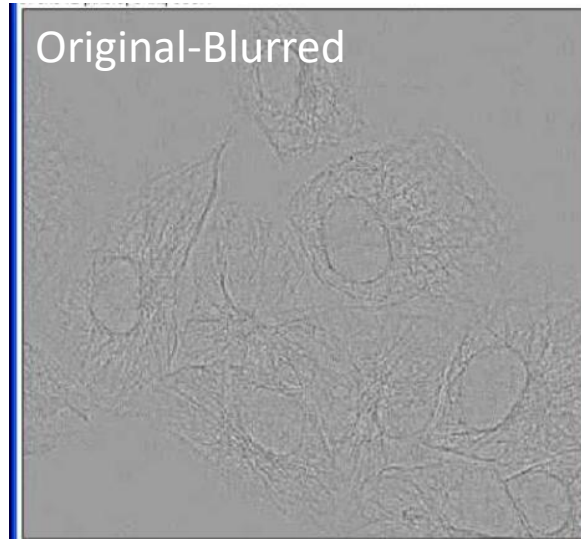
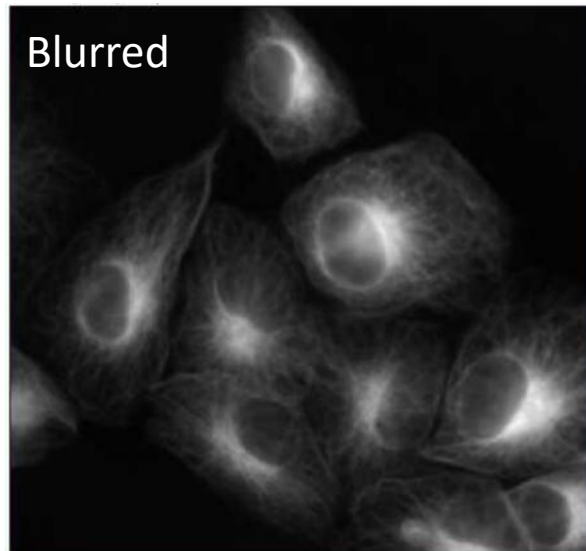
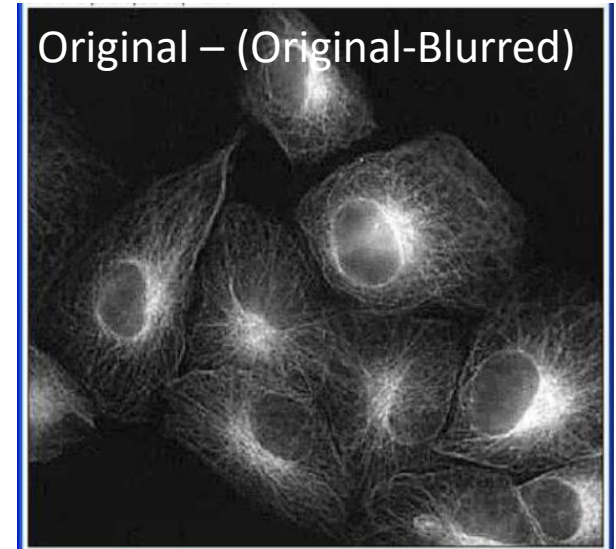
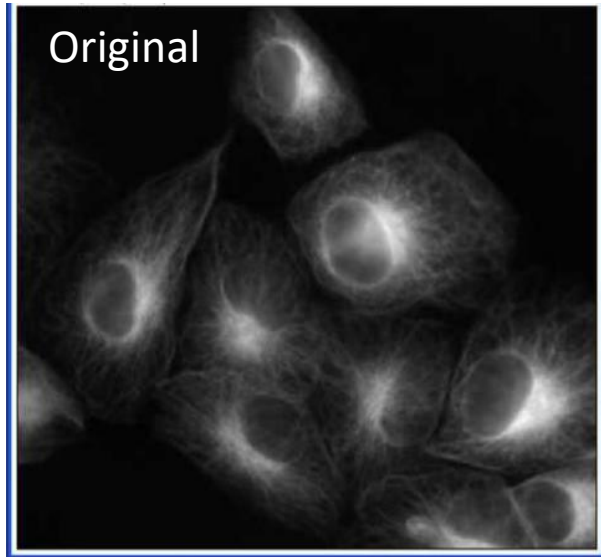
IV/ Quantification: Maths on images

Source image (<i>img1</i>):		Destination image (<i>img2</i>):	
Operator	Result	Operator	Result
Add: $img1 = img1 + img2$		Min: $img1 = \min(img1, img2)$	
Subtract: $img1 = img1 - img2$		Max: $img1 = \max(img1, img2)$	
Multiply: $img1 = img1 \times img2$		Average: $img1 = (img1 + img2) / 2$	
Divide: $img1 = img1 \div img2$		Difference: $img1 = img1 - img2 $	
AND: $img1 = img1 \wedge img2$		Copy: $img1 = img2$	
OR: $img1 = img1 \vee img2$		Transparent--zero	
XOR: $img1 = img1 \oplus img2$			

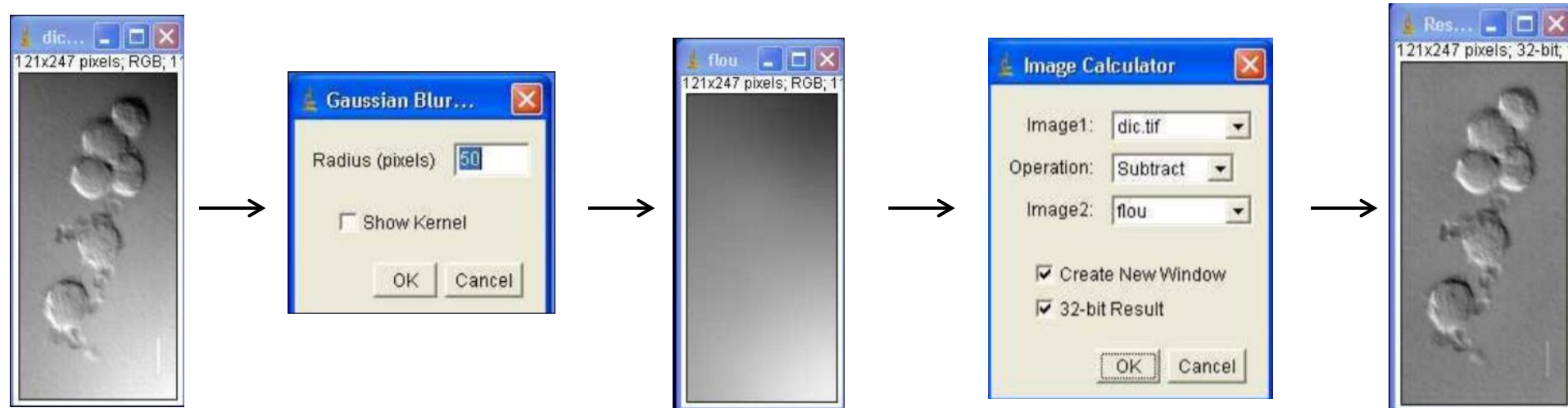
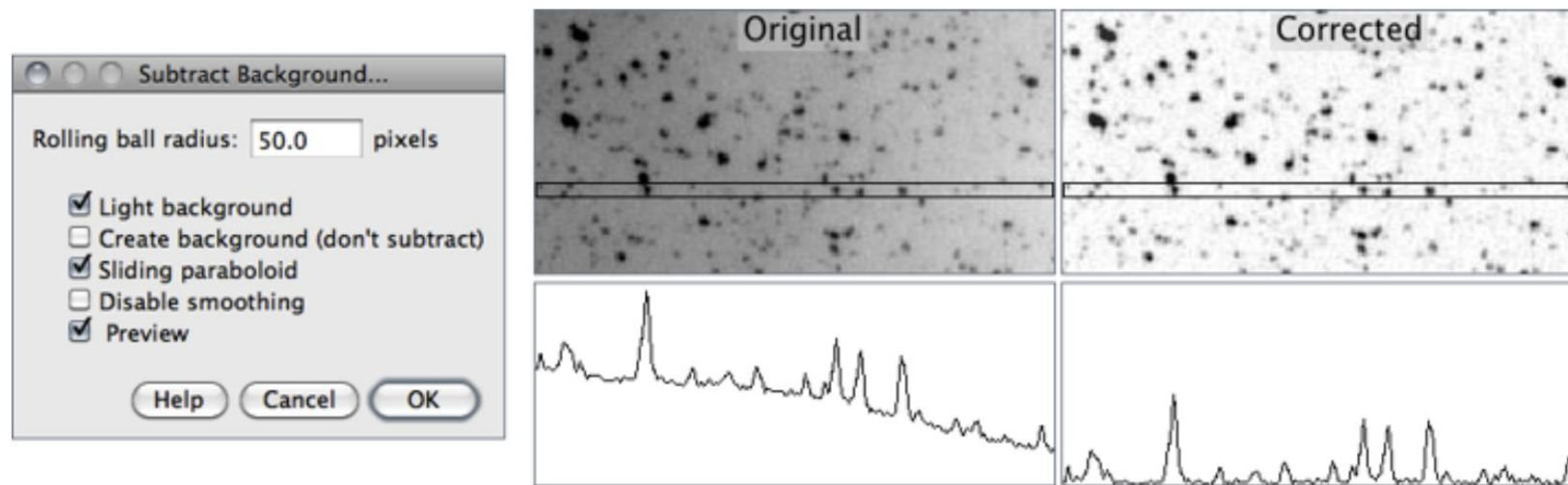
IV/ Quantification: Maths: Masking images



IV/ Quantification: Maths: Sharpening an image

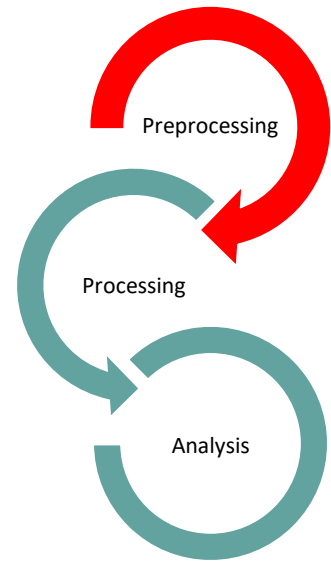
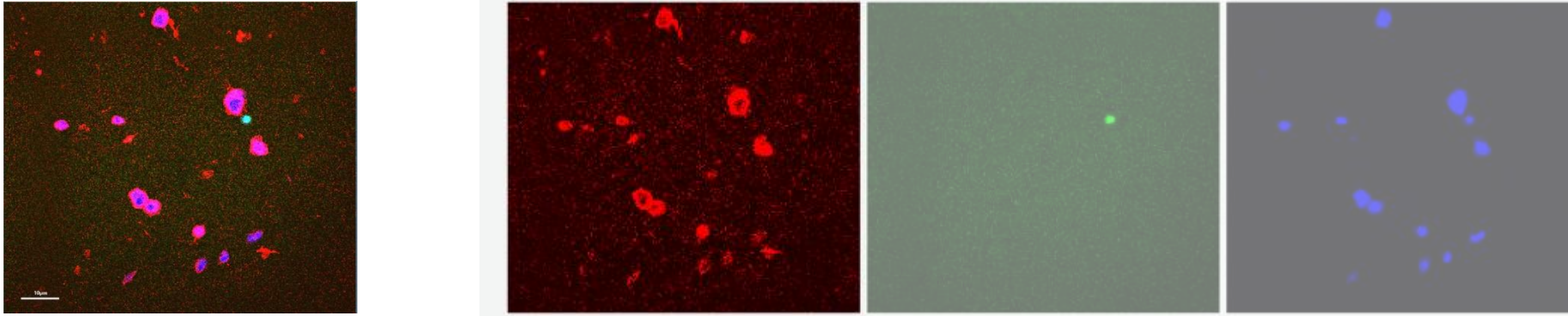


IV/ Quantification: Maths: Subtract background



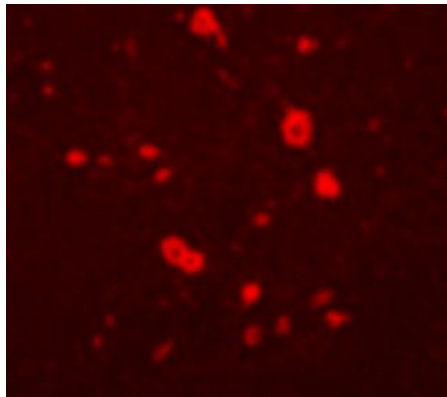
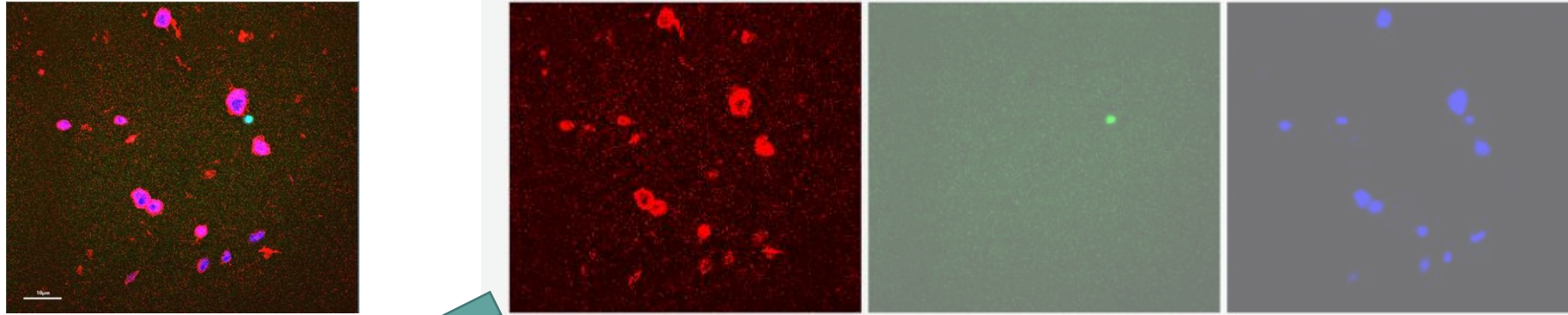
IV/ Quantification: Typical segmentation workflow

Typical workflow: segment cells in the first channel

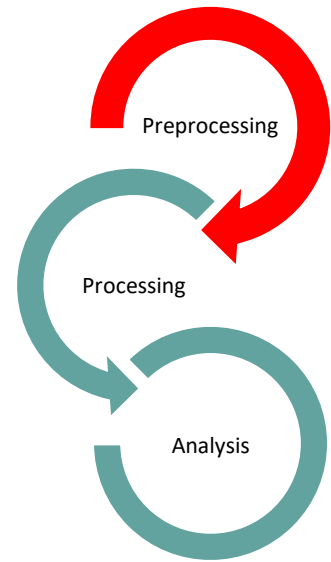


IV/ Quantification: Typical segmentation workflow

Typical workflow: segment cells in the first channel

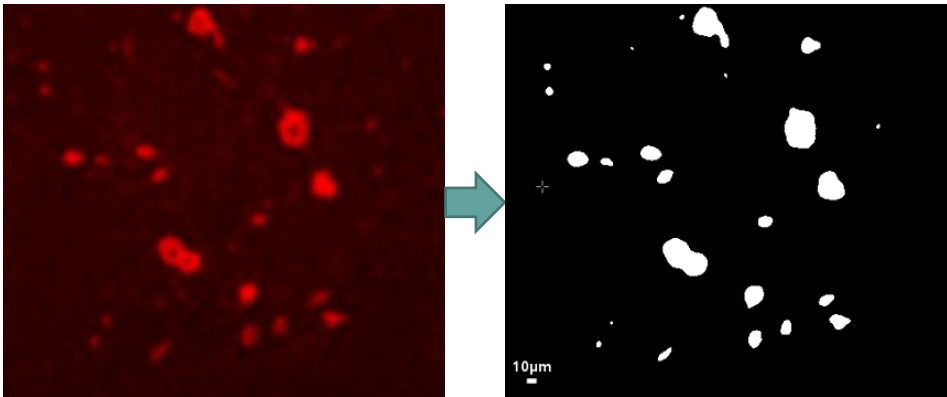
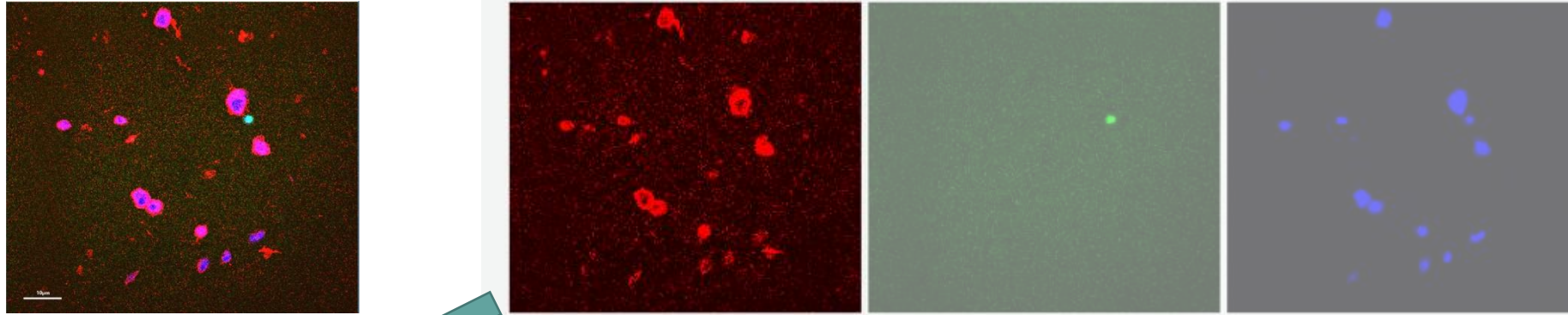


Smooth Image
(Gaussian Filter Sigma 3=
convolution with gaussian
kernel sigma 3)



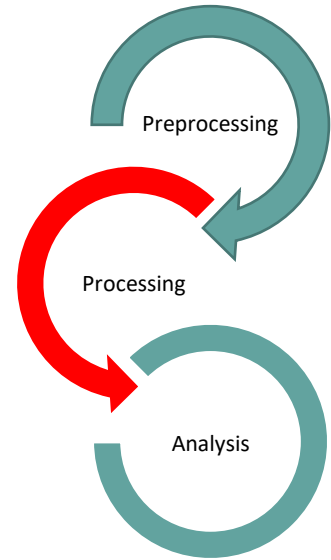
IV/ Quantification: Typical segmentation workflow

Typical workflow: segment cells in the first channel



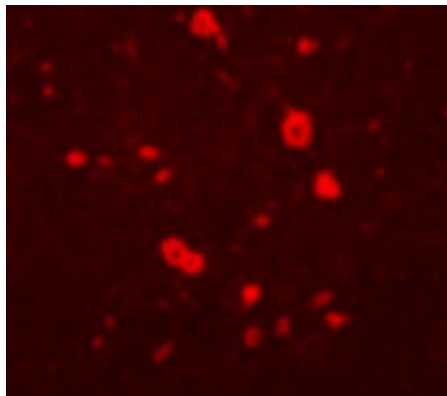
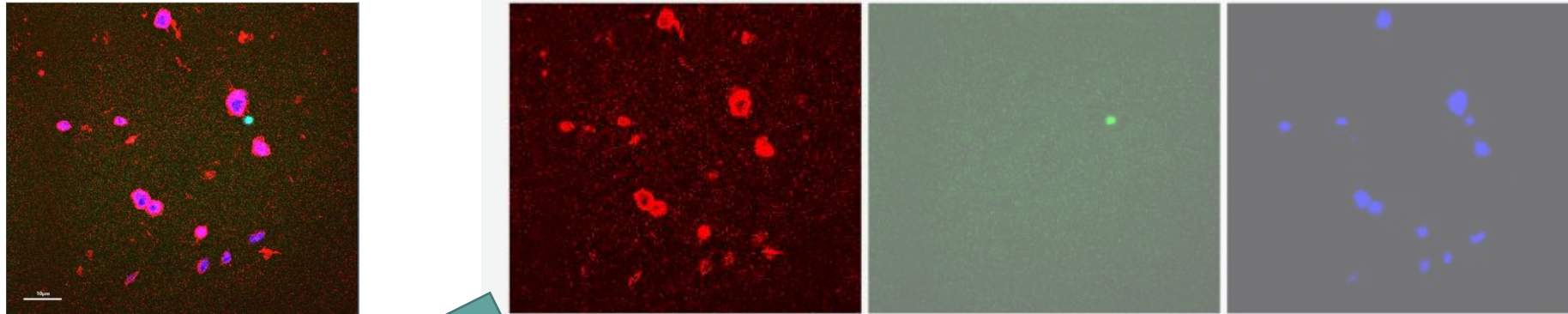
Smooth Image
(Gaussian Filter Sigma 3=
convolution with gaussian
kernel sigma 3)

Select a threshold or
threshold method
Ex: Otsu

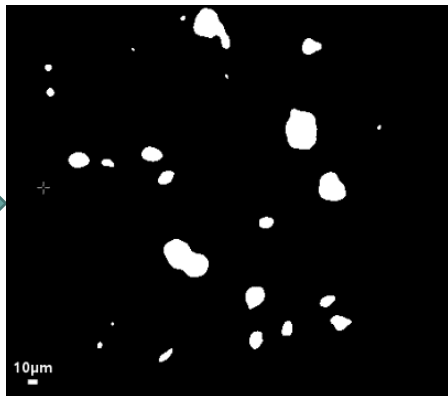


IV/ Quantification: Typical segmentation workflow

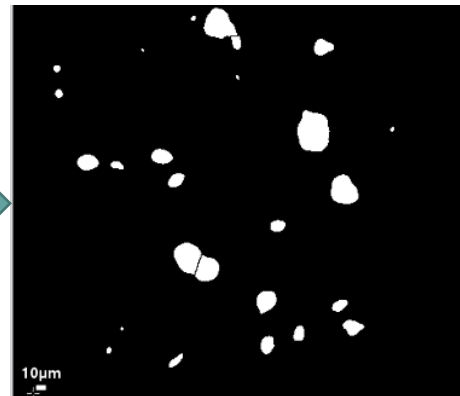
Typical workflow: segment cells in the first channel



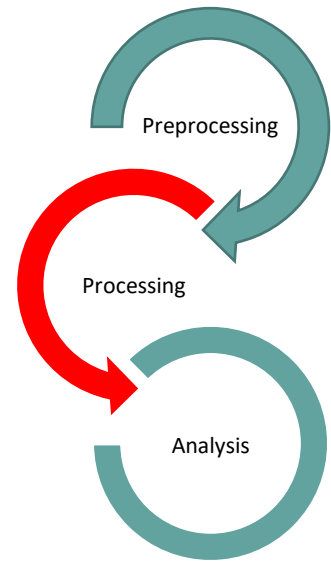
Smooth Image
(Gaussian Filter Sigma 3=
convolution with gaussian
kernel sigma 3)



Select a threshold or
threshold method
Ex: Otsu

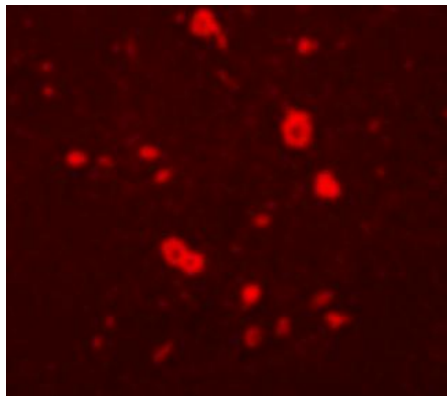
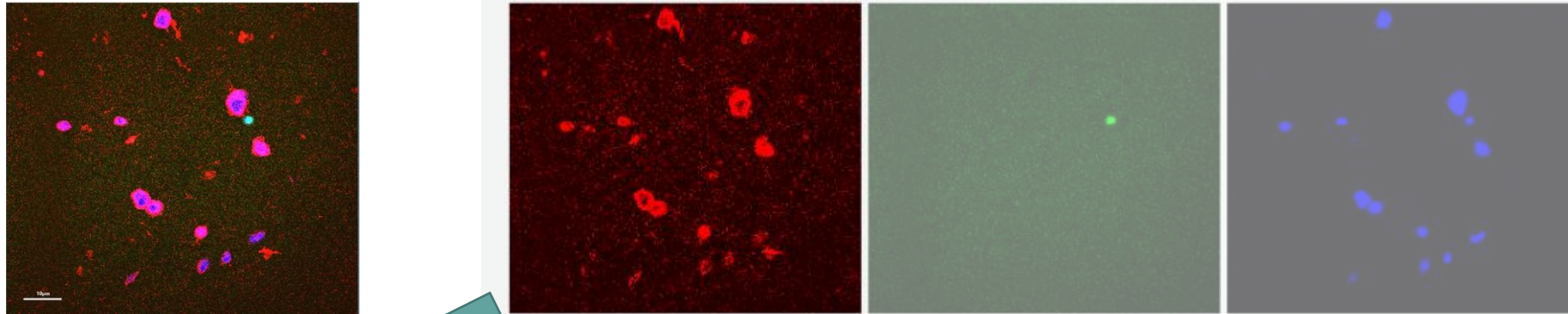


Separate touching objects:
Watershed

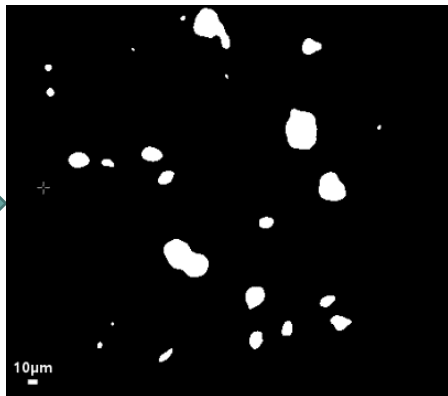


IV/ Quantification: Typical segmentation workflow

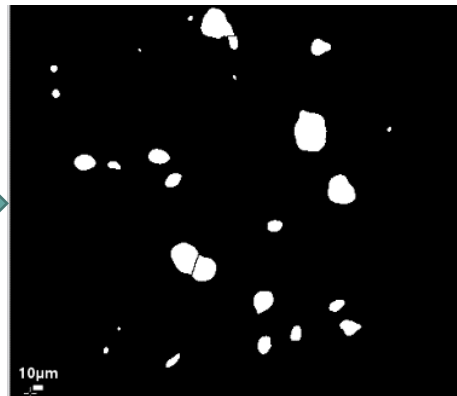
Typical workflow: segment cells in the first channel



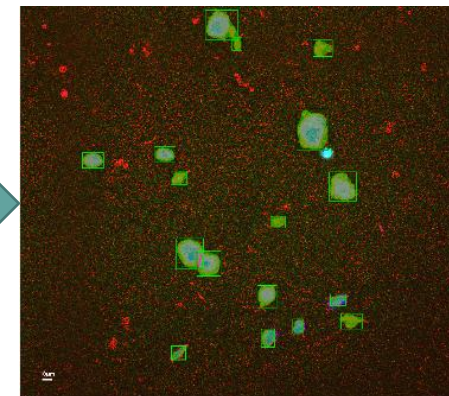
Smooth Image
(Gaussian Filter Sigma 3=
convolution with gaussian
kernel sigma 3)



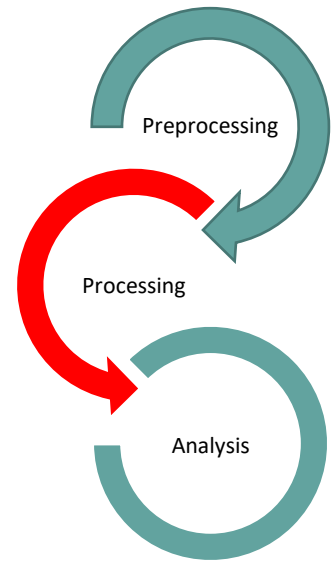
Select a threshold or
threshold method
Ex: Otsu



Separate touching objects:
Watershed

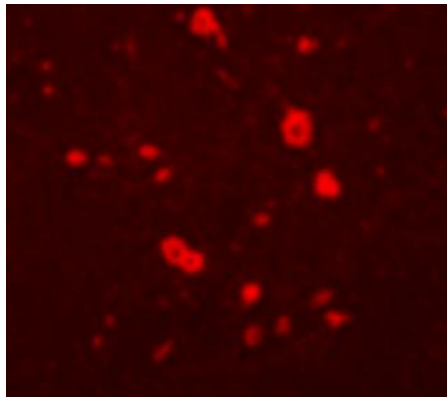
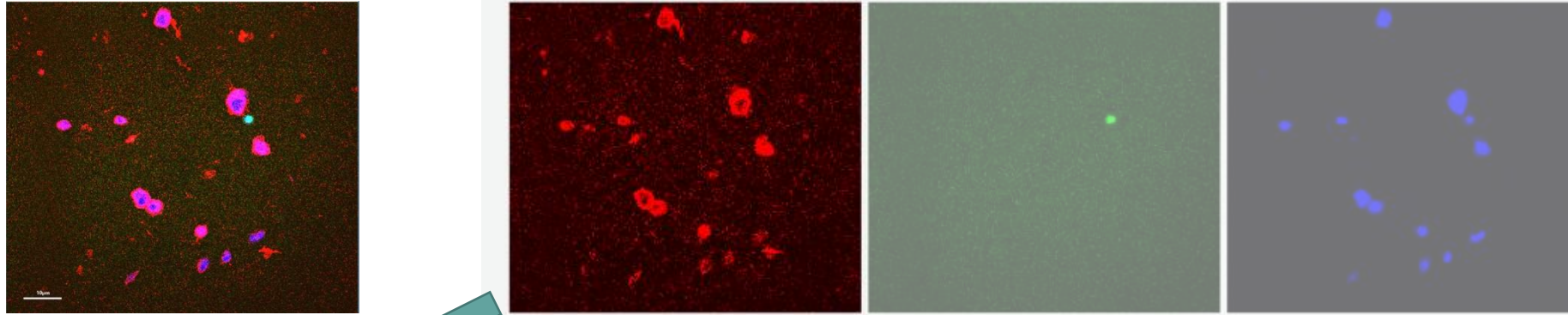


Remove small objects
Minimum size of object to
keep: set threshold for size
to 60 pixels

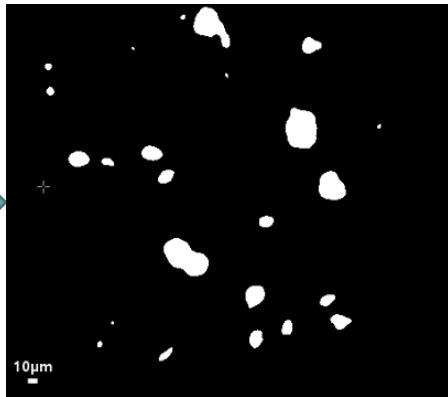


IV/ Quantification: Typical segmentation workflow

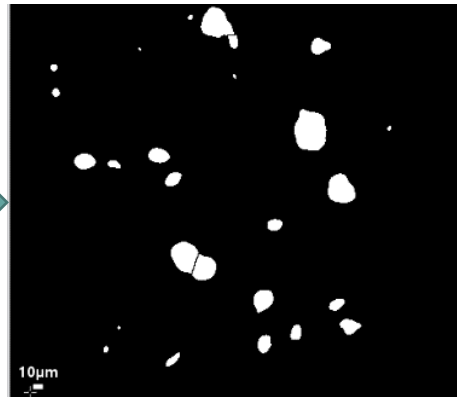
Typical workflow: segment cells in the first channel



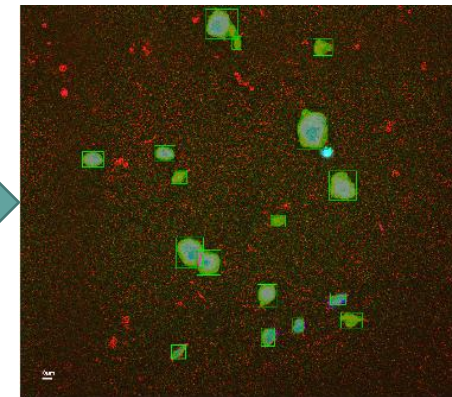
Smooth Image
(Gaussian Filter Sigma 3=
convolution with gaussian
kernel sigma 3)



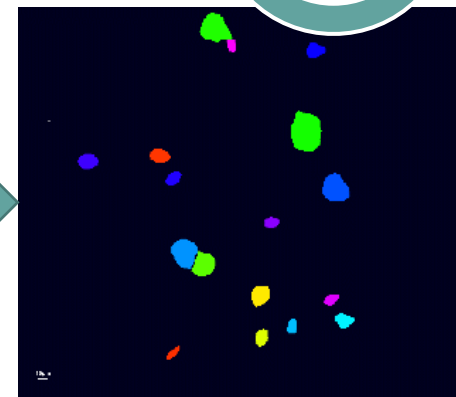
Select a threshold or
threshold method
Ex: Otsu



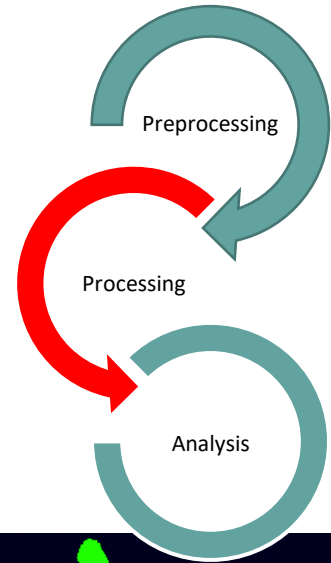
Separate touching objects:
Watershed



Remove small objects
Minimum size of object to
keep: set threshold for size
to 60 pixels

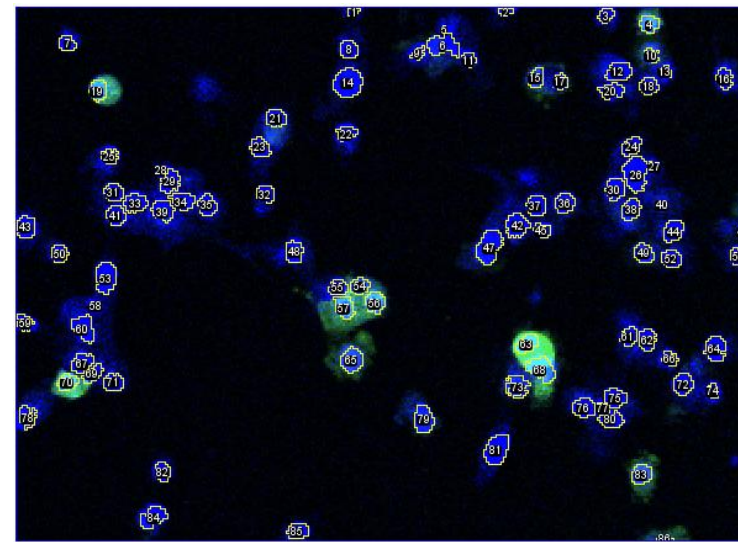
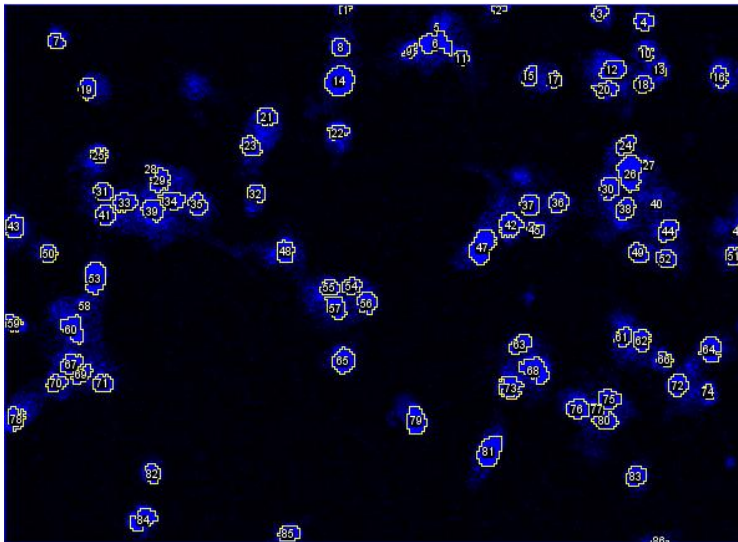
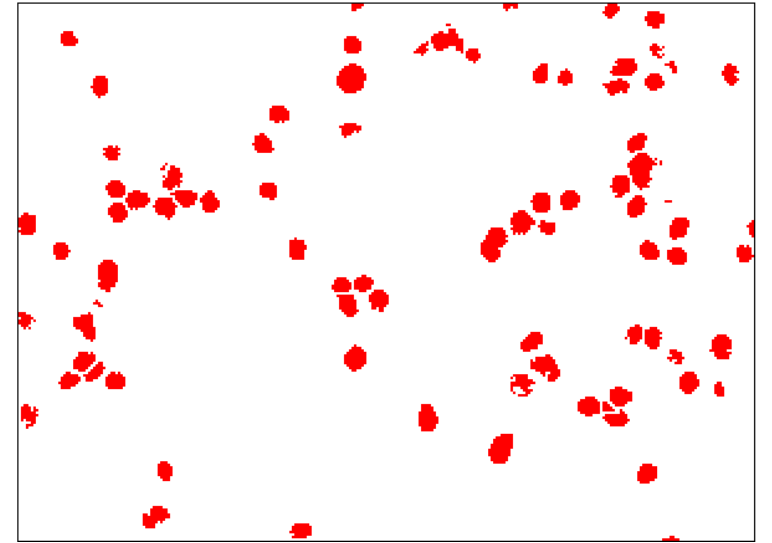
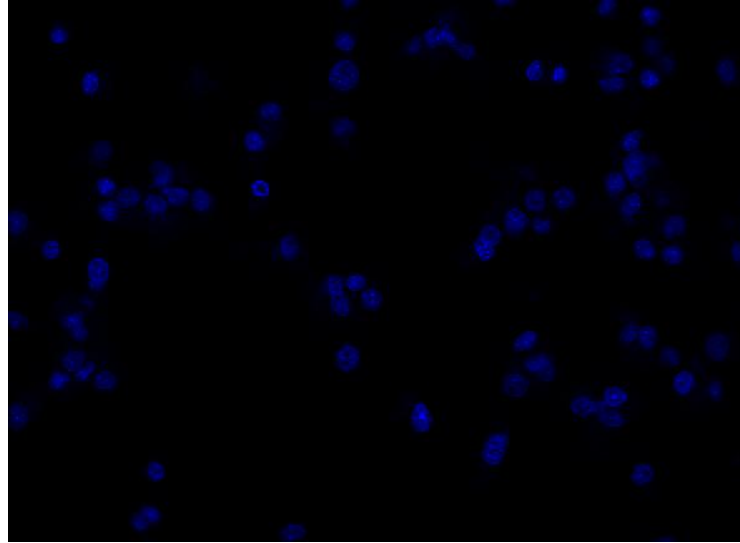
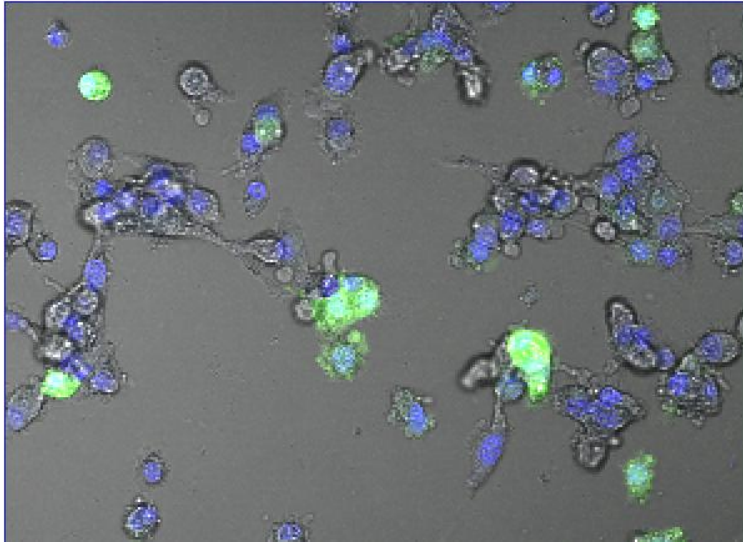


Label image



Hands-on 1: Count nuclei and determine which ones are green

Hands-on 1: Count nuclei and determine which ones are green



Hands-on 1: Count nuclei and determine which ones are green

Image: Melano-1.tif

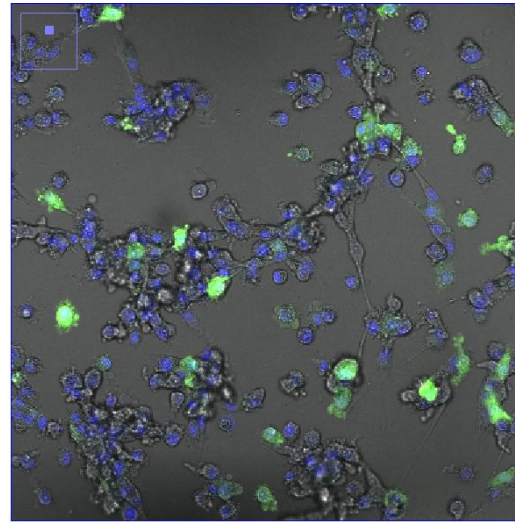
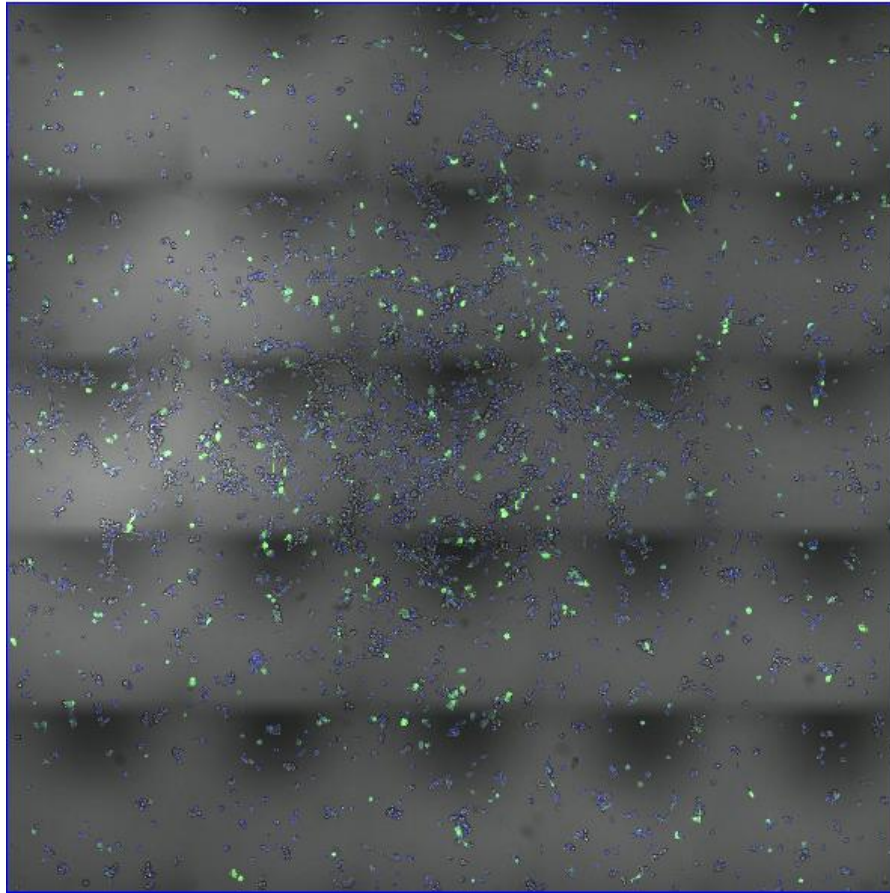
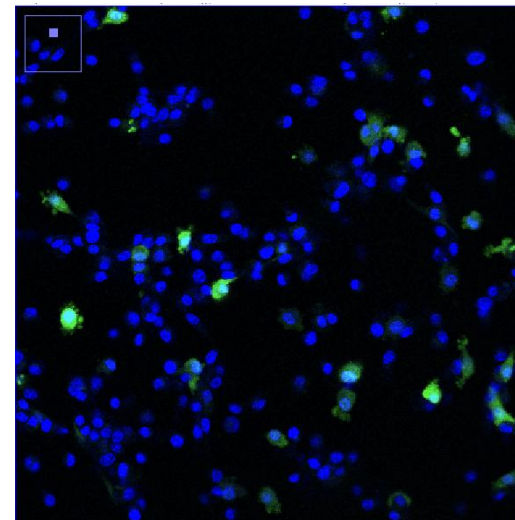


Image › Color › Channels Tool...



Hands-on 1: Count nuclei and determine which ones are green

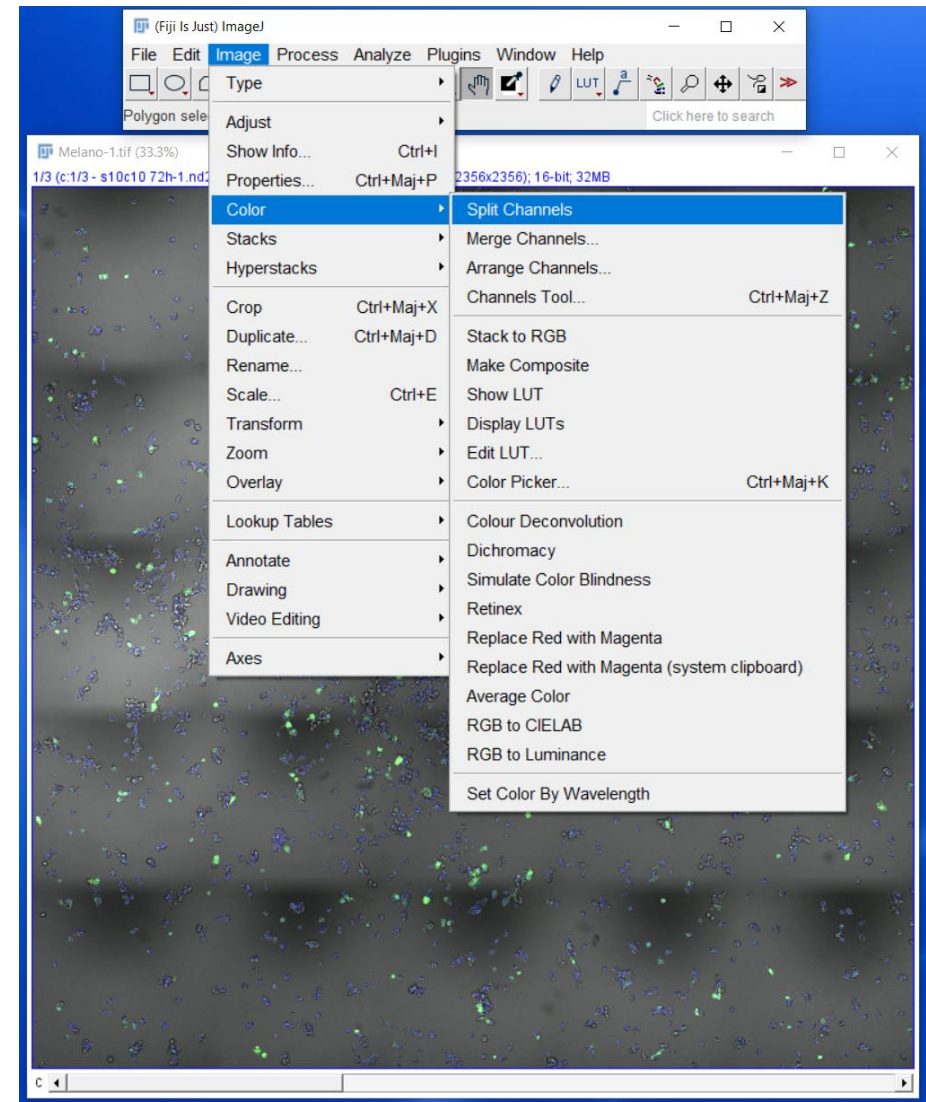
Image: Melano-1.tif

Step 1

Split Channels

Step 2

Close C3

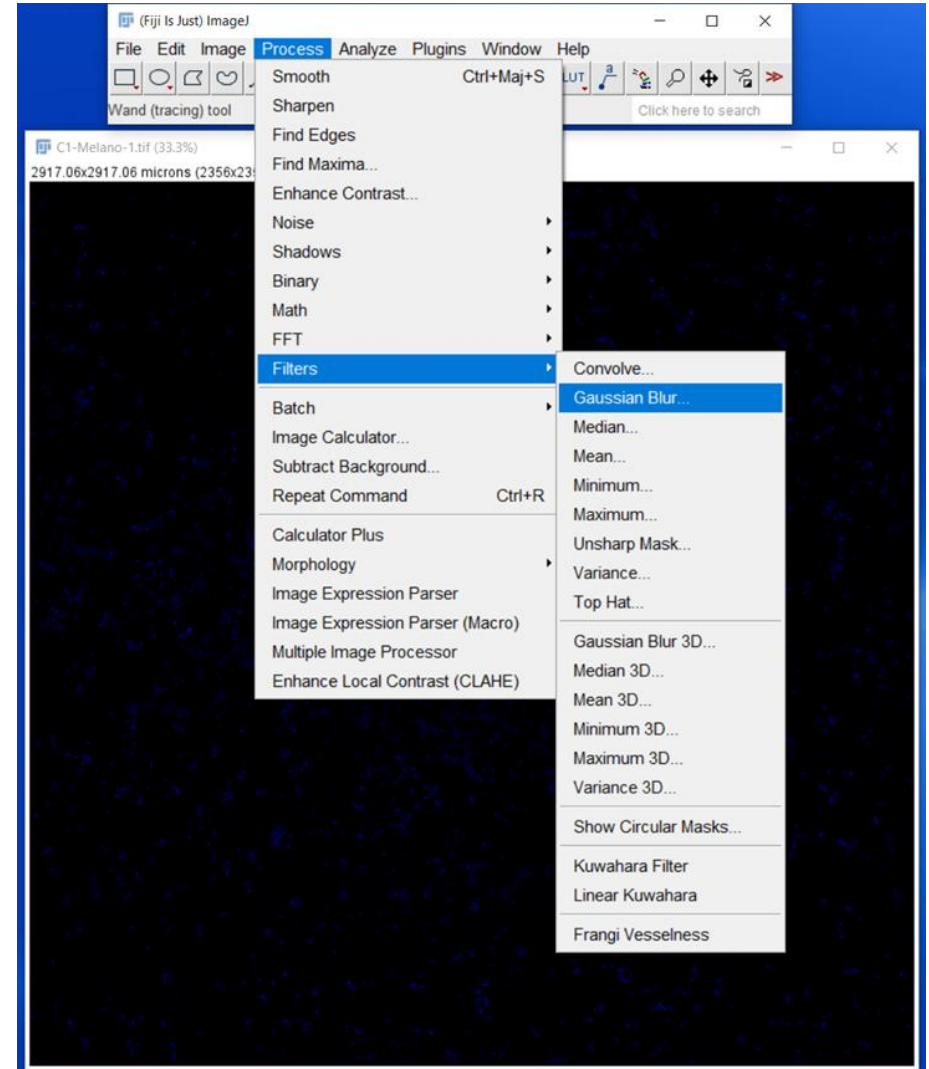
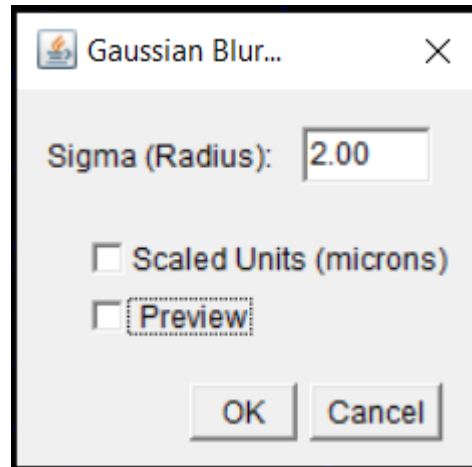


Hands-on 1: Count nuclei and determine which ones are green

Image: Melano-1.tif

Step 3

On C1, Gaussian filter

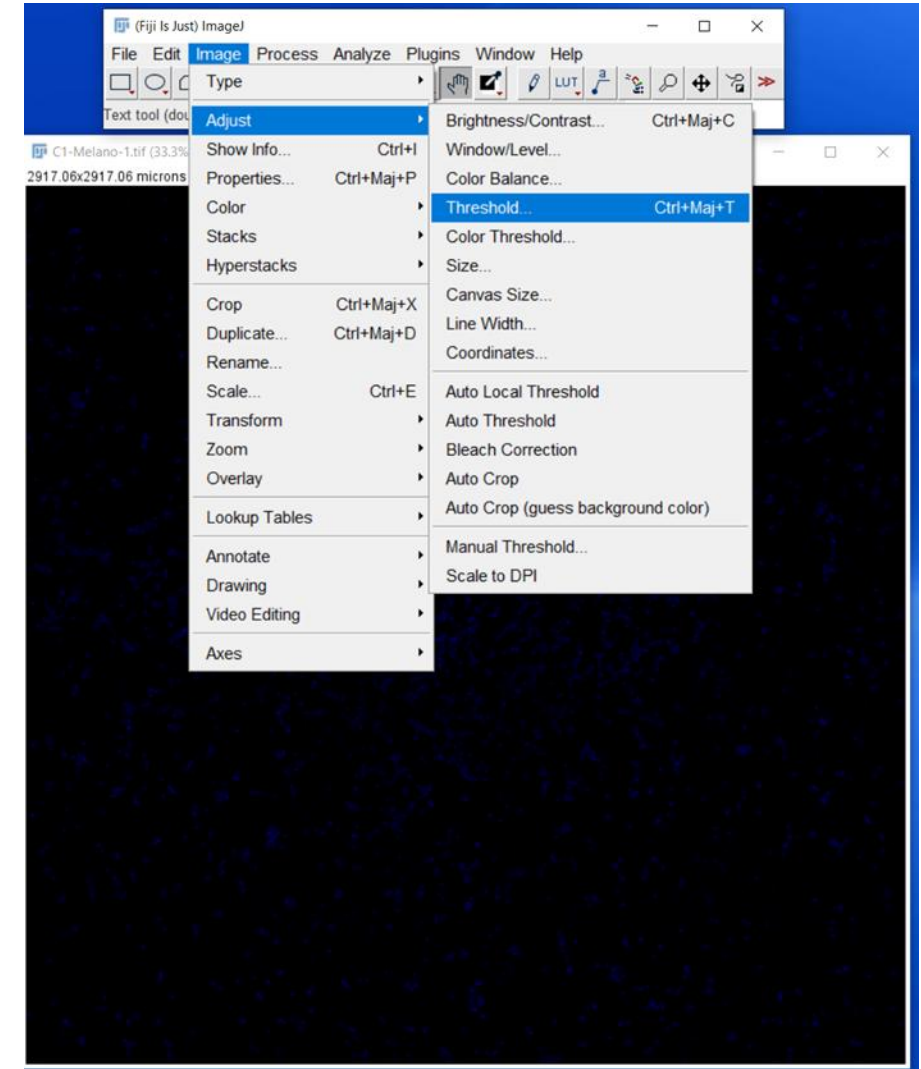
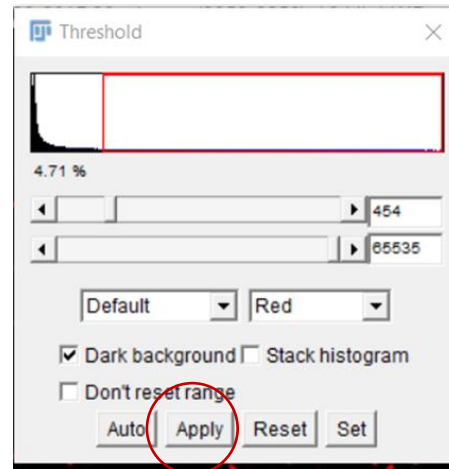


Hands-on 1: Count nuclei and determine which ones are green

Image: Melano-1.tif

Step 4

Thresholding



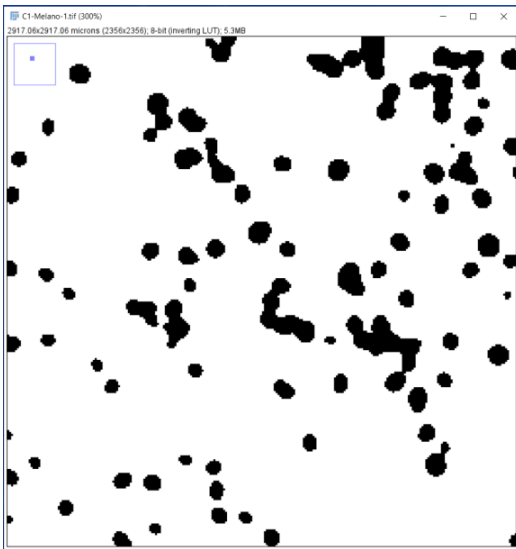
Hands-on 1: Count nuclei and determine which ones are green

Image: Melano-1.tif

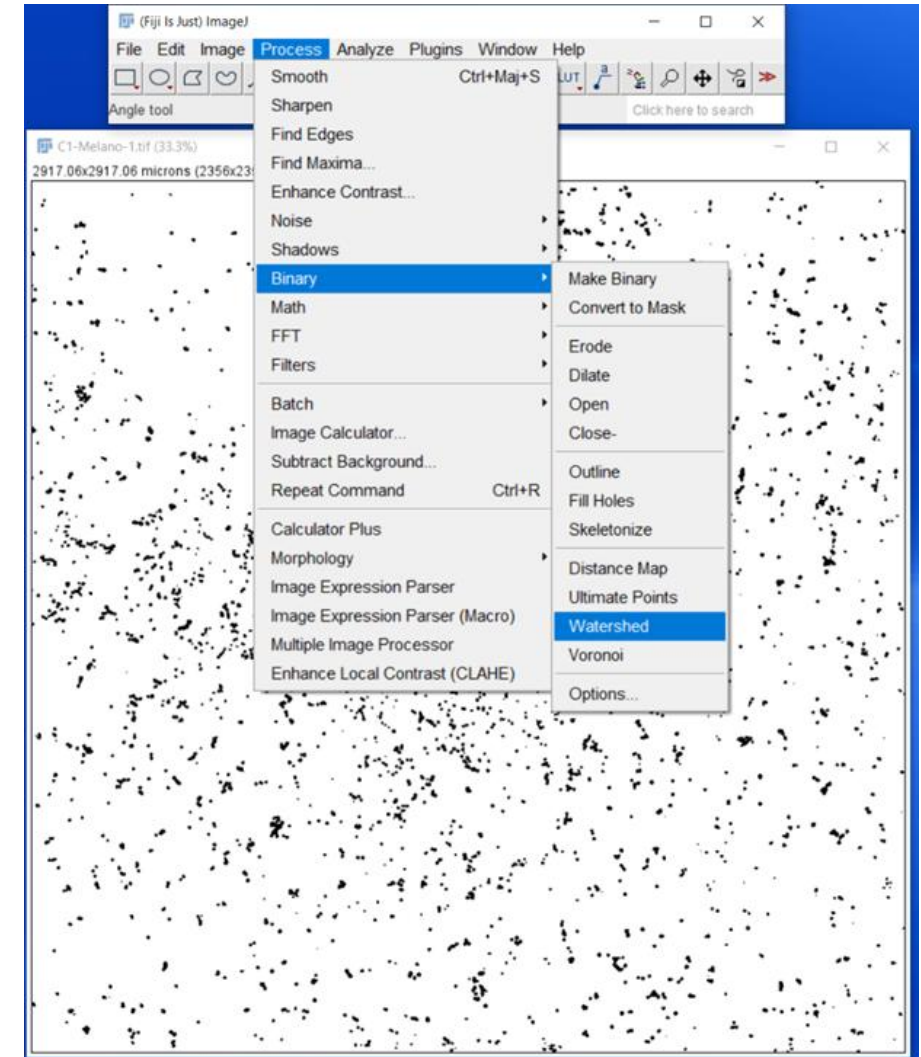
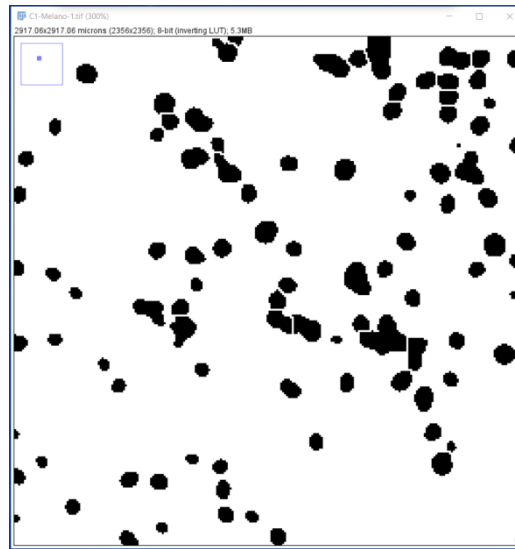
Step 5

Watershed

Before



After

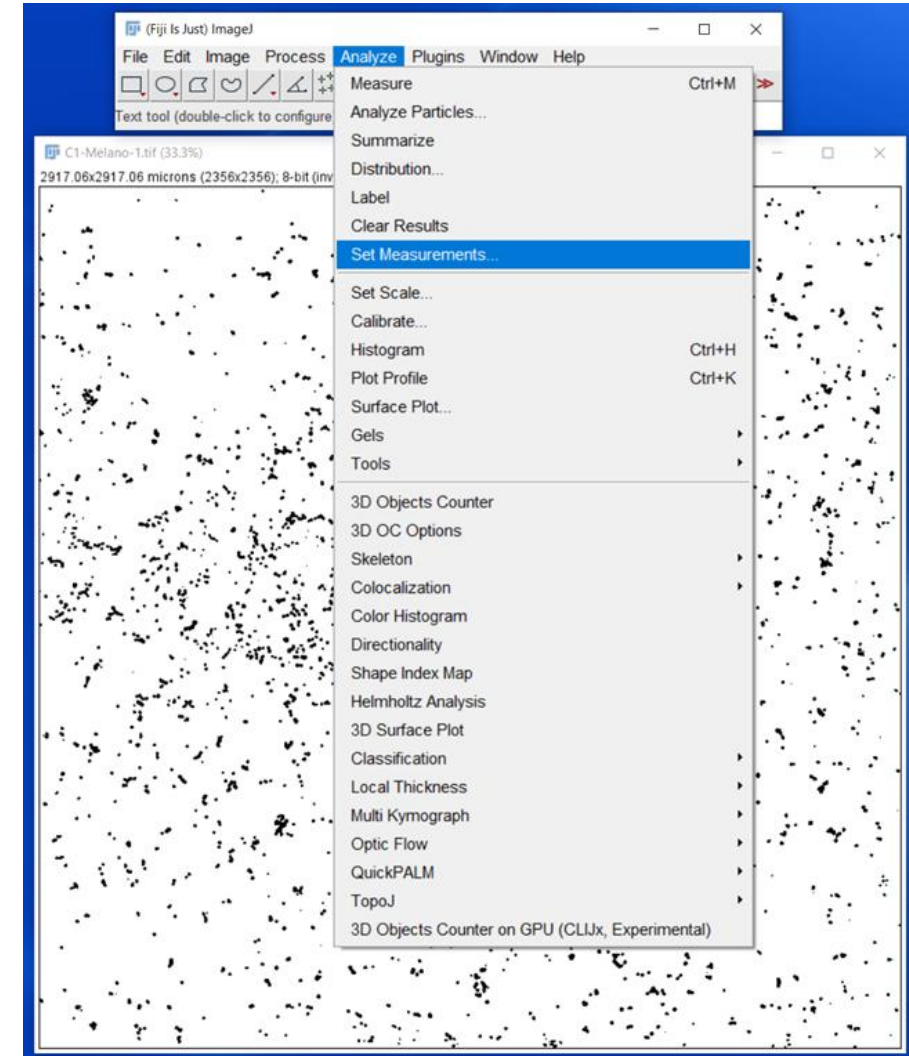
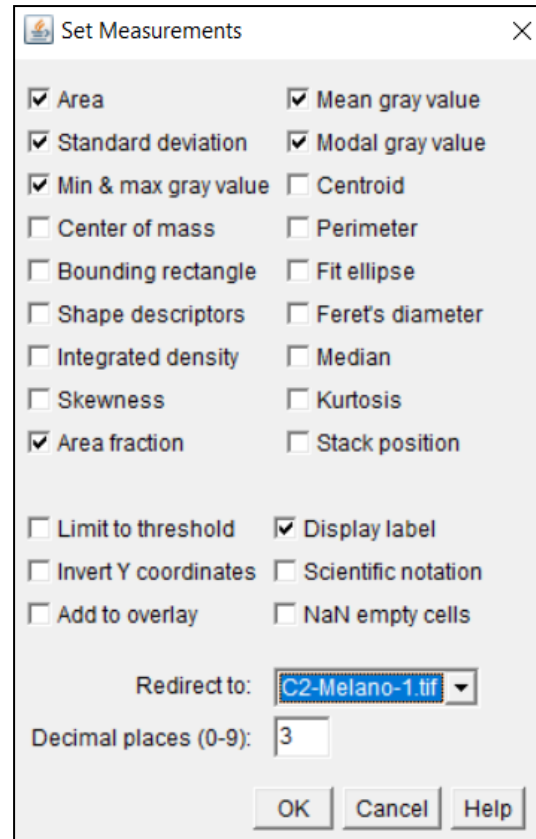


Hands-on 1: Count nuclei and determine which ones are green

Image: Melano-1.tif

Step 6

Set Measurement options, redirect to C2



Hands-on 1: Count nuclei and determine which ones are green

Image: Melano-1.tif

Step 7

Analyze Particles

Analyze Particles

Size (micron^2): 3-Infinity

☐ Pixel units

Circularity: 0.00-1.00

Show: Nothing

☒ Display results

☐ Exclude on edges

☒ Clear results

☐ Include holes

☐ Summarize

☐ Record starts

☐ Add to Manager

☐ In situ Show

OK

Cancel

Help

Results

	Label	Area	Mean	StdDev	Mode	Min	Max	%Area
3072	C2-Melano-1.tif	45.990	139.533	77.661	51	51	329	100
3073	C2-Melano-1.tif	128.771	136.179	41.476	127	75	279	100
3074	C2-Melano-1.tif	22.995	160.067	78.550	160	89	325	100
3075	C2-Melano-1.tif	78.183	101.804	35.583	77	50	191	100
3076	C2-Melano-1.tif	96.578	128.651	46.869	113	54	259	100
3077	C2-Melano-1.tif	110.375	163.708	53.549	159	64	307	100
3078	C2-Melano-1.tif	130.304	156.482	66.056	189	52	419	100
3079	C2-Melano-1.tif	88.913	137.379	50.600	177	59	330	100
3080	C2-Melano-1.tif	121.106	172.063	49.064	142	57	302	100
3081	C2-Melano-1.tif	76.650	337.660	89.820	262	112	496	100
3082	C2-Melano-1.tif	50.589	270.273	90.961	246	69	456	100
3083	C2-Melano-1.tif	4.599	98.000	17.521	80	80	115	100

(Fiji Is Just) ImageJ

File Edit Image Process Analyze Plugins Window Help

Measure Ctrl+M

Analyze Particles...

Summarize

Distribution...

Label

Clear Results

Set Measurements...

Set Scale...

Calibrate...

Histogram Ctrl+H

Plot Profile Ctrl+K

Surface Plot...

Gels

Tools

3D Objects Counter

3D OC Options

Skeleton

Colocalization

Color Histogram

Directionality

Shape Index Map

Helmholtz Analysis

3D Surface Plot

Classification

Local Thickness

Multi Kymograph

Optic Flow

QuickPALM

TopoJ

3D Objects Counter on GPU (CLJX, Experimental)

Hands-on 1: Count nuclei and determine which ones are green

Optional: play with the ROIs

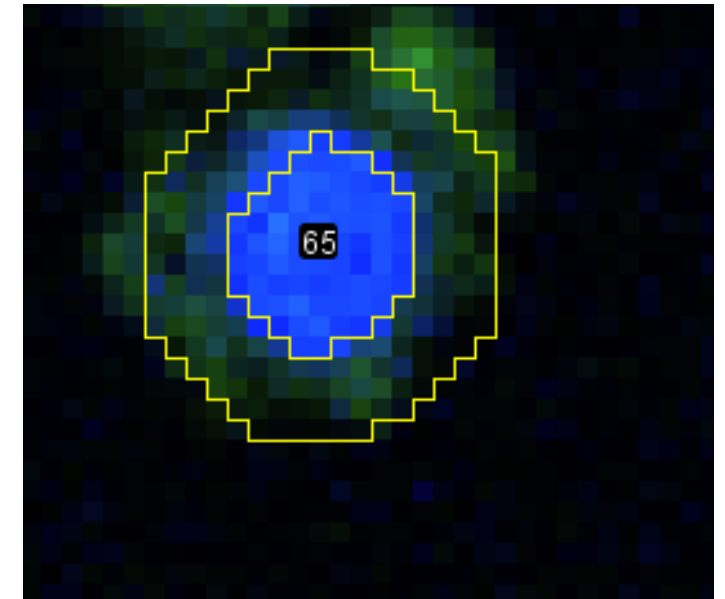
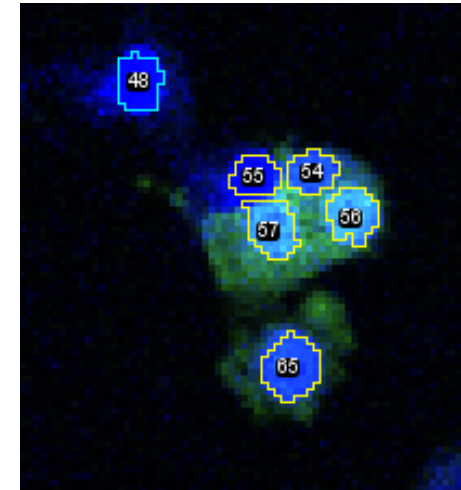
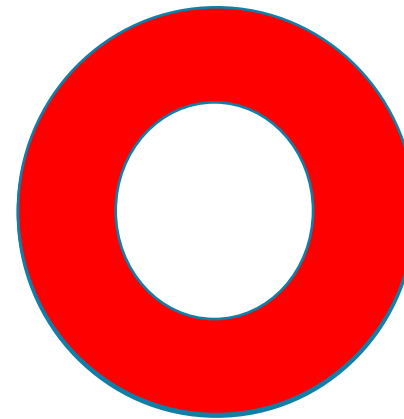
Results		
File	Edit	Font Results
	Label	Mean
1	Melano-1.tif:c:2/3 - s10c10 72h-1.nd2 (series 1)	176.37780
2	Melano-1.tif:0065-0142:c:2/3 - s10c10 72h-1.nd2 (series 1)	1278.13043
3	Melano-1.tif:0048-0098:c:2/3 - s10c10 72h-1.nd2 (series 1)	160.32000

Tools to play with ROIs:

Select **Original ROI**

Edit > Selection > Enlarge: +xx dilates the ROI, -xx shrinks it

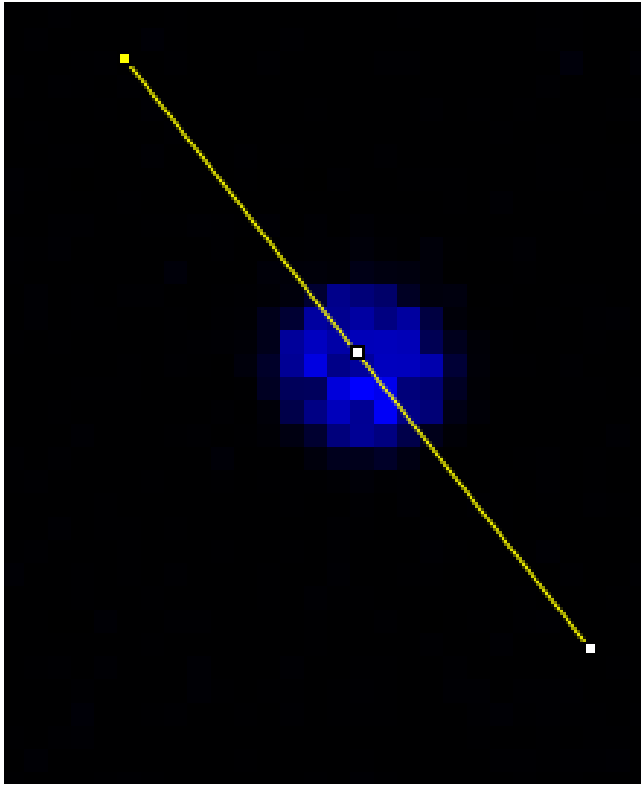
Edit > Selection > Make Band : we will measure only in the band



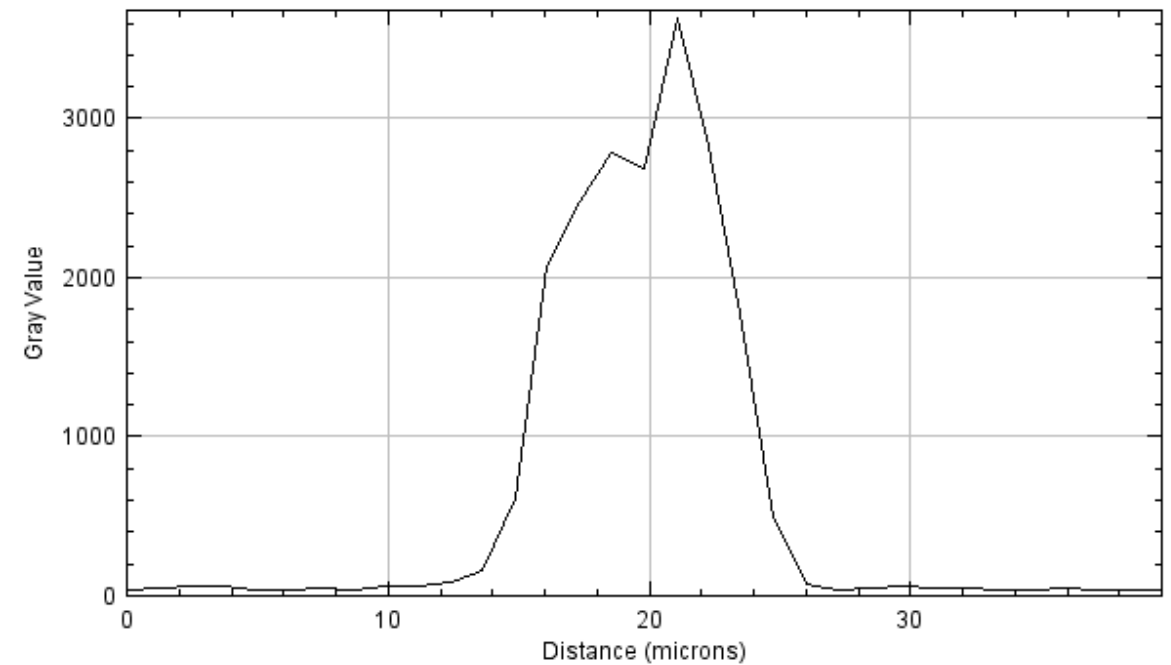
IV/ Quantification: Plot profile



Draw a line



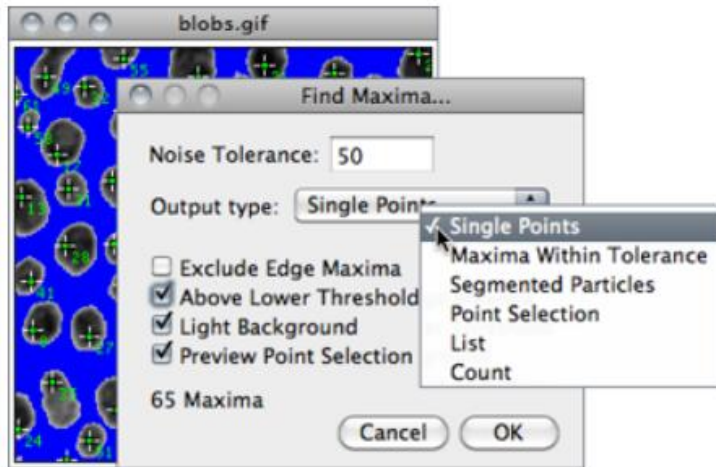
Analyze > Plot Profile (k)



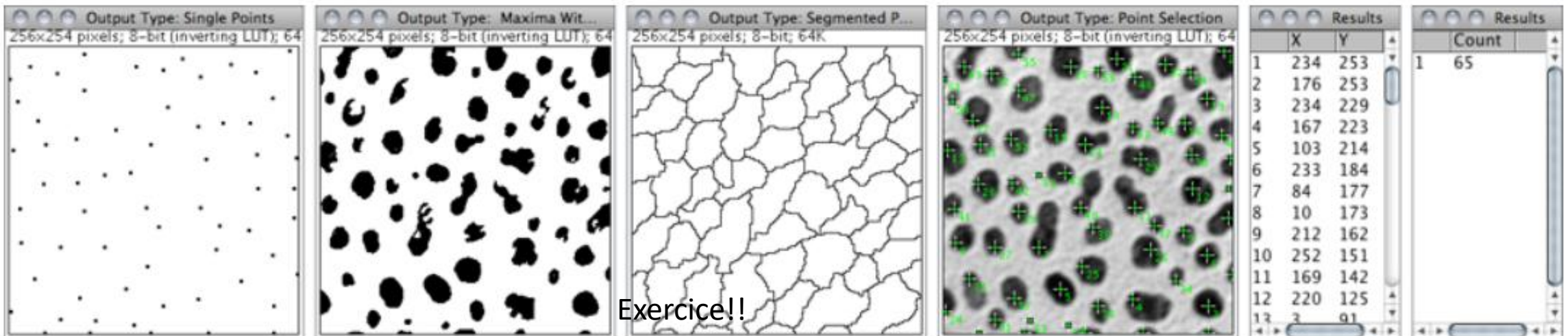
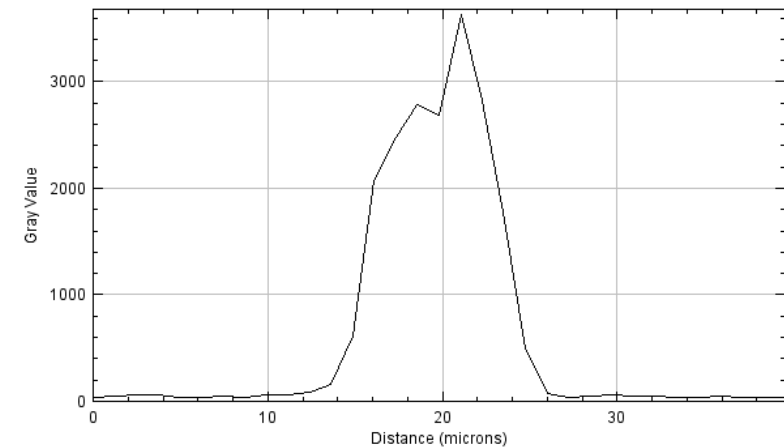
IV/ Quantification: Find maxima



Process > Find Maxima...



... detects the bumps!





IV/ Quantification: Find maxima: *Hands-on!*

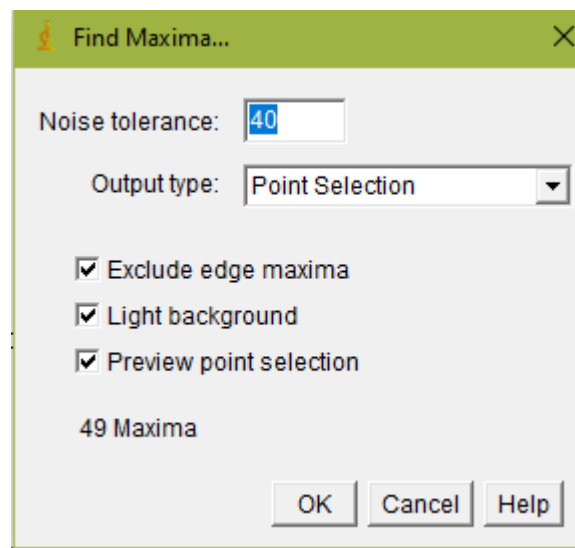
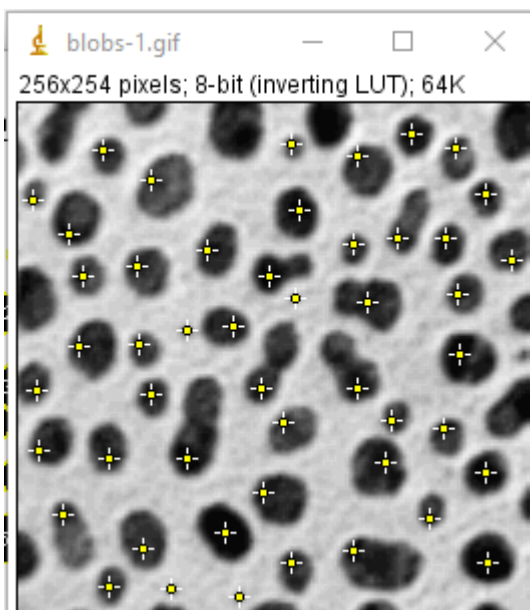
Exercise 2a

Step 1

Open the blobs

Step 2

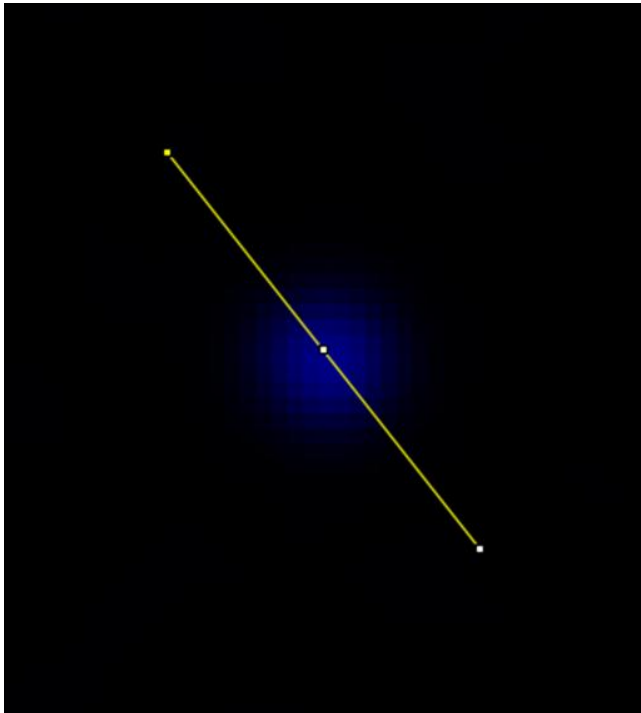
Process > Find Maxima... Try different parameters!



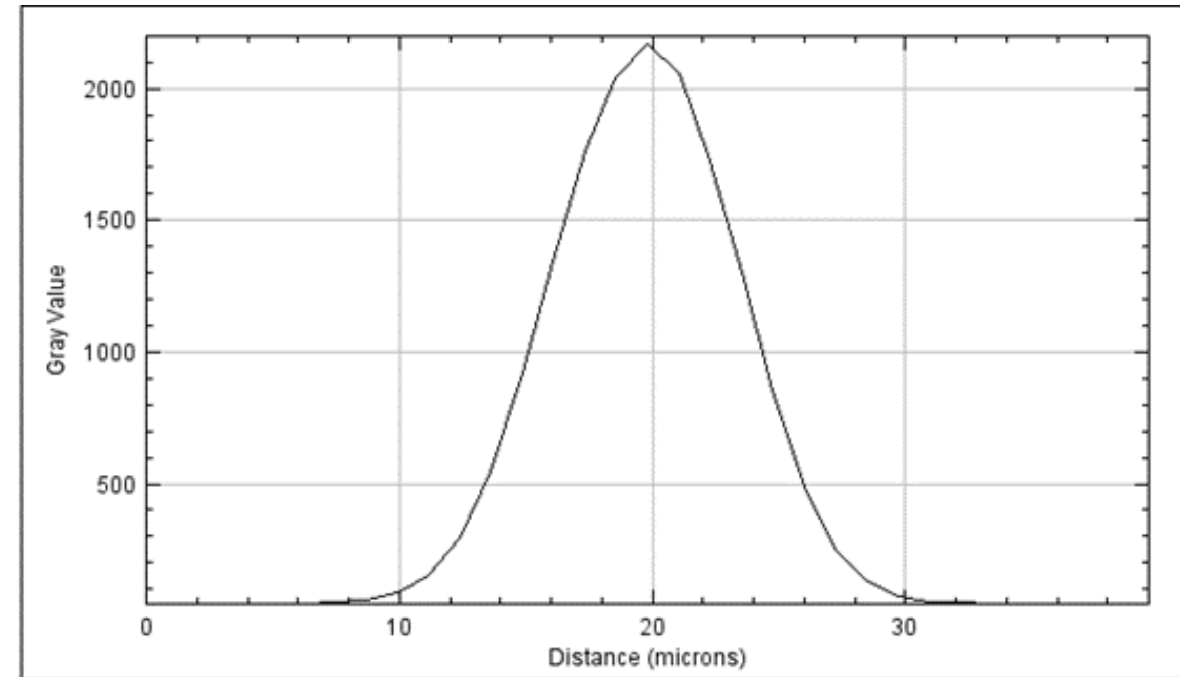
IV/ Quantification: Find Maxima



Even more efficient
after a gaussian blur

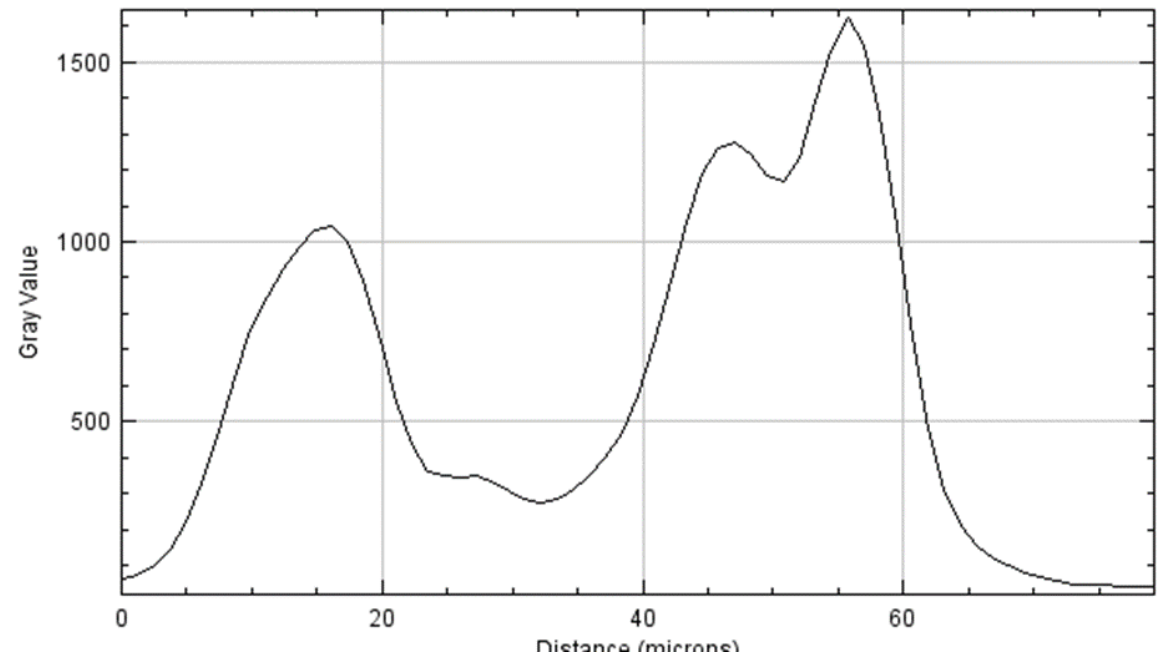
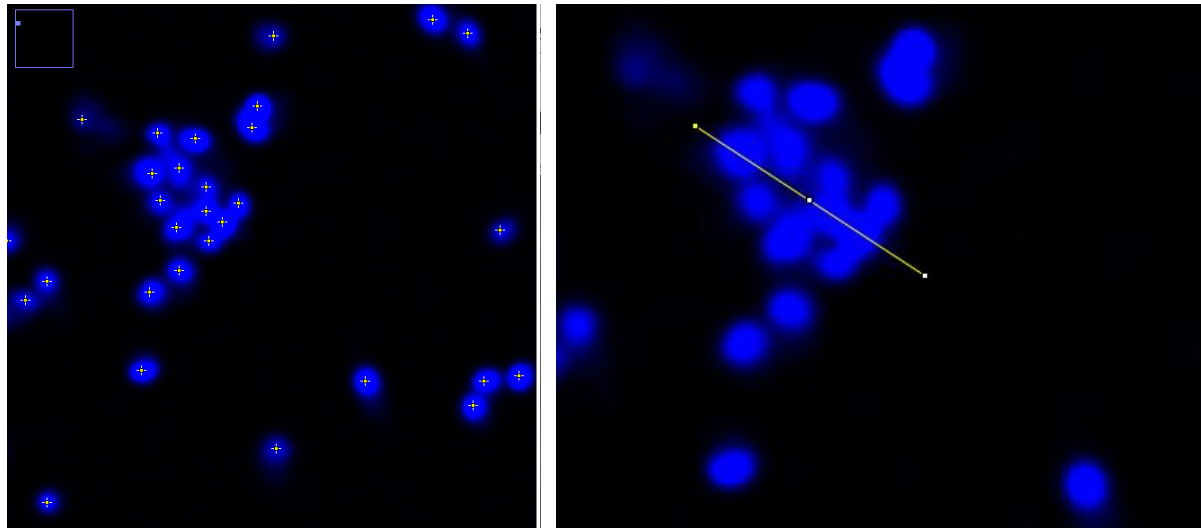


Analyze > Plot Profile (k)



IV/ Quantification: Find Maxima

Even more efficient
after a gaussian blur





IV/ Quantification: Find maxima: *Hands-on!*

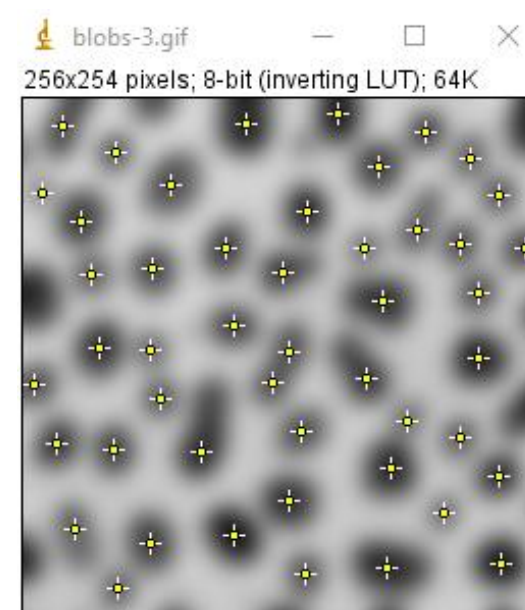
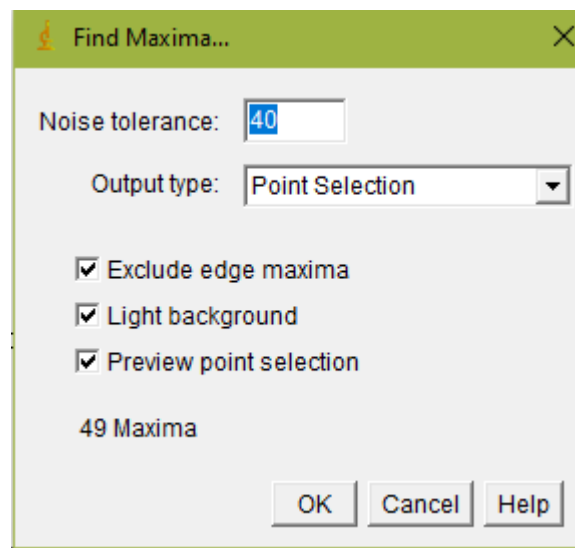
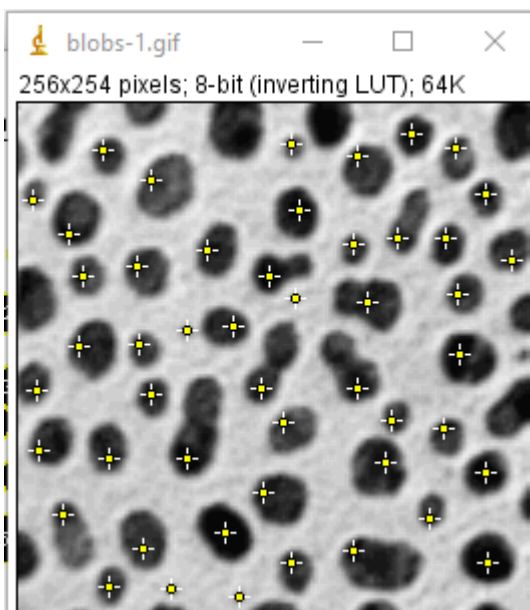
Exercise 2b

Step 3

Process > Filters > Gaussian Blur...
sigma = to radius of objects

Step 4

Process > Find Maxima... Try
different parameters!



Hands-on 2: Quantify the number of red spots per nucleus

Hands-on 2: Quantify the number of red spots per nucleus

Image: PLA.tif

Deciding: 3D or 2D? If 2D max intensity projection?

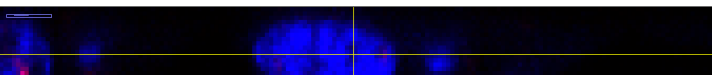
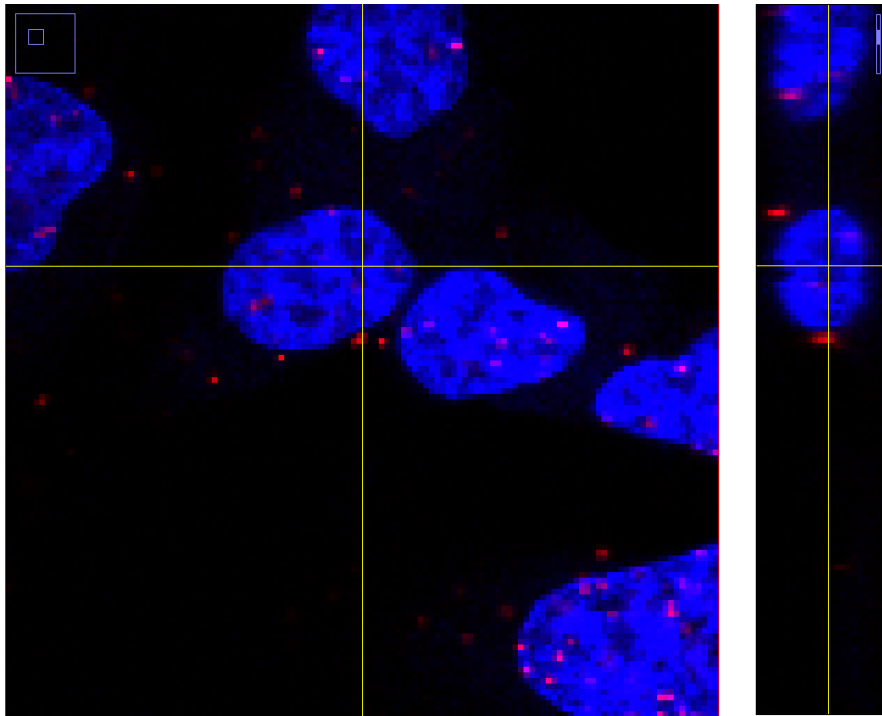


Image › Stacks › Orthogonal Views

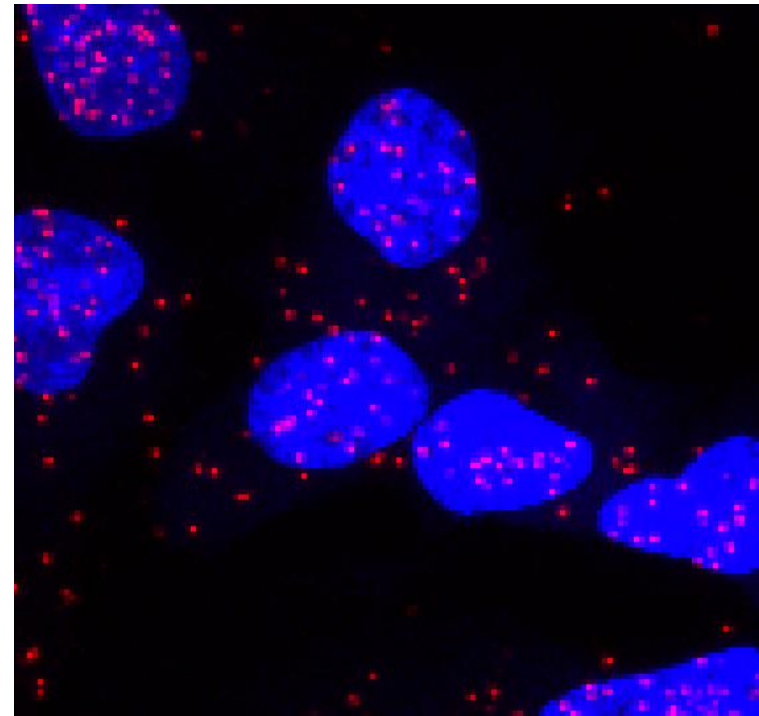


Image › Stacks › Z Project...
(max projection)

Hands-on 2: Quantify the number of red spots per nucleus

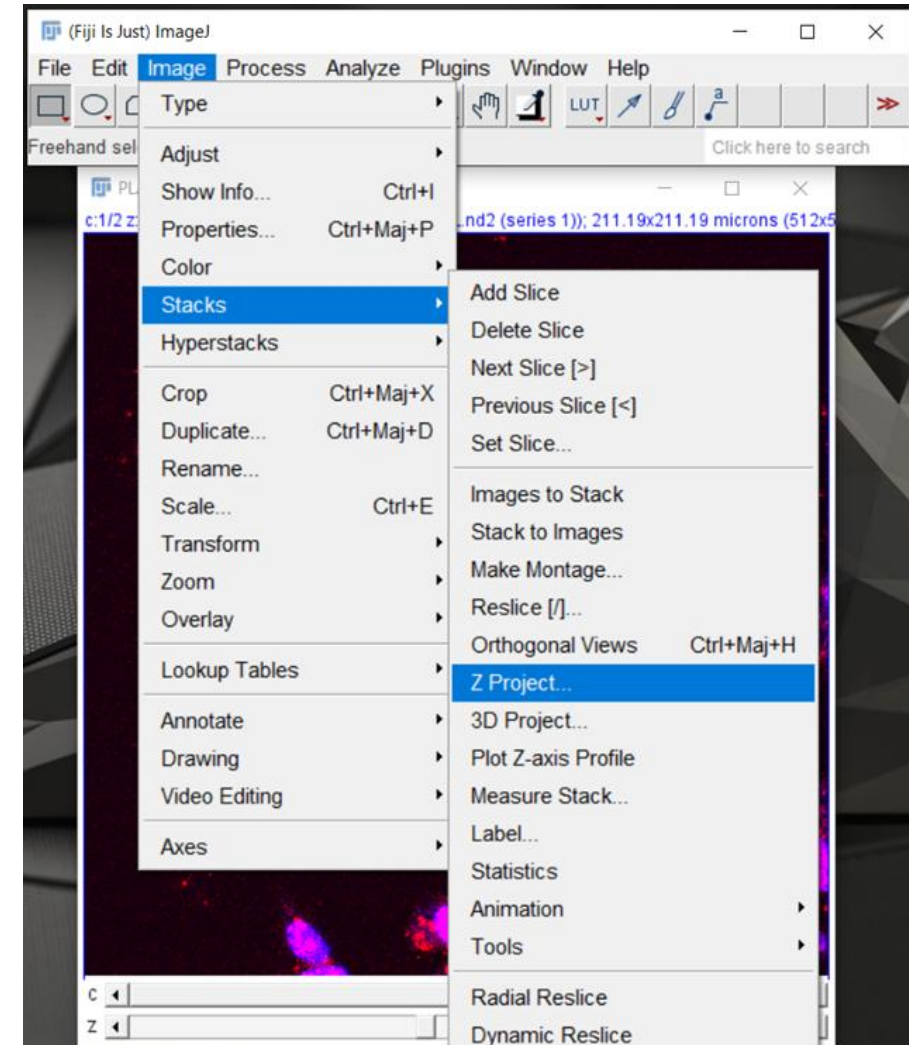
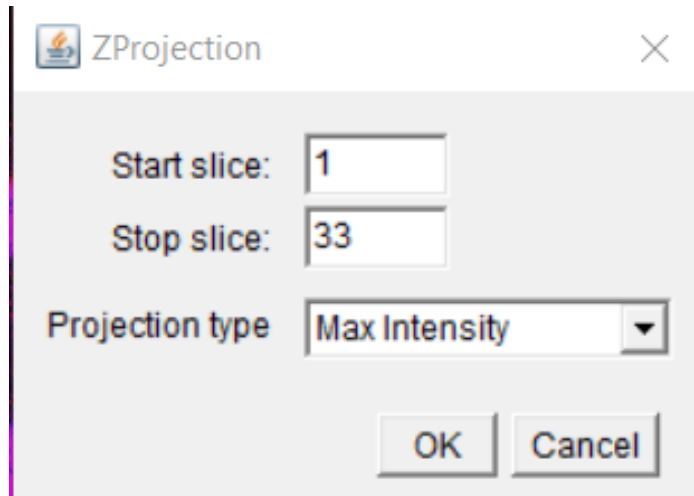
Image: PLA.tif

Step 1

Z projection in maximum intensity

Step 2

Close original image

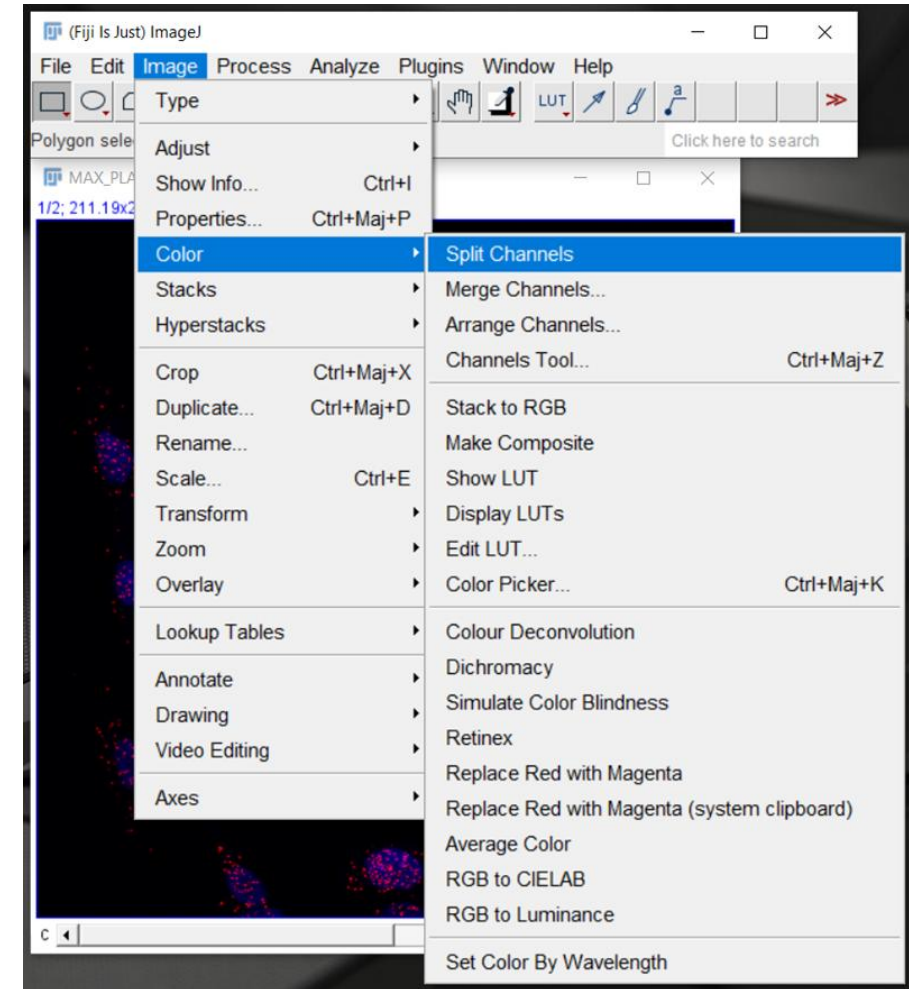


Hands-on 2: Quantify the number of red spots per nucleus

Image: PLA.tif

Step 3

Split Channels



Hands-on 2: Quantify the number of red spots per nucleus

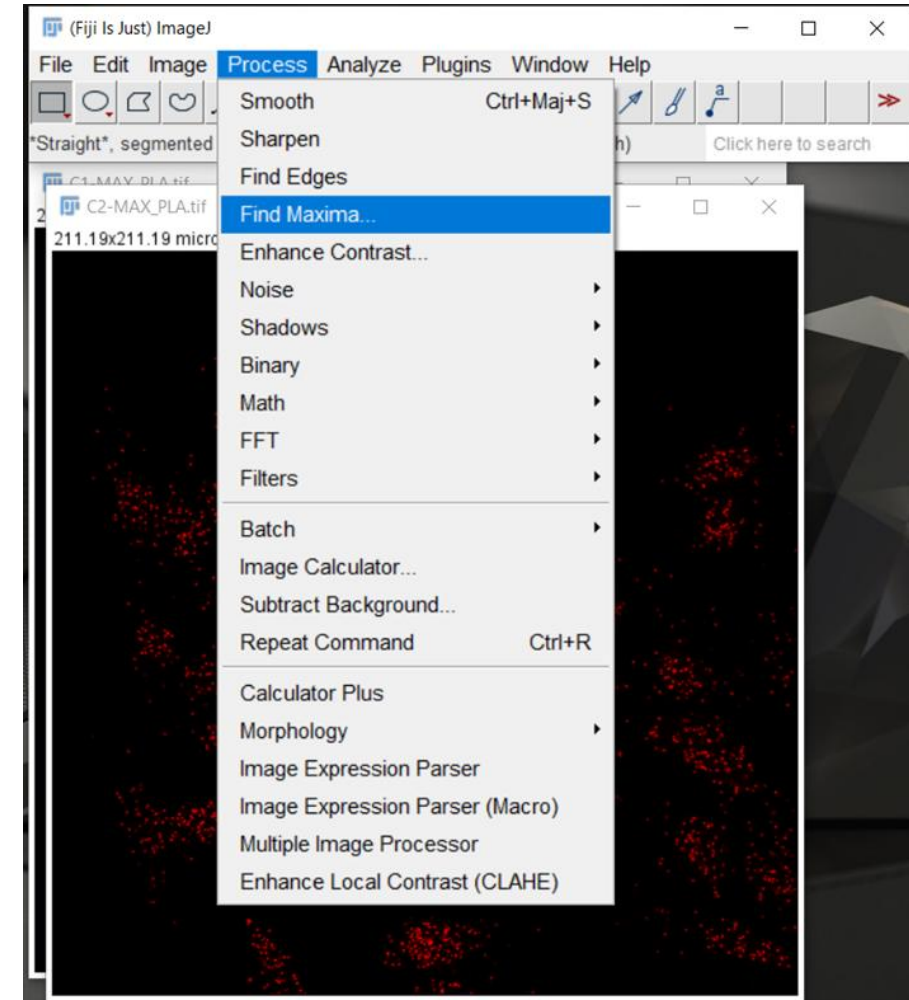
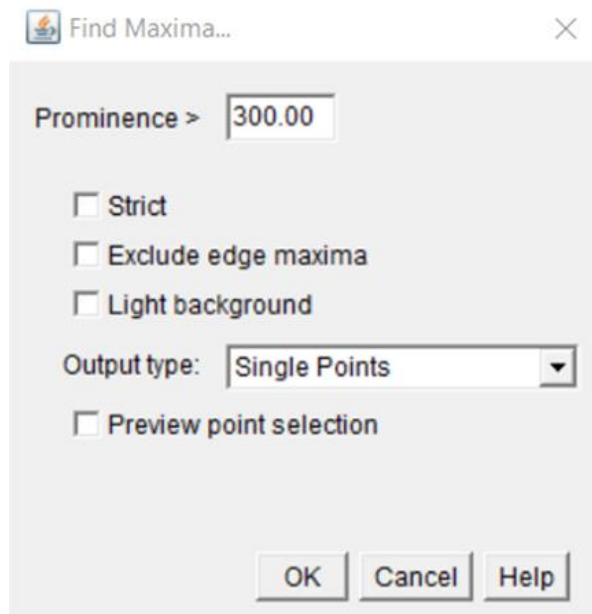
Image: PLA.tif

Step 4

Find Maxima on C2, with Single Points image as output

Step 5

Close C2

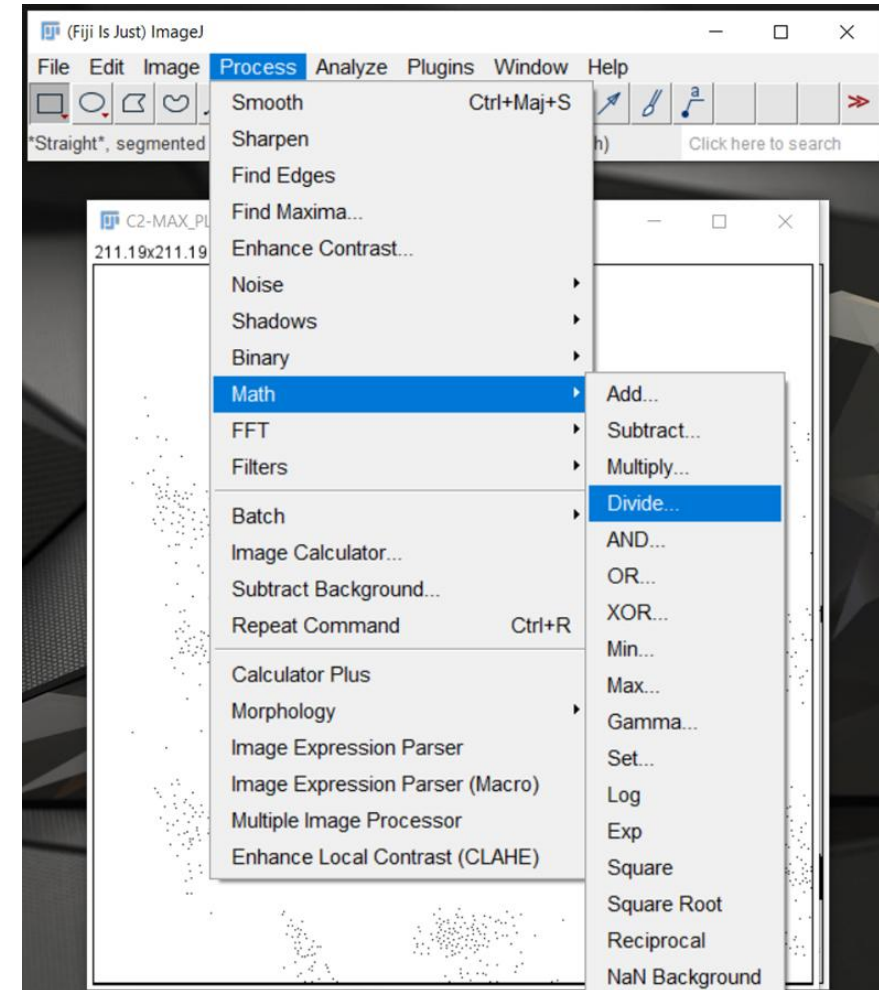
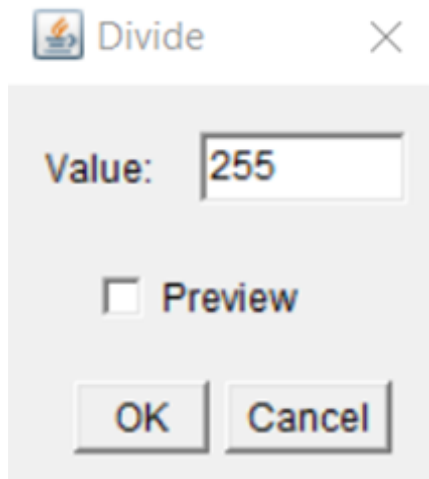


Hands-on 2: Quantify the number of red spots per nucleus

Image: PLA.tif

Step 6

Divide values by 255

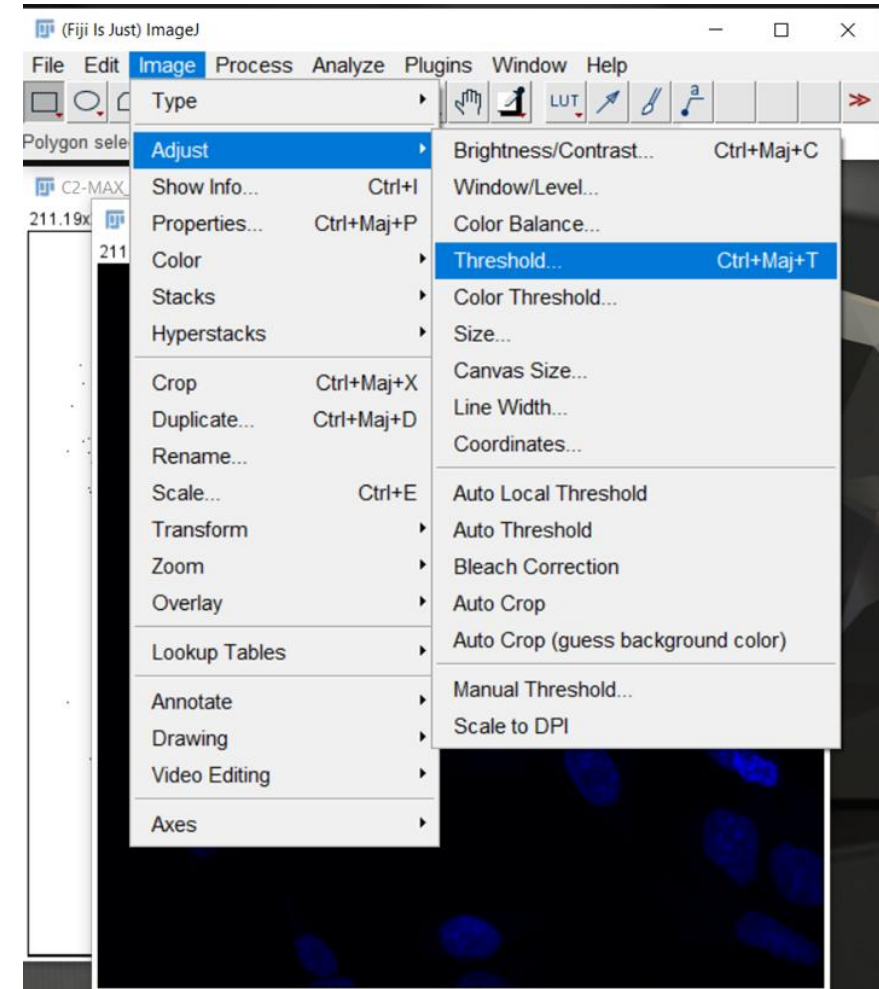
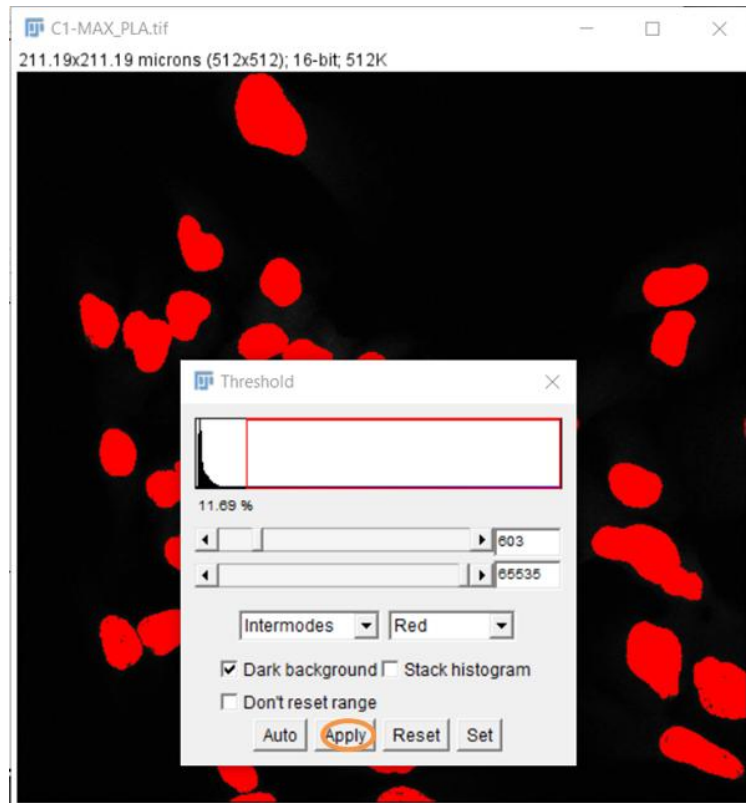


Hands-on 2: Quantify the number of red spots per nucleus

Image: PLA.tif

Step 7

Segment C1

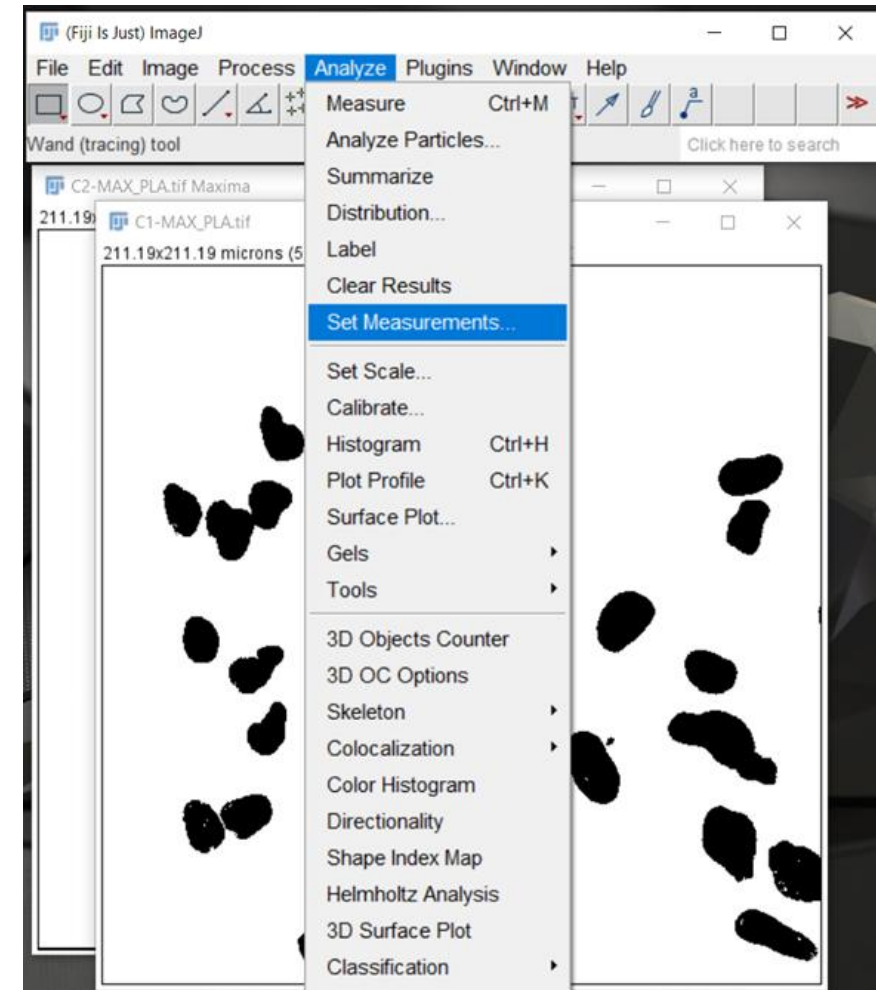
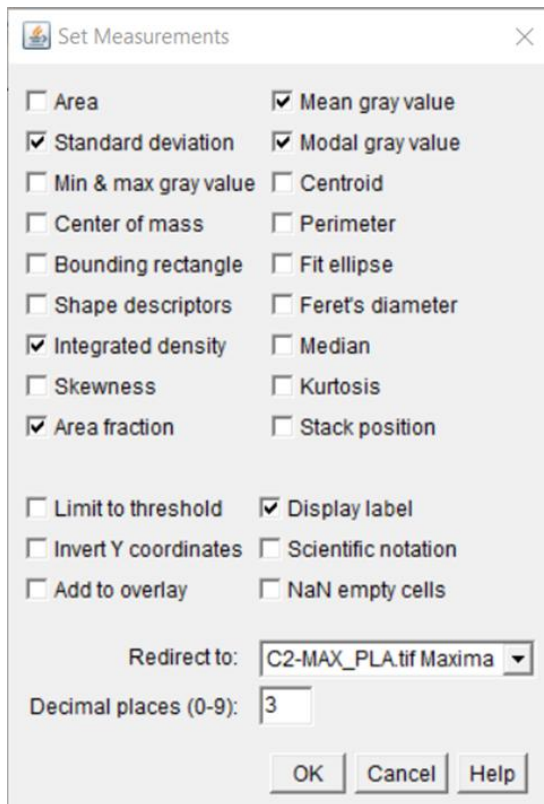


Hands-on 2: Quantify the number of red spots per nucleus

Image: PLA.tif

Step 8

Set Measurements with Integrated density and redirect towards C2



Hands-on 2: Quantify the number of red spots per nucleus

Image: PLA.tif

Step 9

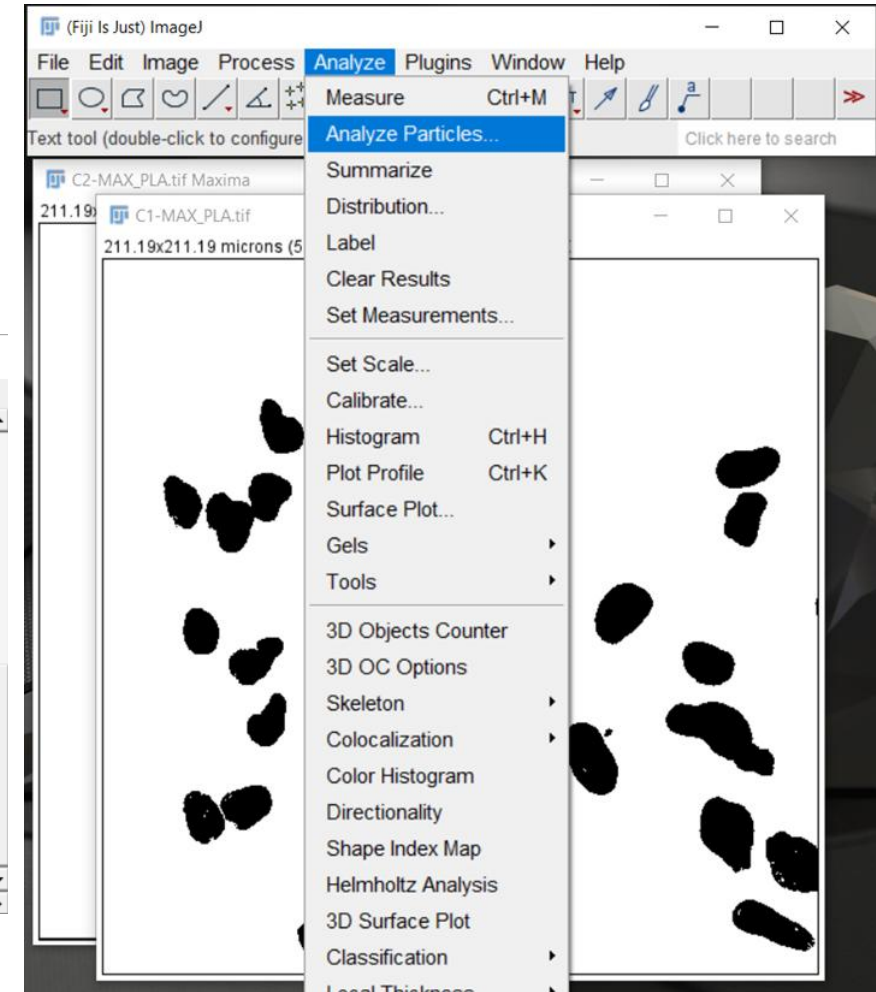
Analyze Particles

The 'Analyze Particles' dialog box is shown on the left, with the following settings:

- Size (micron^2): 0-Infinity
- ☐ Pixel units
- Circularity: 0.00-1.00
- Show: Nothing
- ☒ Display results
- ☐ Exclude on edges
- ☒ Clear results
- ☐ Include holes
- ☐ Summarize
- ☐ Record starts
- ☐ Add to Manager
- ☐ In situ Show

The 'Results' table is shown on the right, with a red box highlighting the 'RawIntDen' column:

File	Edit	Font	Results				
	Label	Mean	StdDev	Mode	IntDen	%Area	RawIntDen
18	C2-MAX_PLA.tif Maxima	0.042	0.202	0	11.910	4.242	70
19	C2-MAX_PLA.tif Maxima	0.049	0.217	0	5.955	4.930	35
20	C2-MAX_PLA.tif Maxima	0.054	0.227	0	7.146	5.440	42
21	C2-MAX_PLA.tif Maxima	0.030	0.172	0	10.889	3.042	64
22	C2-MAX_PLA.tif Maxima	0.033	0.180	0	8.167	3.347	48.000
23	C2-MAX_PLA.tif Maxima	0.000	0.000	0	0.000	0.000	0.000
24	C2-MAX_PLA.tif Maxima	0.043	0.203	0	12.080	4.303	71.000
25	C2-MAX_PLA.tif Maxima	0.030	0.171	0	7.826	3.022	46.000
26	C2-MAX_PLA.tif Maxima	0.034	0.181	0	7.316	3.373	43.000
27	C2-MAX_PLA.tif Maxima	0.035	0.183	0	6.805	3.466	40.000
28	C2-MAX_PLA.tif Maxima	0.066	0.249	0	11.569	6.628	68.000
29	C2-MAX_PLA.tif Maxima	0.045	0.208	0	4.594	4.515	27.000
30	C2-MAX_PLA.tif Maxima	0.033	0.179	0	1.531	3.309	9.000
31	C2-MAX_PLA.tif Maxima	0.032	0.177	0	0.851	3.205	5.000



Plan

I/ Fiji

II/ Fundamentals

III/ Quantification

Hands-on 1: Count nuclei and determine which ones are green

Hands-on 2: Quantify the number of red spots per nucleus

IV/ Intro to automation

IV/ Initiation to image analysis automation

IV/ Automation: Macro Recorder: *Hands-on!*

Step 1

Open an image

Step 2

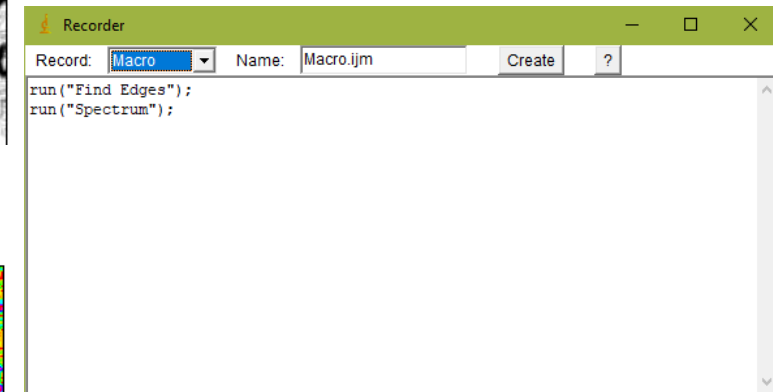
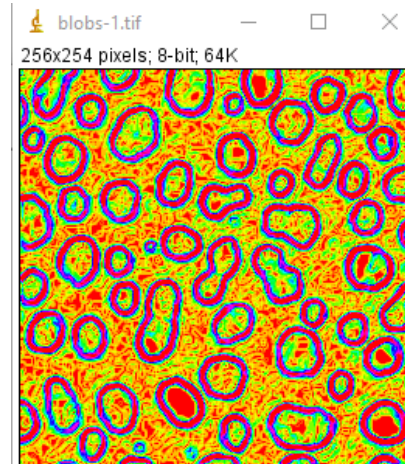
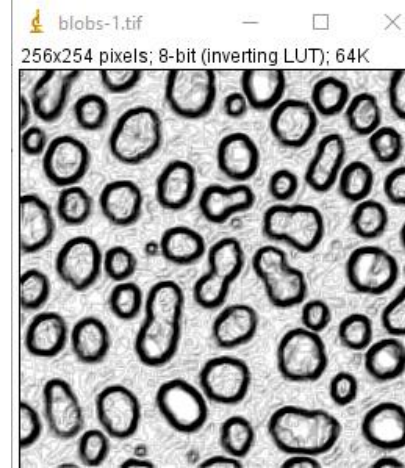
Plugins>Macros>Record...

Step 3

Apply a filter, e.g. Process > Find Edges

Step 4

Change the LUT, e.g. « Spectrum »



IV/ Automation: Macro Recorder: *Hands-on!*



Imagine that you want to apply the same processing to all the images in a folder...

Step 1

Copy the lines from the Recorder

Step 2

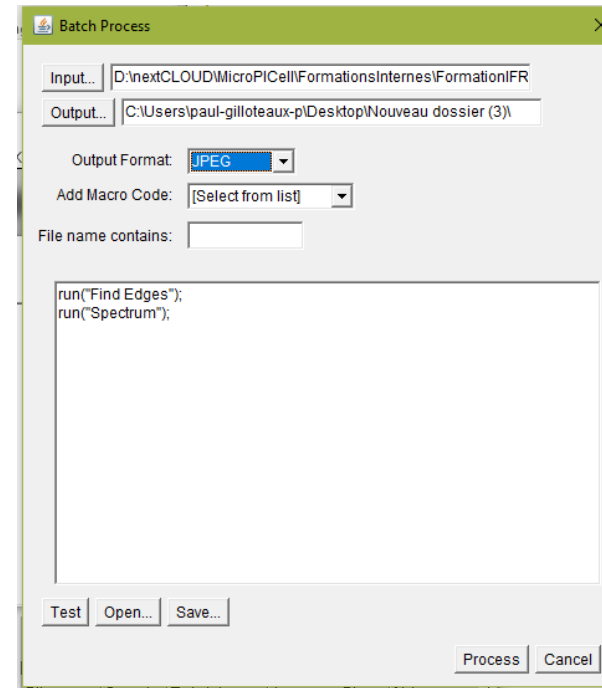
Process > Batch > Macro

Step 3

Paste the lines and select parameters

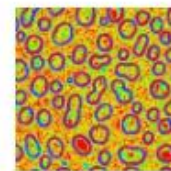
Step 4

Click on Process and close message if any pops up

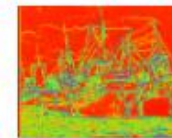


Batch Process opens all the images containing the specified expression or extension, applies the commands listed, and saves the last active image in its final state, with the specified extension. Careful to not overwrite your OG files by selecting the same output folder and extensions as your input!

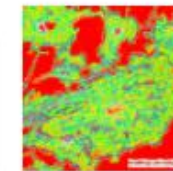
> Bureau > Nouveau dossier (3)



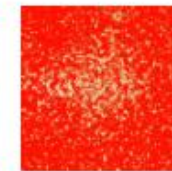
blobs.jpg



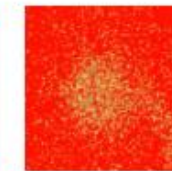
boats.jpg



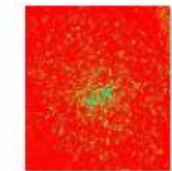
FluorescentCells.j



Melano-1.jpg



Melano-2.jpg



mitosis.jpg

IV/ Automation: Macro Recorder: *Hands-on!*

Now redo all the steps of Example 1 with the Recorder on!

Step 1

Create a directory with both melano images

Step 2

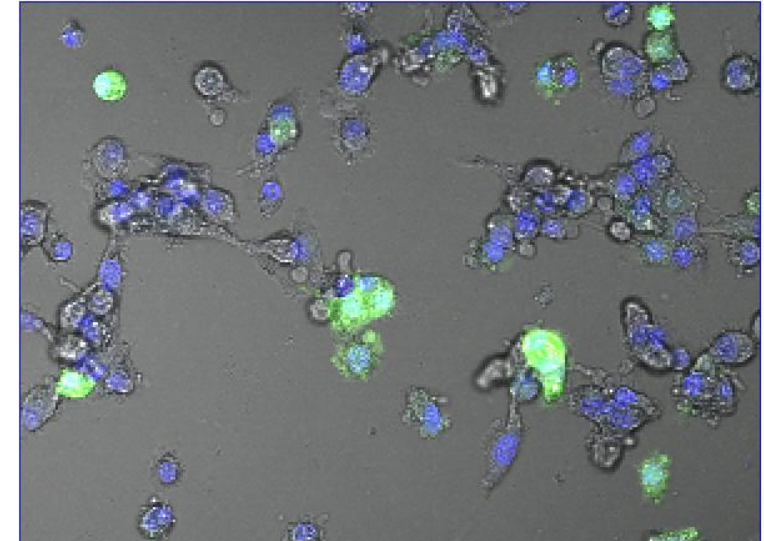
Record all steps of Exemple 1

Step 3

Create a macro, clean it

Step 4

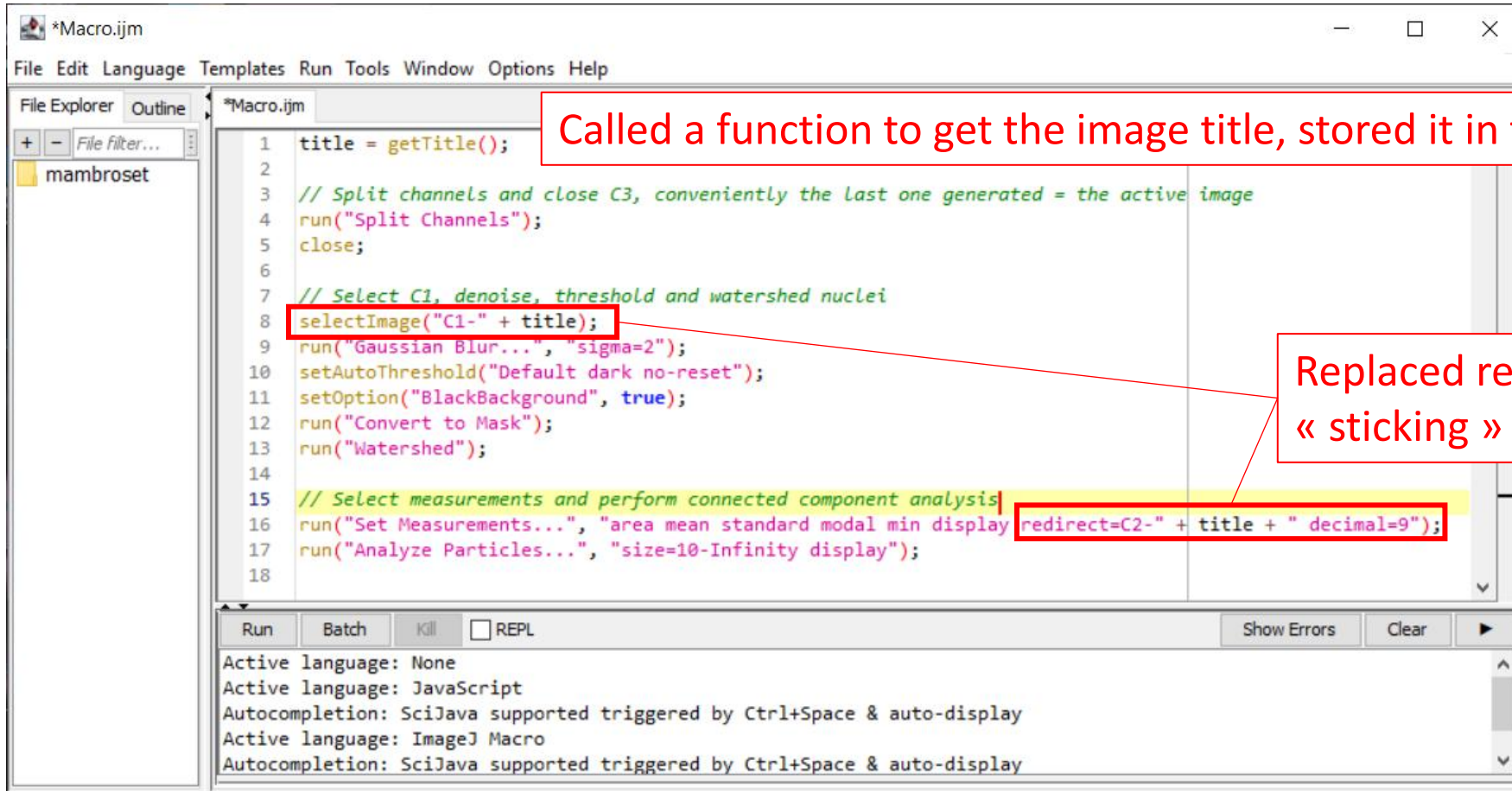
What do you think will pose a problem?



VI/ Automation: Comments and variables

Comments (//) make the code more readable.

Variables (“boxes” where to store information) make the code more general



The screenshot shows the ImageJ Macro Editor window titled '*Macro.ijm'. The menu bar includes File, Edit, Language, Templates, Run, Tools, Window, Options, and Help. The left sidebar shows a File Explorer with a folder named 'mambroset'. The main editor area contains the following macro code:

```
1 title = getTitle();
2
3 // Split channels and close C3, conveniently the last one generated = the active image
4 run("Split Channels");
5 close;
6
7 // Select C1, denoise, threshold and watershed nuclei
8 selectImage("C1-" + title);
9 run("Gaussian Blur...", "sigma=2");
10 setAutoThreshold("Default dark no-reset");
11 setOption("BlackBackground", true);
12 run("Convert to Mask");
13 run("Watershed");
14
15 // Select measurements and perform connected component analysis
16 run("Set Measurements...", "area mean standard modal min display redirect=C2-" + title + " decimal=9");
17 run("Analyze Particles...", "size=10-Infinity display");
18
```

Annotations in red boxes and lines:

- A red box around line 8, `selectImage("C1-" + title);`, with a callout box stating: "Called a function to get the image title, stored it in the « title » variable".
- A red box around line 16, `redirect=C2-" + title + " decimal=9");`, with a callout box stating: "Replaced recorded image name by « sticking » the variable with + signs".

The bottom status bar shows: Run, Batch, Kill, ☐ REPL, Show Errors, Clear, and a right arrow button. The active language is None, and the active language is JavaScript. Autocompletion is SciJava supported triggered by Ctrl+Space & auto-display. The active language is ImageJ Macro, and the autocompletion is SciJava supported triggered by Ctrl+Space & auto-display.

VI/ Automation: Macro functions reference

Info about coding macros:

<https://imagej.nih.gov/ij/developer/macro/macros.html>

Macro Functions:

- <https://imagej.nih.gov/ij/developer/macro/functions.html>
- Help > Macro Functions...

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Built-in Macro Functions

[\[A\]](#)[\[B\]](#)[\[C\]](#)[\[D\]](#)[\[E\]](#)[\[F\]](#)[\[G\]](#)[\[H\]](#)[\[I\]](#)[\[J\]](#)[\[K\]](#)[\[L\]](#)[\[M\]](#)[\[N\]](#)[\[O\]](#)[\[P\]](#) [Print List](#)
[\[Q\]](#)[\[R\]](#)[\[S\]](#)[\[T\]](#)[\[U\]](#)[\[V\]](#)[\[W\]](#)[\[X\]](#)[\[Y\]](#)[\[Z\]](#)

[A](#) [\[Top \]](#)

abs(n)

Returns the absolute value of *n*.

acos(n)

Returns the inverse cosine (in radians) of *n*.

Array Functions

These functions operate on arrays. Refer to the [ArrayFunctions](#) macro for examples.

Array.concat(array1,array2) - Returns a new array created by joining two or more arrays or values ([examples](#)). Requires 1.46c.

Array.copy(array) - Returns a copy of *array*.

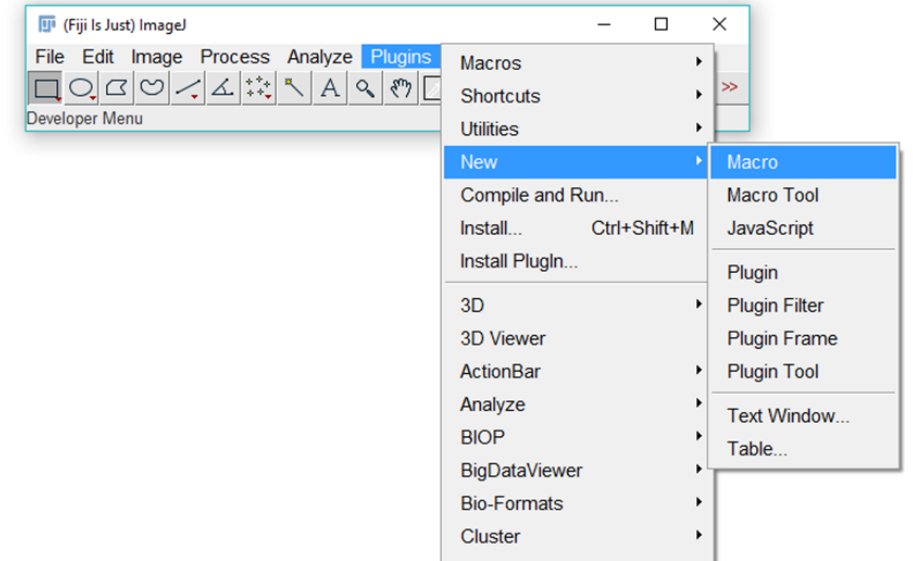
Array.fill(array, value) - Assigns the specified numeric value to each element of *array*.

Array.findMaxima(array, tolerance) - Returns an array holding the peak positions (sorted with descending strength). Tolerance is the minimum amplitude difference needed to separate two peaks. There is an optional 'excludeOnEdges' argument that defaults to 'true'. [Examples](#). Requires 1.48c.

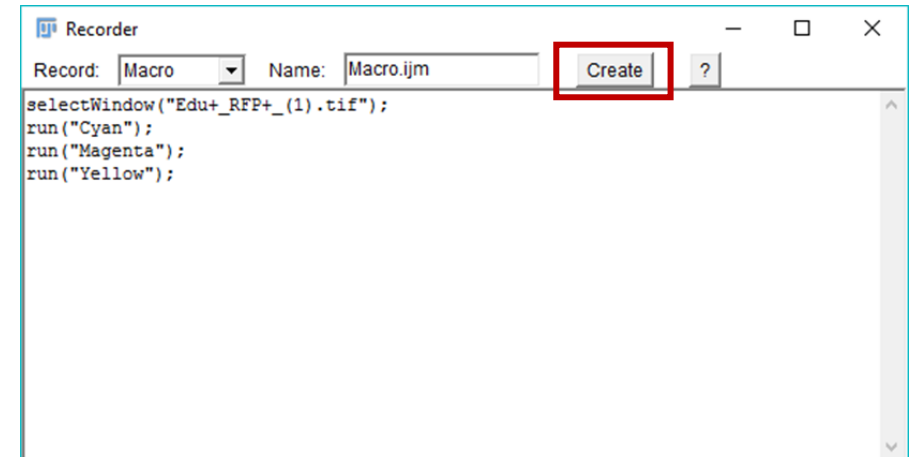
Some notions of scripting

Scripting: opening the script editor

To open the script editor:
Plugins > New > Macro



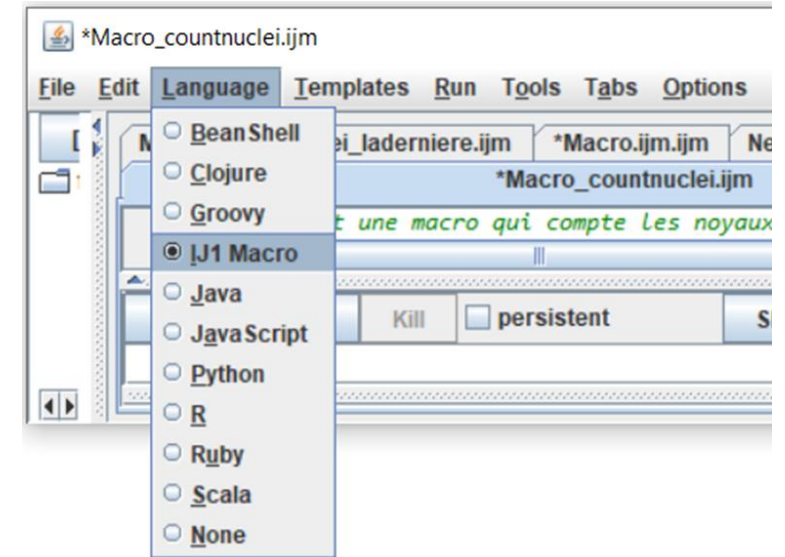
Or click on Create in the Recorder



Scripting: opening the script editor

Be sure to select the correct language!

This enables autocompletion and facilitates scripting



Scripting: *Hands-on!*

Goal: Print « Hello World! » to the Log window

Step 1

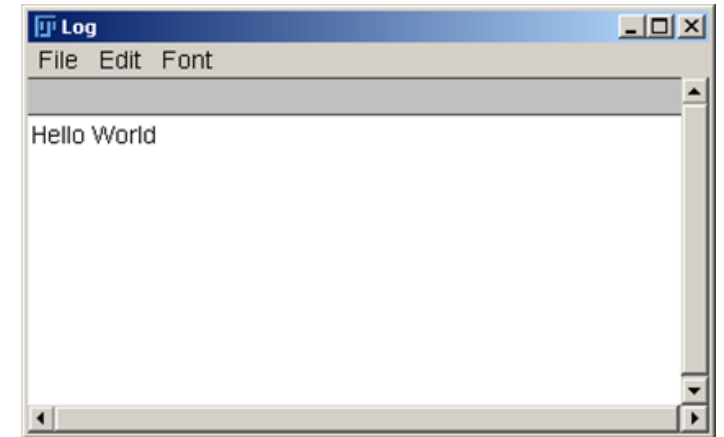
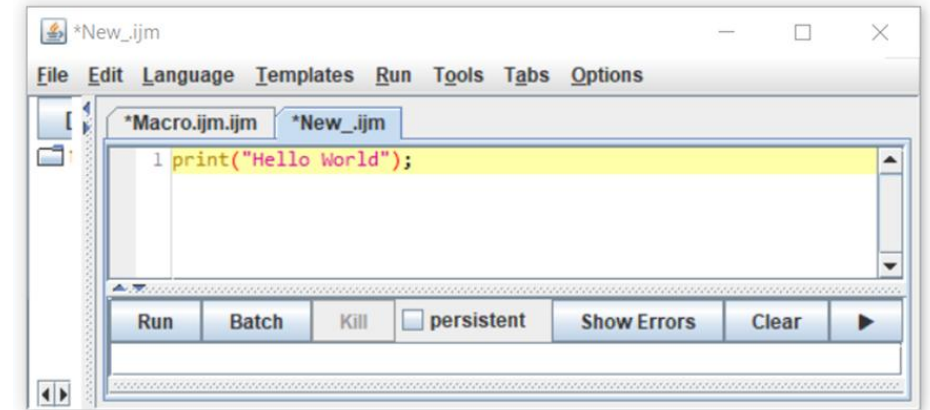
Create a New Macro:
Plugins > New > Macro

Step 2

Write `print("Hello World!");`

Step 3

Click on Run!



Scripting: Functions, parameters and data type

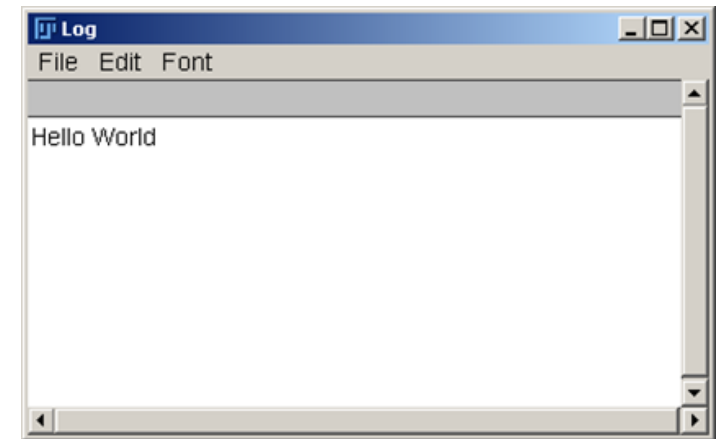
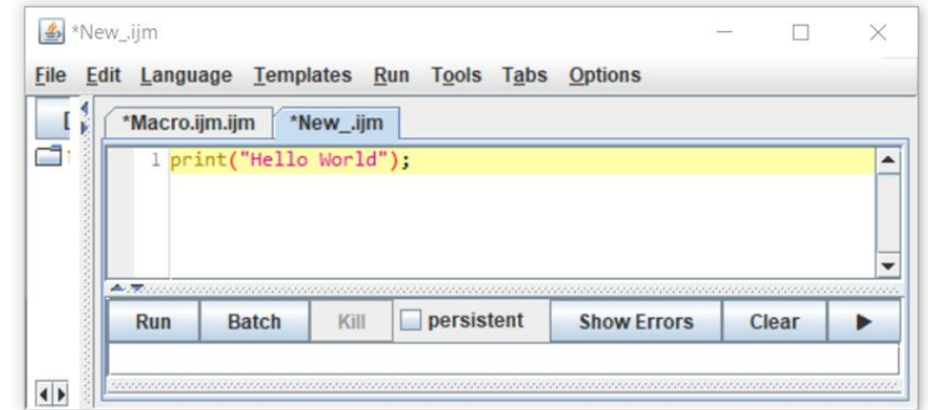
```
print("Hello World!");
```

Name of the function: `print()`

Parameter of the function: `"Hello World!"`

`"Hello World!"` is a `"string"` of characters and `"strings"` are always inside quotes

The semicolon indicates that the function call ends here



Scripting: Functions, parameters and data type

We use **functions** to do things

e.g. `run("Next Slice [>]");`

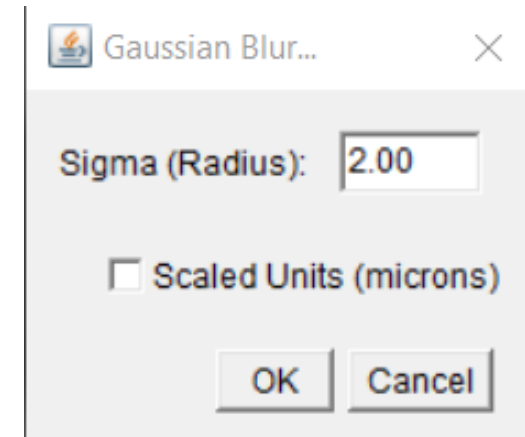
Sometimes they need **parameters** as well

e.g. whenever a graphical user interface (GUI) is used to input additional parameters, it would translate like this in a macro:

`run("Gaussian Blur...", "sigma=2");`

Information that change are contained in variables

e.g. `nbChannels = 3;`



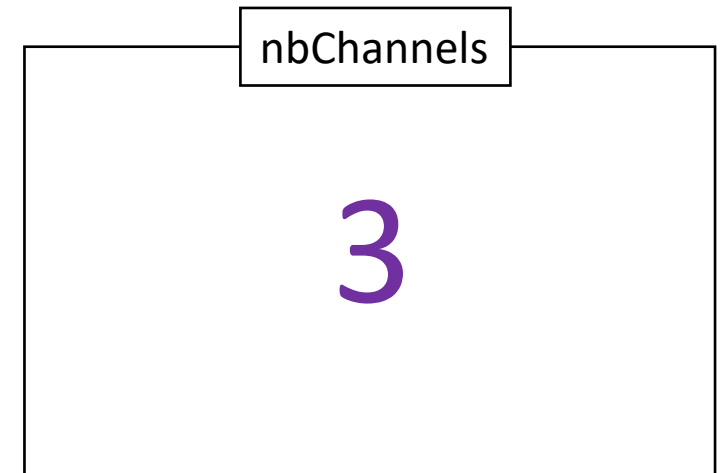
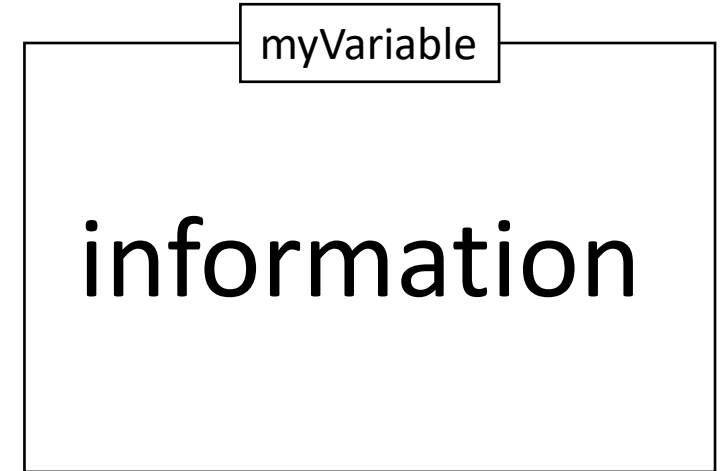
Scripting: Functions, parameters and data type

Information that change are contained in **variables**

e.g. nbChannels = 3;

Variables are like boxes with a name in which you store information

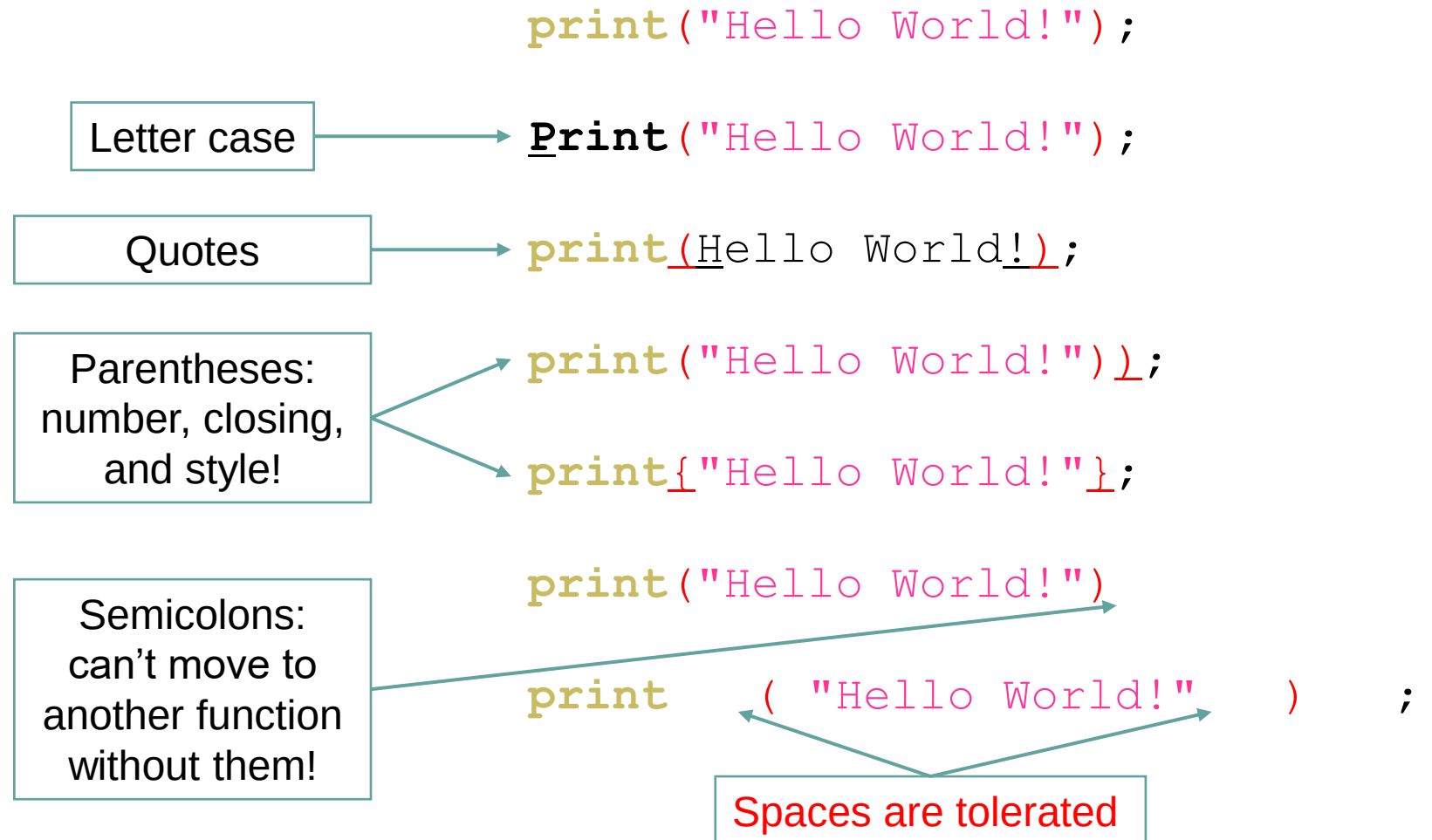
Function and variable names should be explicit about what they refer to!



Scripting: Functions, parameters and data type

The script interpreter is a bit picky about how you write your code!

Careful of:



Scripting: *Hands-on!*

Goal 1: Print « Hello World » to the Log window using a variable

Step 1

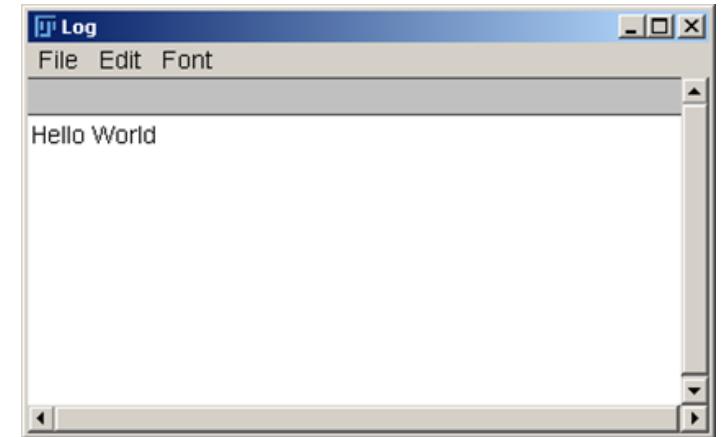
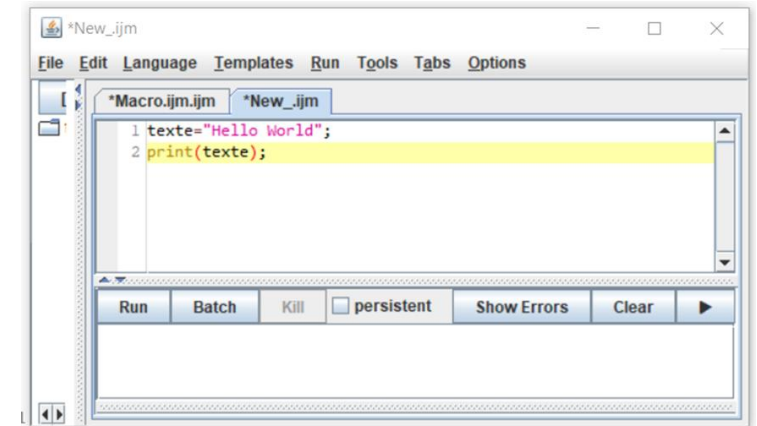
Use a variable to store the text
(Hint: the text should be a string)

Step 2

Put the variable as a parameter
of the print function

Step 3

Click on Run!



Scripting: *Hands-on!*

Goal 2: Print « Hello World » to the Log window using **two** variables

Step 1

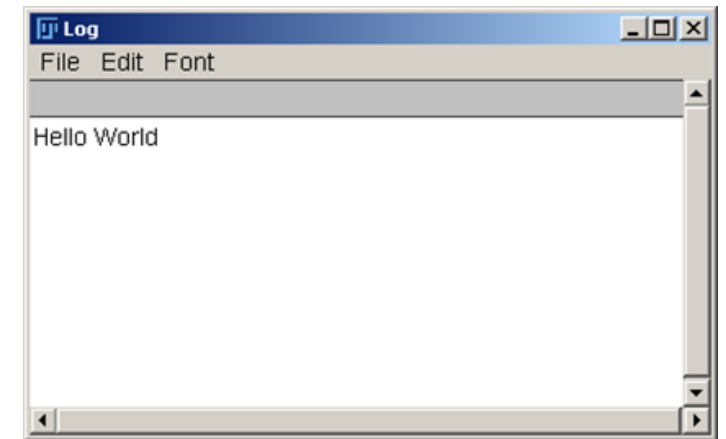
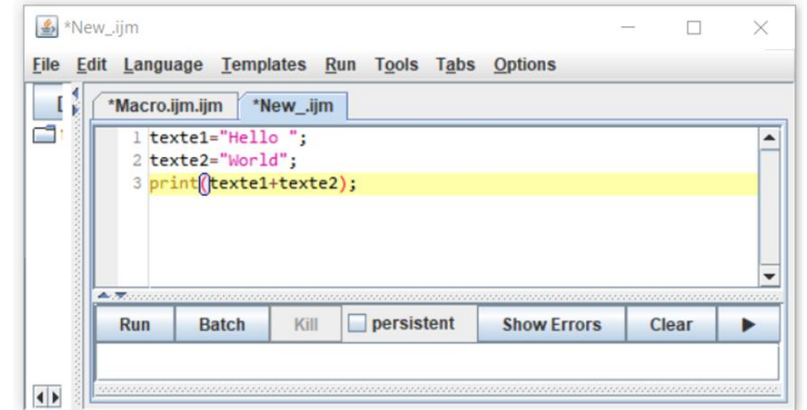
Use TWO variables to store different parts of the text

Step 2

Put the variables as parameters of the print function (Hint: the « + » sign allows to **concatenate** variables)

Step 3

Click on Run!



Scripting: *Hands-on!*

Goal 3: Print « Hello World !» to the Log window using **two** variables

Step 1

Modify the script, without adding variables, to print « Hello World ! »

Step 2

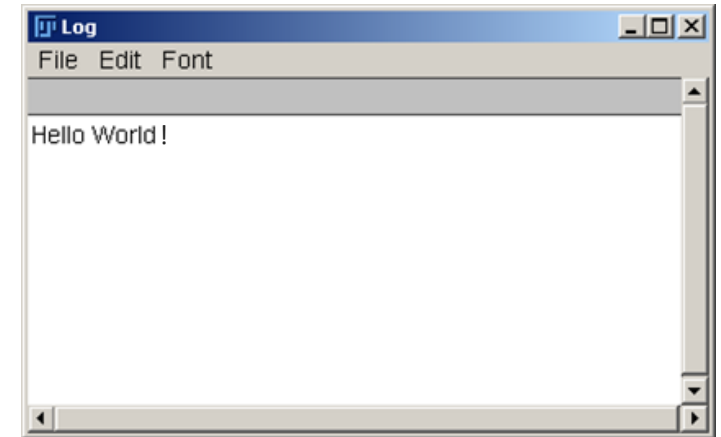
Try to find several solutions

Step 3

Click on Run!

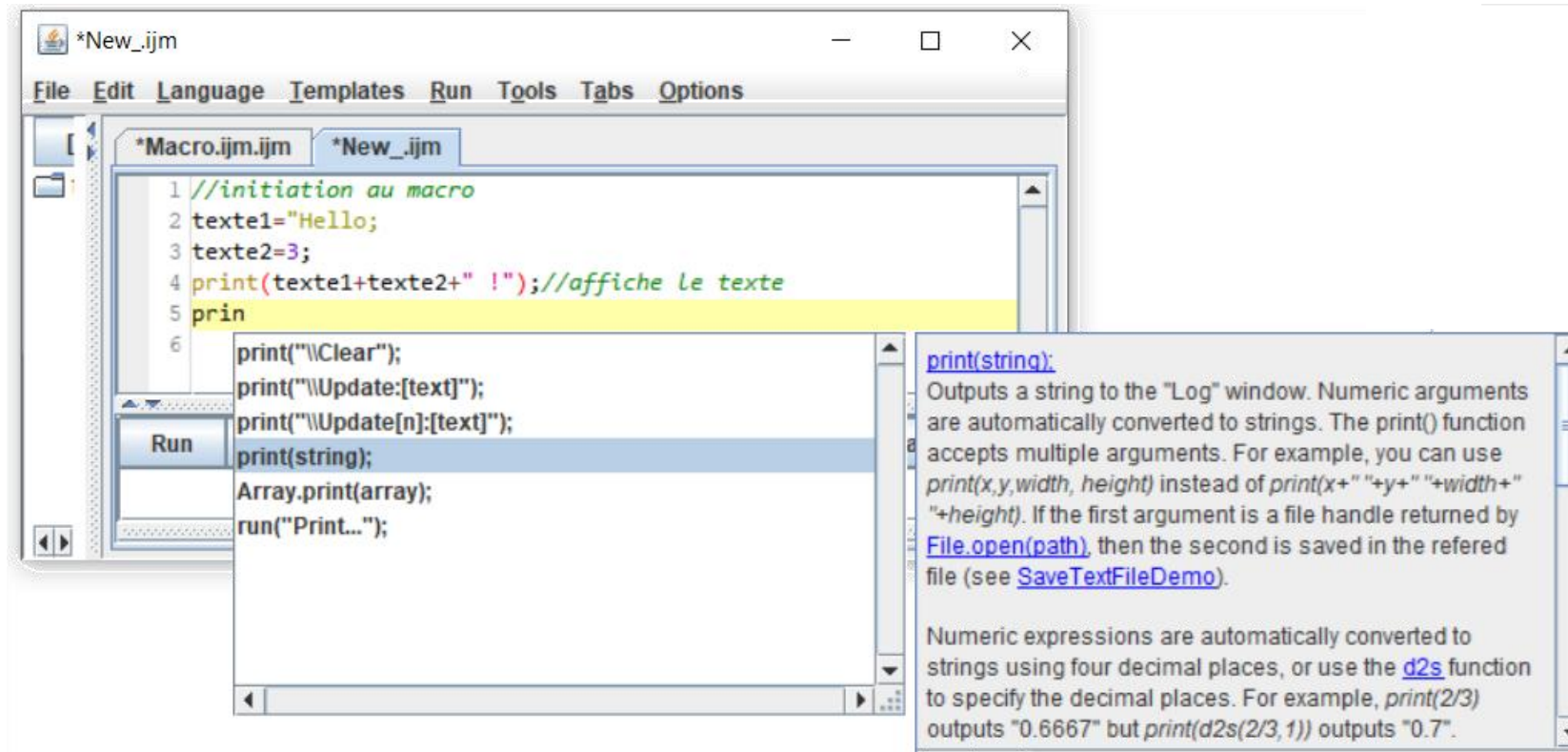


```
*New_.ijm
File Edit Language Templates Run Tools Tabs Options
*ChangeLUT3Canaux.ijm.ijm *New_.ijm
1 //initiation au macro
2 text1="Hello"
3 text2=" World"
4 print(text1+text2+" !"); //affiche le texte
5
6
7
```



Scripting: Autocompletion

Autocompletion: Helps with avoiding spelling mistakes!



Offers to complete as you type with **functions** it knows or **variables** you have created previously

Scripting: Colors

Colors: Helps with detecting scripting mistakes!

Variable

Function

String

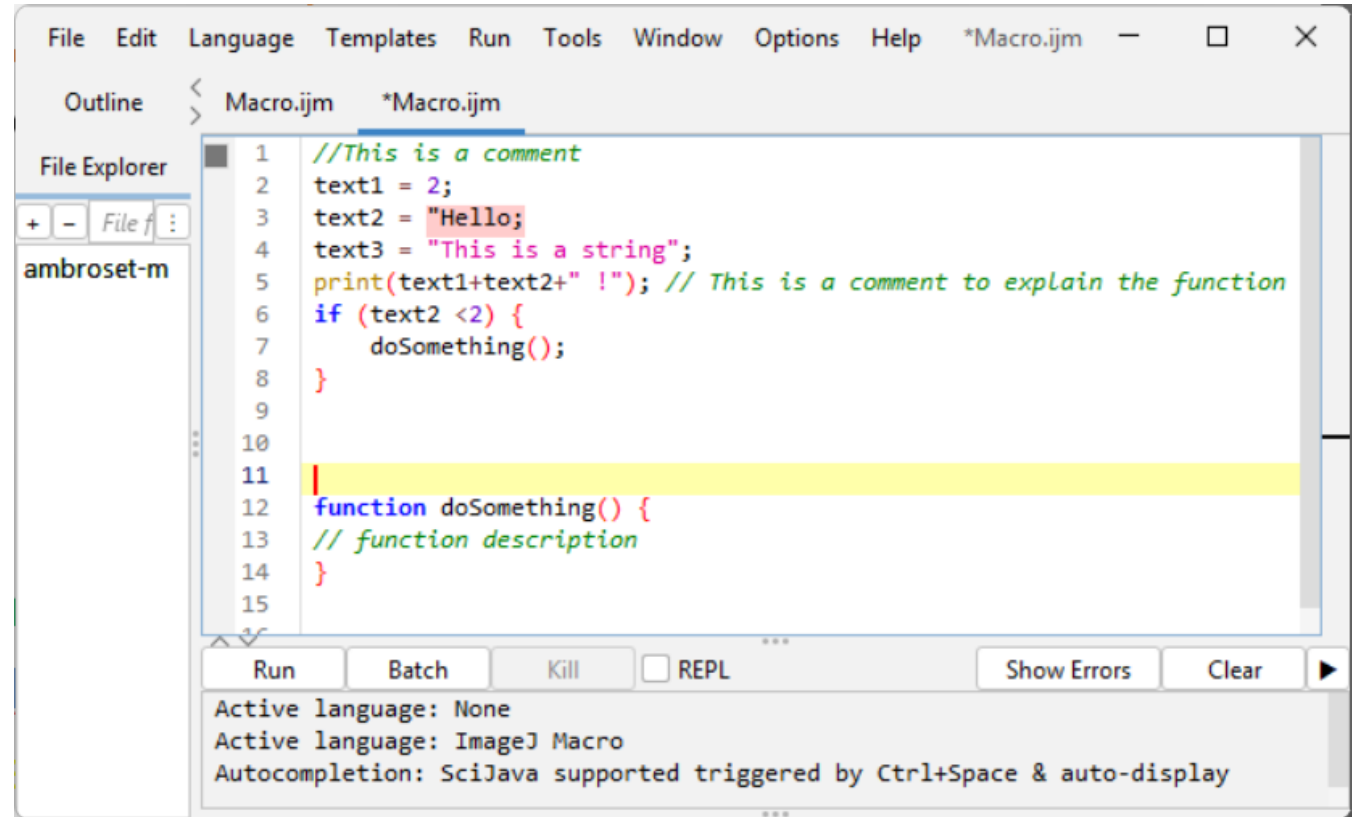
Number

Comment

Control structures: for, if, while ...

Quoting mistake

Parentheses, brackets and braces



The screenshot shows the SciJava IDE interface. The main editor window displays a file named `Macro.ijm` with the following code:

```
1 //This is a comment
2 text1 = 2;
3 text2 = "Hello;
4 text3 = "This is a string";
5 print(text1+text2+" !"); // This is a comment to explain the function
6 if (text2 <2) {
7     doSomething();
8 }
9
10
11
12 function doSomething() {
13     // function description
14 }
15
```

The code is syntax-highlighted: comments are green, strings are pink, numbers are blue, and function names are blue. The word `function` on line 12 is highlighted in yellow. The IDE also shows a menu bar (File, Edit, Language, Templates, Run, Tools, Window, Options, Help), a toolbar, and a status bar at the bottom with the text: "Active language: None", "Active language: ImageJ Macro", and "Autocompletion: SciJava supported triggered by Ctrl+Space & auto-display".

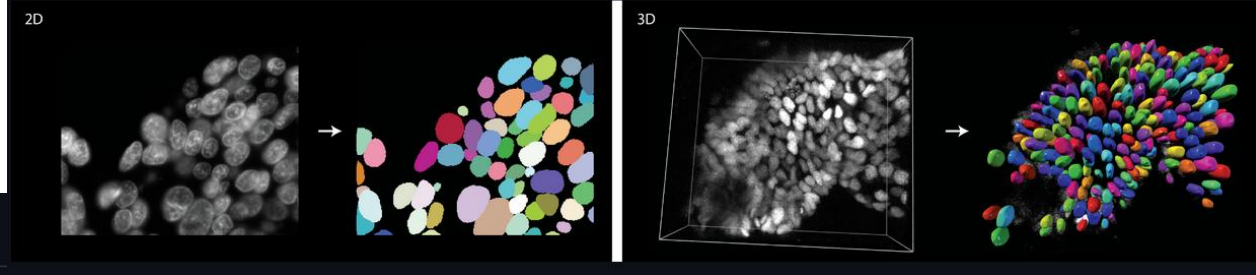
Bonus: Intro to Machine Learning: Segmentation: StarDist



<https://github.com/qupath/qupath-extension-stardist>

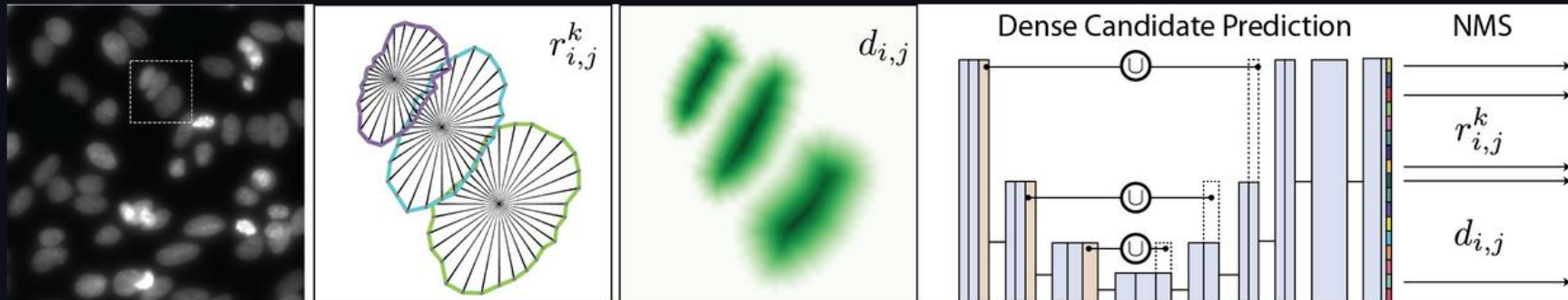
<https://github.com/stardist/stardist>

StarDist - Object Detection with Star-convex Shapes



Overview

The following figure illustrates the general approach for 2D images. The training data consists of corresponding pairs of input (i.e. raw) images and fully annotated label images (i.e. every pixel is labeled with a unique object id or 0 for background). A model is trained to densely predict the distances (r) to the object boundary along a fixed set of rays and object probabilities (d), which together produce an overcomplete set of candidate polygons for a given input image. The final result is obtained via non-maximum suppression (NMS) of these candidates.



If you use this extension, you should cite the original StarDist publication
•Uwe Schmidt, Martin Weigert, Coleman Broaddus, and Gene Myers.

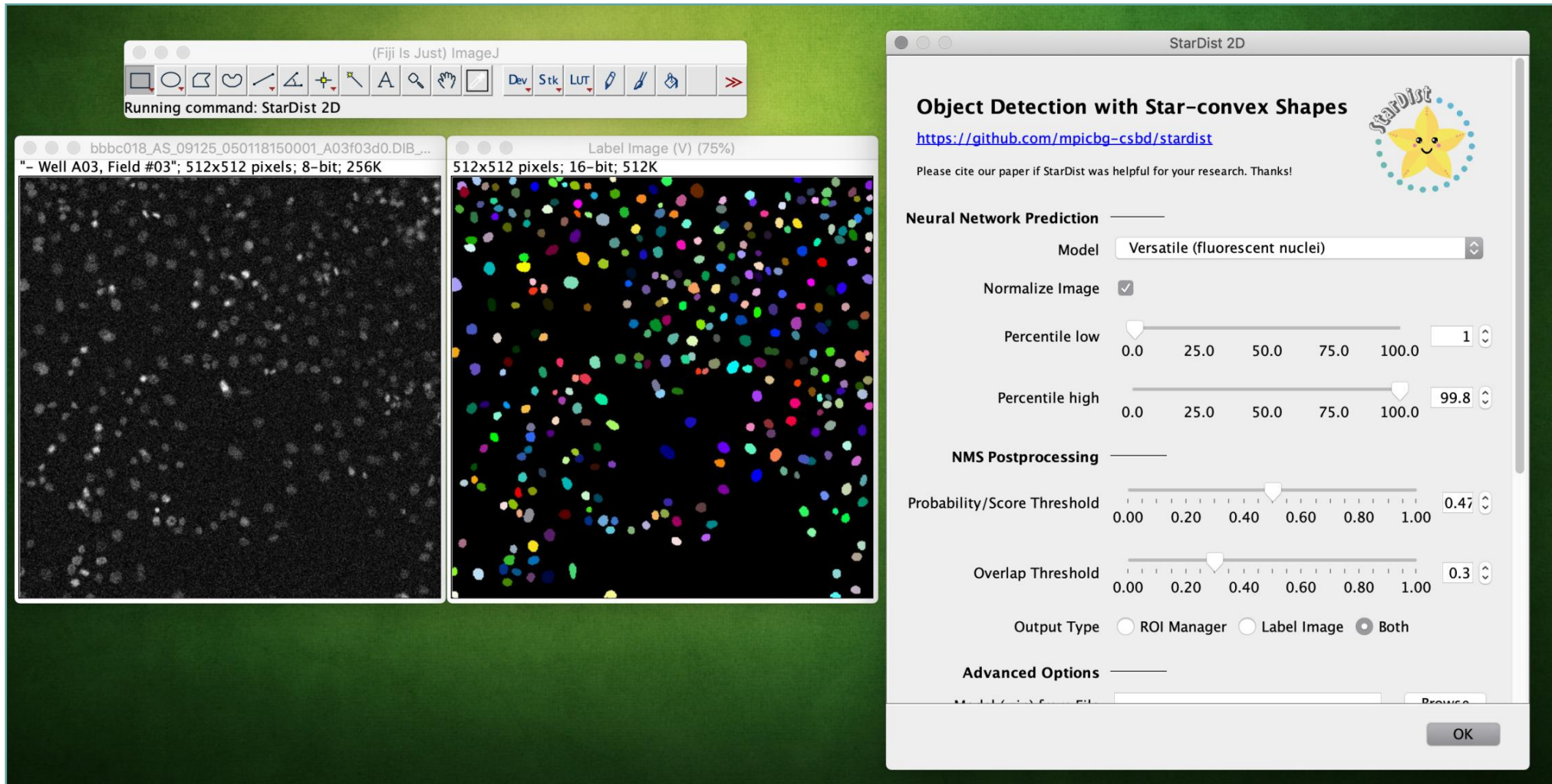
[Cell Detection with Star-convex Polygons.](#)

International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Granada, Spain, September 2018.

Bonus: Intro to Machine Learning: Segmentation: StarDist



<https://imagej.net/plugins/stardist>

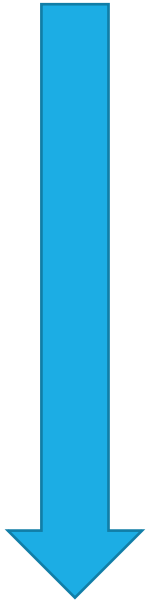


Comparison of ad hoc vs ML methods in image analysis

Differences: Paradigm shift

Human time

Traitement



Annotation

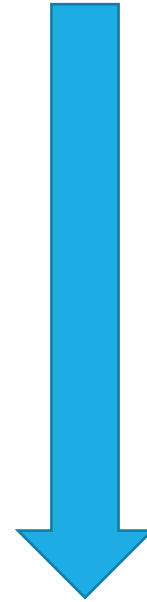
Algorithm

Machine Learning

Deep Learning

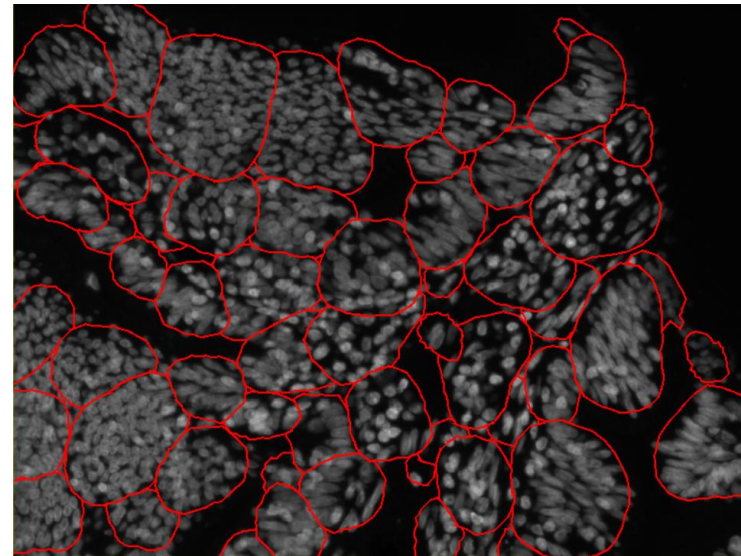
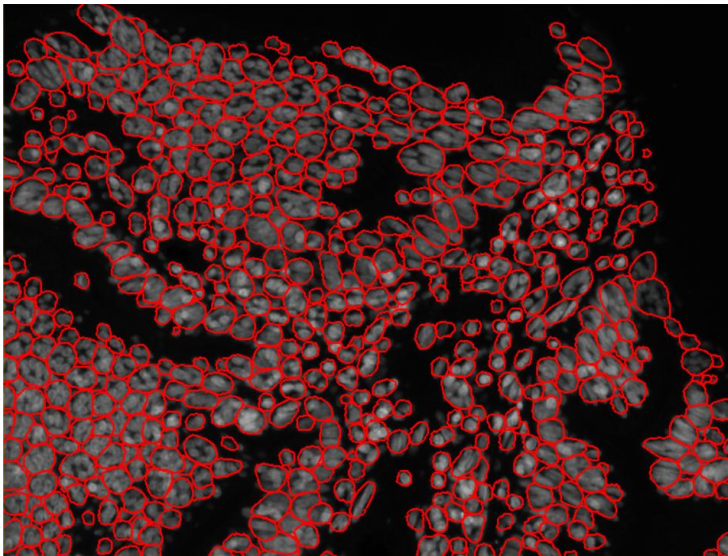
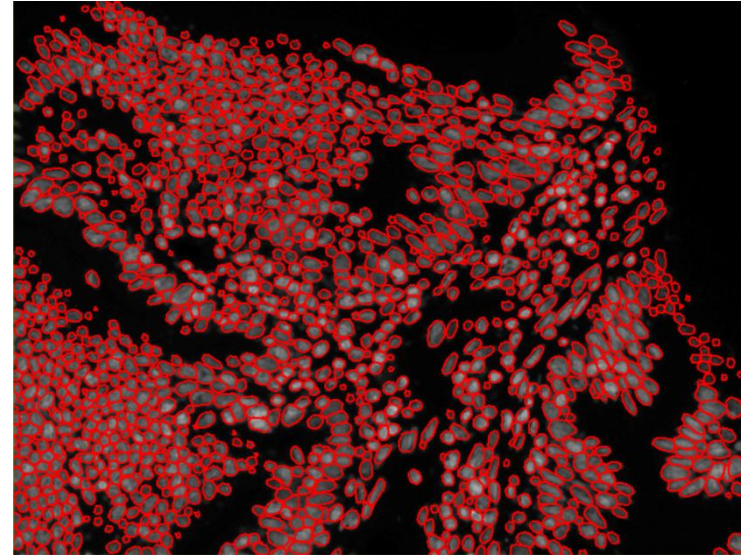
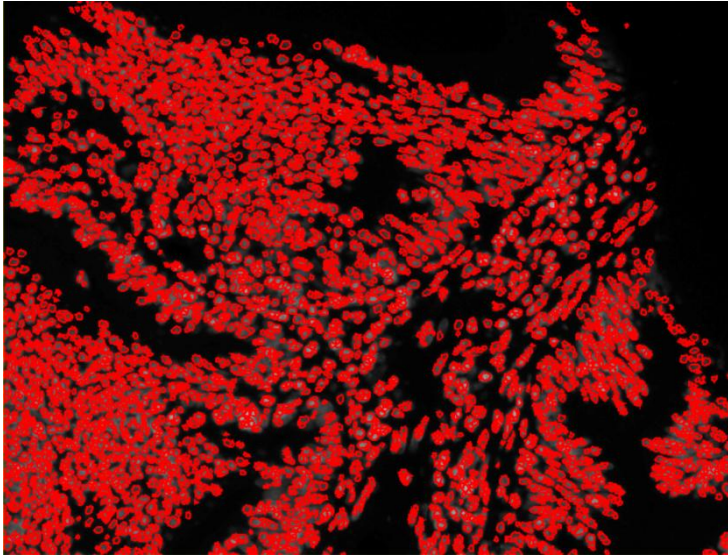
Image quantity

-



+

Warning



Warning

